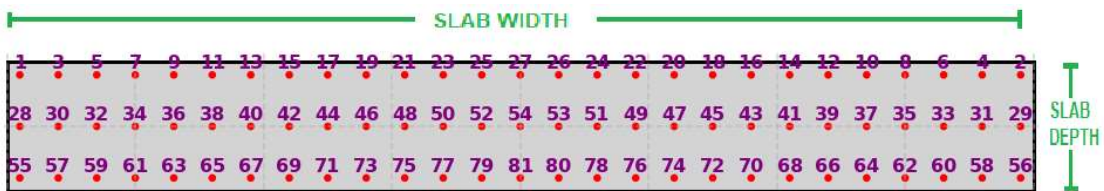
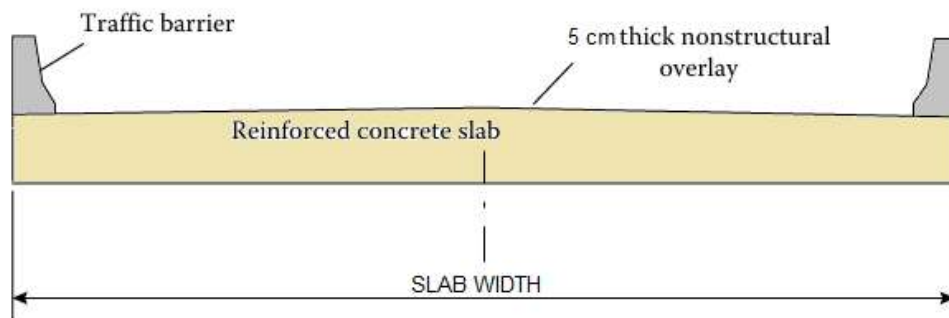
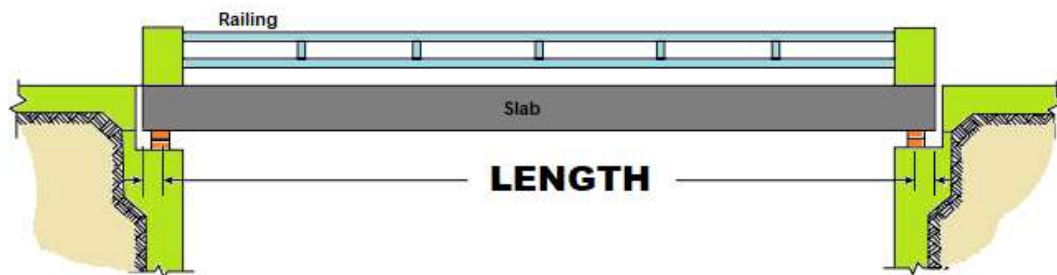


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

# **COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A CONCRETE SLAB SUPERSTRUCTURE SIMPLY SUPPORTED SHORT SPAN BRIDGE: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS**

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)





$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

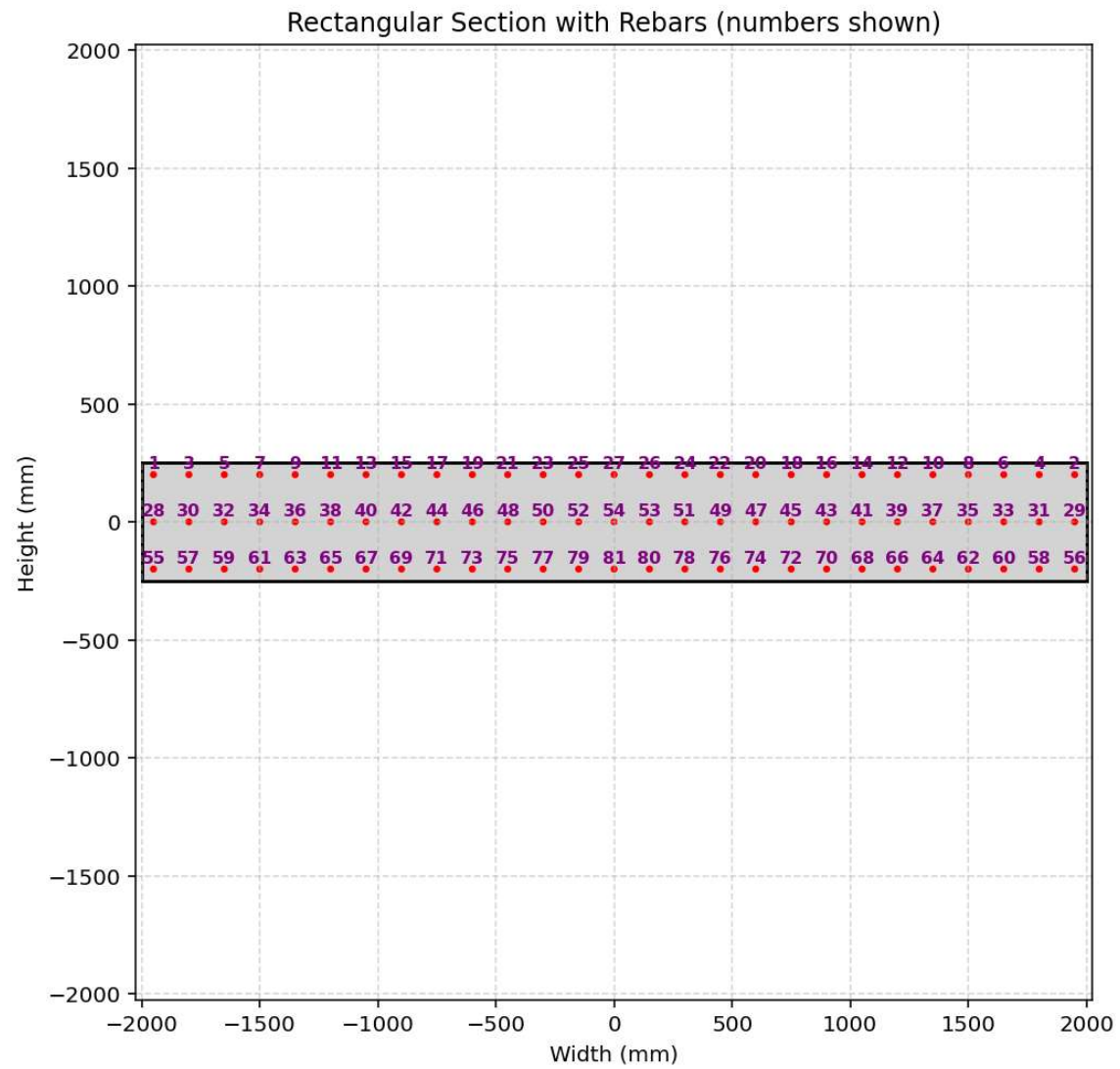
$$\text{Structure Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

$$\rho A \frac{\partial^2 v(x,t)}{\partial t^2} + EI \frac{\partial^4 v(x,t)}{\partial x^4} = P(t)$$

boundary conditions

$$v(0,t) = v(L,t) = 0, \quad \left. \frac{\partial^2 v(x,t)}{\partial x^2} \right|_{x=0} = \left. \frac{\partial^2 v(x,t)}{\partial x^2} \right|_{x=L} = 0.$$







Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES\_FILES\+EIGHT\_ANALY...IMPLY\_SUPPORTED\CONCRETE\_REC\_SECTION\_FUN\_EXTRA.py

P(t)\_BRIDGE\_CONCRETE\_SIMPLY\_SUPPORTED.py X CONCRETE\_REC\_SECTION\_FUN\_EXTRA.py X

```
126 # (diameter [mm], y-coord [mm], x-coord [mm])
127 # The coordinates below are the *centre* of each bar.
128 # You can modify them or the cover value as needed.
129 REBAR_DIA = 16.0 # [mm] Rebar Diameter
130 rebars = [
131     (REBAR_DIA, Hsec/2 - cover, -Bsec/2 + cover), # 1
132     (REBAR_DIA, Hsec/2 - cover, Bsec/2 - cover), # 2
133     (REBAR_DIA, Hsec/2 - cover, -Bsec/2 + cover + 1*150), # 3
134     (REBAR_DIA, Hsec/2 - cover, Bsec/2 - cover - 1*150), # 4
135     (REBAR_DIA, Hsec/2 - cover, -Bsec/2 + cover + 2*150), # 5
136     (REBAR_DIA, Hsec/2 - cover, Bsec/2 - cover - 2*150), # 6
137     (REBAR_DIA, Hsec/2 - cover, -Bsec/2 + cover + 3*150), # 7
138     (REBAR_DIA, Hsec/2 - cover, Bsec/2 - cover - 3*150), # 8
139     (REBAR_DIA, Hsec/2 - cover, -Bsec/2 + cover + 4*150), # 9
140     (REBAR_DIA, Hsec/2 - cover, Bsec/2 - cover - 4*150), # 10
141     (REBAR_DIA, Hsec/2 - cover, -Bsec/2 + cover + 5*150), # 11
142     (REBAR_DIA, Hsec/2 - cover, Bsec/2 - cover - 5*150), # 12
143     (REBAR_DIA, Hsec/2 - cover, -Bsec/2 + cover + 6*150), # 13
144     (REBAR_DIA, Hsec/2 - cover, Bsec/2 - cover - 6*150), # 14
145     (REBAR_DIA, Hsec/2 - cover, -Bsec/2 + cover + 7*150), # 15
146     (REBAR_DIA, Hsec/2 - cover, Bsec/2 - cover - 7*150), # 16
147     (REBAR_DIA, Hsec/2 - cover, -Bsec/2 + cover + 8*150), # 17
148     (REBAR_DIA, Hsec/2 - cover, Bsec/2 - cover - 8*150), # 18
149     (REBAR_DIA, Hsec/2 - cover, -Bsec/2 + cover + 9*150), # 19
150     (REBAR_DIA, Hsec/2 - cover, Bsec/2 - cover - 9*150), # 20
151     (REBAR_DIA, Hsec/2 - cover, -Bsec/2 + cover + 10*150), # 21
152     (REBAR_DIA, Hsec/2 - cover, Bsec/2 - cover - 10*150), # 22
153     (REBAR_DIA, Hsec/2 - cover, -Bsec/2 + cover + 11*150), # 23
154     (REBAR_DIA, Hsec/2 - cover, Bsec/2 - cover - 11*150), # 24
155     (REBAR_DIA, Hsec/2 - cover, -Bsec/2 + cover + 12*150), # 25
156     (REBAR_DIA, Hsec/2 - cover, Bsec/2 - cover - 12*150), # 26
157     (REBAR_DIA, Hsec/2 - cover, 0.0), # 27 - x=0.0
158
159     (REBAR_DIA, 0.0, -Bsec/2 + cover), # 28
```

Rectangular Section with Rebars (numbers shown)

Height (mm)

Width (mm)

[Python Console] Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 129, Col 43 UTF-8 CRLF RW Mem 40%

# STATIC ANALYSIS



Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES\_FILES\+EIGHT\_ANALY...UPPORTED\p(t)\_BRIDGE\_CONCRETE\_SIMPLY\_SUPPORTED.py

P(t)\_BRIDGE\_CONCRETE\_SIMPLY\_SUPPORTED.py X CONCRETE\_REC\_SECTION\_FUN\_EXTRA.py X

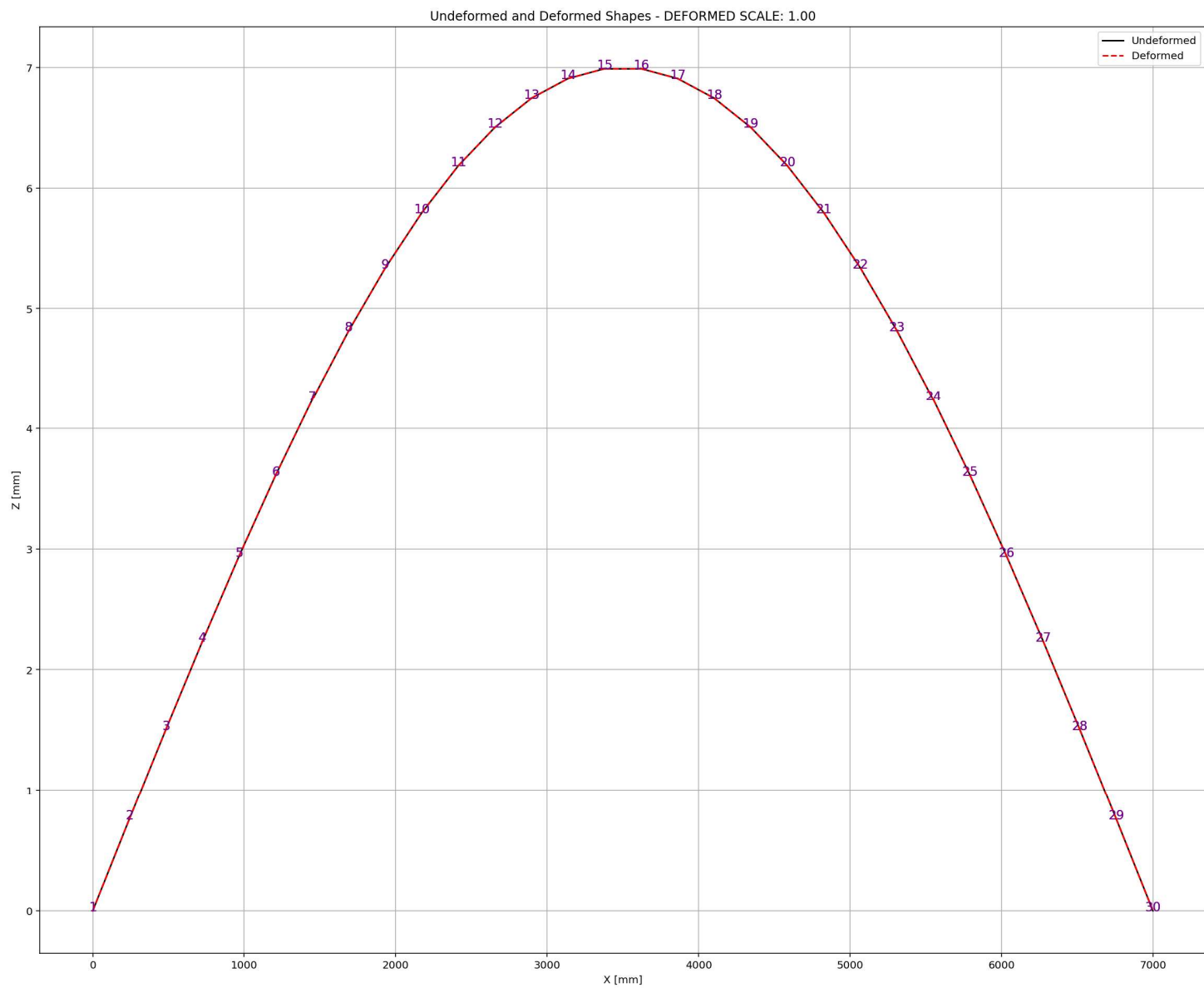
```
1163 # STATIC ANALYSIS
1164 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1165 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1166 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1167 ANAL_TYPE = 'STATIC'
1168
1169 DATA = SHORT_SPAN_SLAB_CONCRETE_BRIDGE(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1170 (reaction, disp_mid,
1171  ele_axialforce, ele_shearforce, ele_momentforce,
1172  node_displacementsX, node_displacementsY,
1173  dispX, dispY) = DATA
1174
1175 S01.PLOT_2D_FRAME(deformed_scale=1.0)
1176
1177 #%%-----
1178 # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1179 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1180 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1181 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1182 ANAL_TYPE = 'PUSHOVER'
1183
1184 DATA = SHORT_SPAN_SLAB_CONCRETE_BRIDGE(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1185 (reaction_PUSH, disp_mid_PUSH,
1186  ele_axialforce_PUSH, ele_shearforce_PUSH, ele_momentforce_PUSH,
1187  node_displacementsX_PUSH, node_displacementsY_PUSH,
1188  dispX_PUSH, dispY_PUSH,
1189  PERIOD_MIN_PUSH, PERIOD_MAX_PUSH,
1190  STRESSb_PUSH, STRAINb_PUSH, STRESSt_PUSH, STRAINt_PUSH, CURVATURE_PUSH) = DATA
1191
1192 XDATA = disp_mid_PUSH
1193 YDATA = reaction_PUSH
1194 XLABEL = 'Displacement in Middle Span [mm]'
1195 YLABEL = 'Base Reaction [N]'
1196 TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
```

unlabeled and Deformed Shapes - DEFORMED SCALE: 1.00

Python Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 81, Col 65 UTF-8 CRLF RW Mem 39%





# **PUSHOVER ANALYSIS**

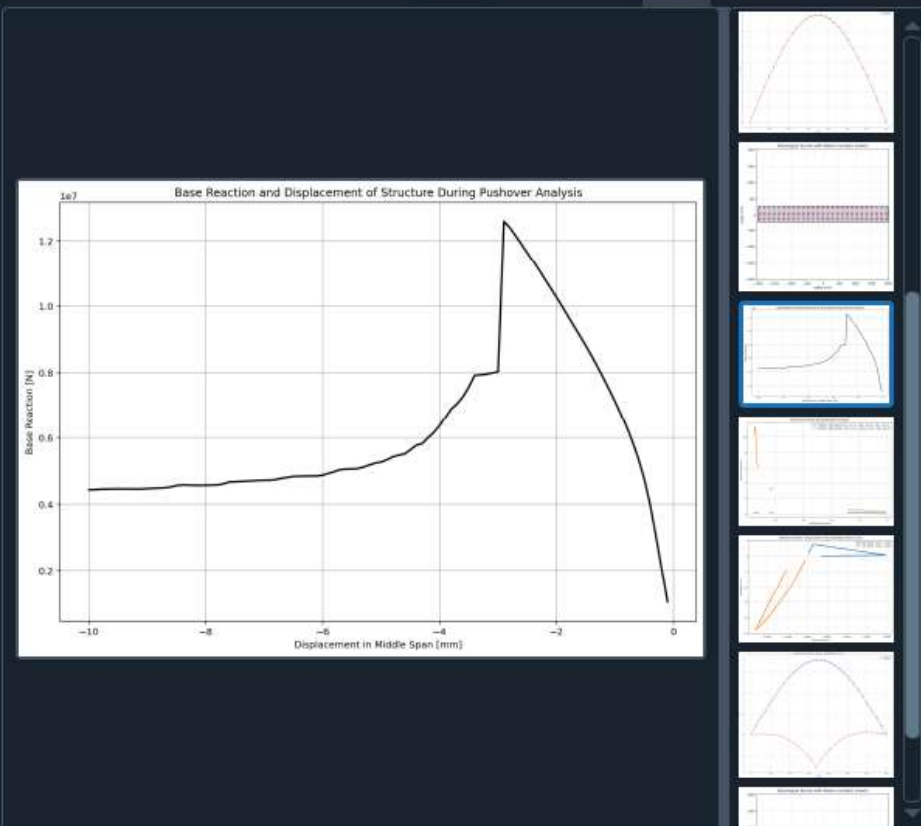
C:\Users\Dell\Desktop\OPENSEES\_FILES\HEIGHT\_ANALY...\_CONCRETE\_SLAB\_SUPERSTRUCTURE\_SIMPLY\_SUPPORTED.py

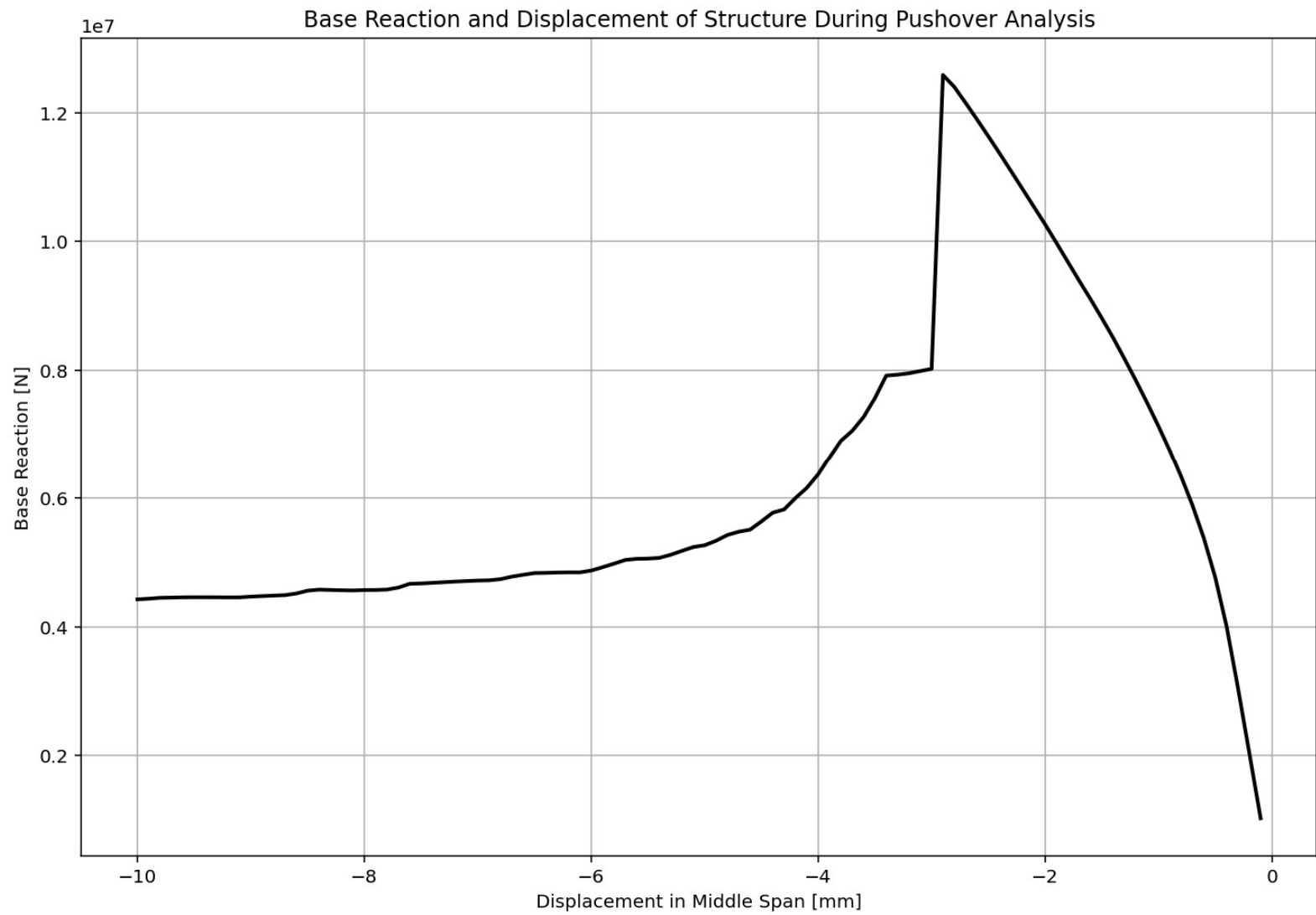
P(t)\_BRIDGE\_CONCRE...IMPLY\_SUPPORTED.py X

```

446 if ANAL_TYPE == 'PUSHOVER': # STATIC TIME-HISTORY ANALYSIS
447     IDctrlDOF = 2 # 1: Horizontal Displacement - 2: Vertical Displacement
448     DINCR = -0.1 # [mm] Incremental Vertical Displacement
449     DMAX = -13.0 # [mm] Max. Vertical Displacement
450     ops.integrator('DisplacementControl', center_node, IDctrlDOF, DINCR)
451     ops.analysis('Static')
452     Nsteps = int(np.abs(DMAX/ DINCR))
453     STEP = 0.0
454     for step in range(Nsteps):
455         OK = ops.analyze(1)
456         S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
457         ops.reactions()
458         reaction.append(ops.nodeReaction(1, 2)+ops.nodeReaction(30, 2)) # SHEAR BASE R
459         disp_mid.append(ops.nodeDisp(center_node, 2)) # DISPLACEMENT
460         dispX.append(ops.nodeDisp(center_node, 1)) # DISPLACEMENT
461         dispY.append(ops.nodeDisp(center_node, 2)) # DISPLACEMENT
462         # Store elements force, strain and stress
463         for ele_id in ele_axialforce.keys():
464             ele_axialforce[ele_id].append(ops.eleResponse(ele_id, 'force')[0])
465             ele_shearforce[ele_id].append(ops.eleResponse(ele_id, 'force')[1])
466             ele_momentforce[ele_id].append(ops.eleResponse(ele_id, 'force')[2])
467         # Store displacements
468         for node_id in node_displacementsX.keys():
469             node_displacementsX[node_id].append(ops.nodeDisp(node_id, 1))
470             node_displacementsY[node_id].append(ops.nodeDisp(node_id, 2))
471         # IN EACH STEP, STRUCTURE PERIOD GOING TO BE CALCULATED
472         #PERIODmin, PERIODmax = S06.RAYLEIGH_DAMPING(3, 0.5*DR, DR, 0, 1)
473         PERIODmin, PERIODmax = S05.EIGENVALUE_ANALYSIS(3, PLOT=True)
474         PERIOD_MIN.append(PERIODmin)
475         PERIOD_MAX.append(PERIODmax)
476         # SECTION STRESS-STRAIN FOR BOTTOM FIBER
477         stressB, strainB = S10.SECTION_STRESS_STRAIN(center_node-1, SEC_TAG, -Depth*0.
478         # Store in separate lists
479         STRESSb.append(stressB)

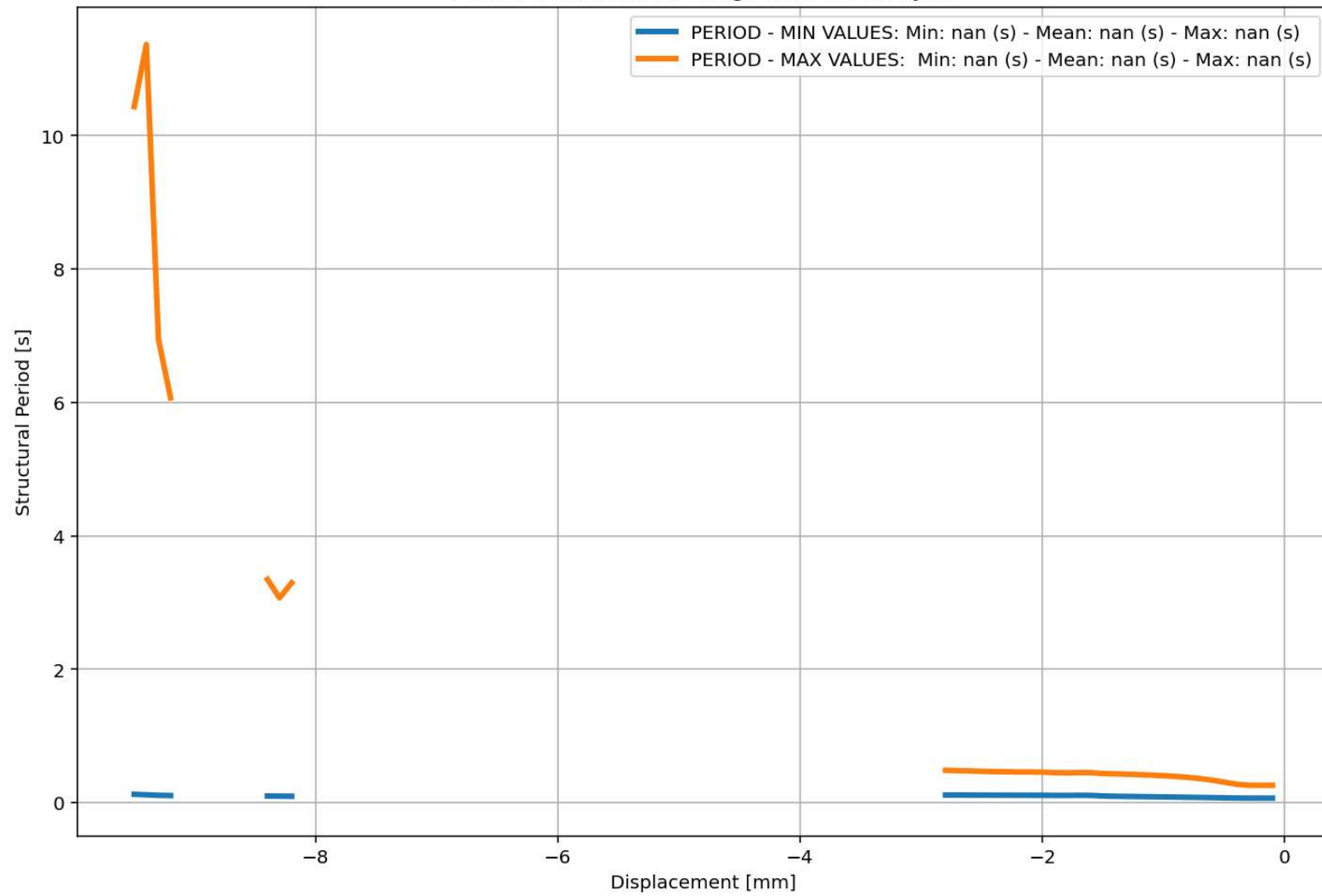
```

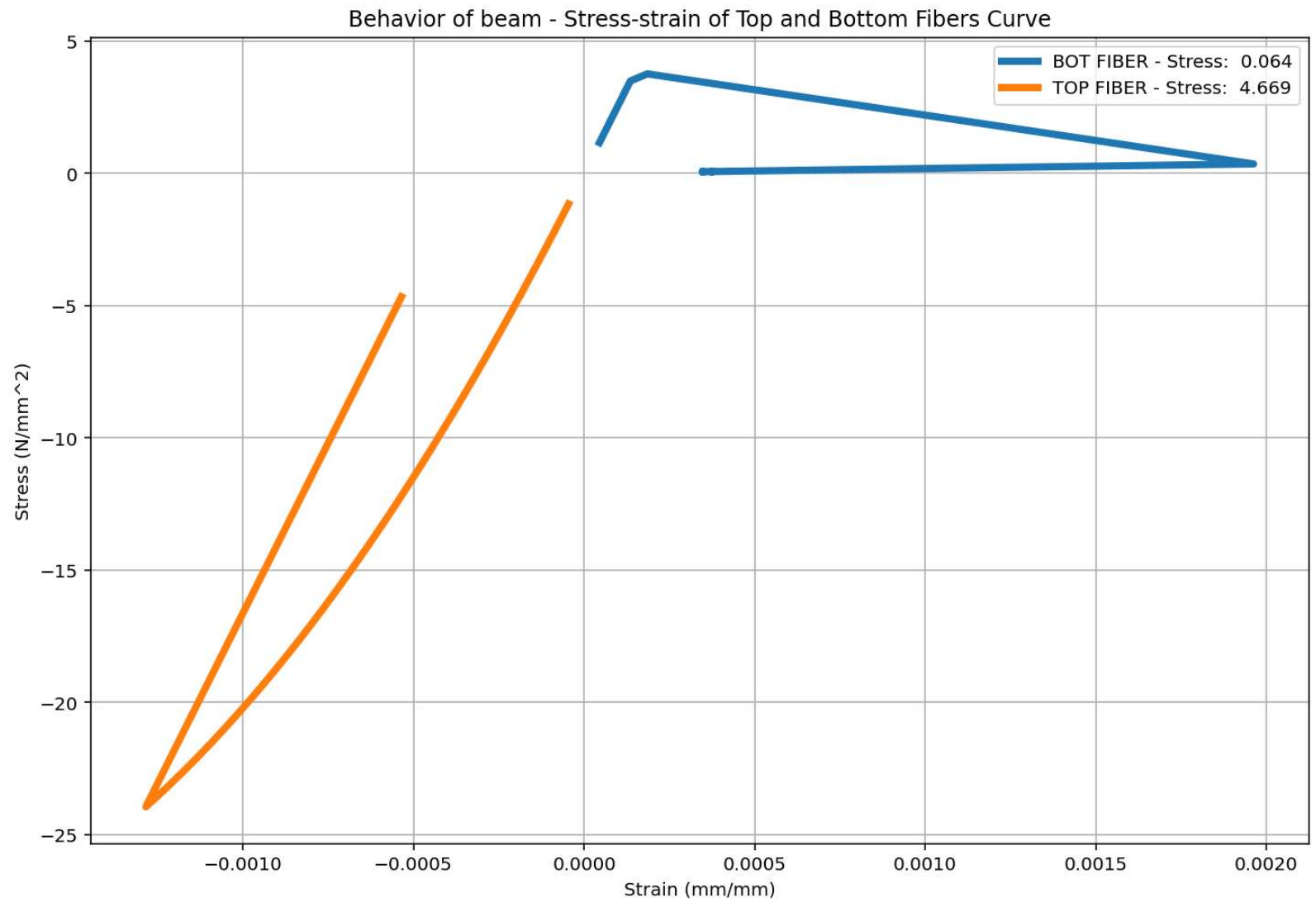


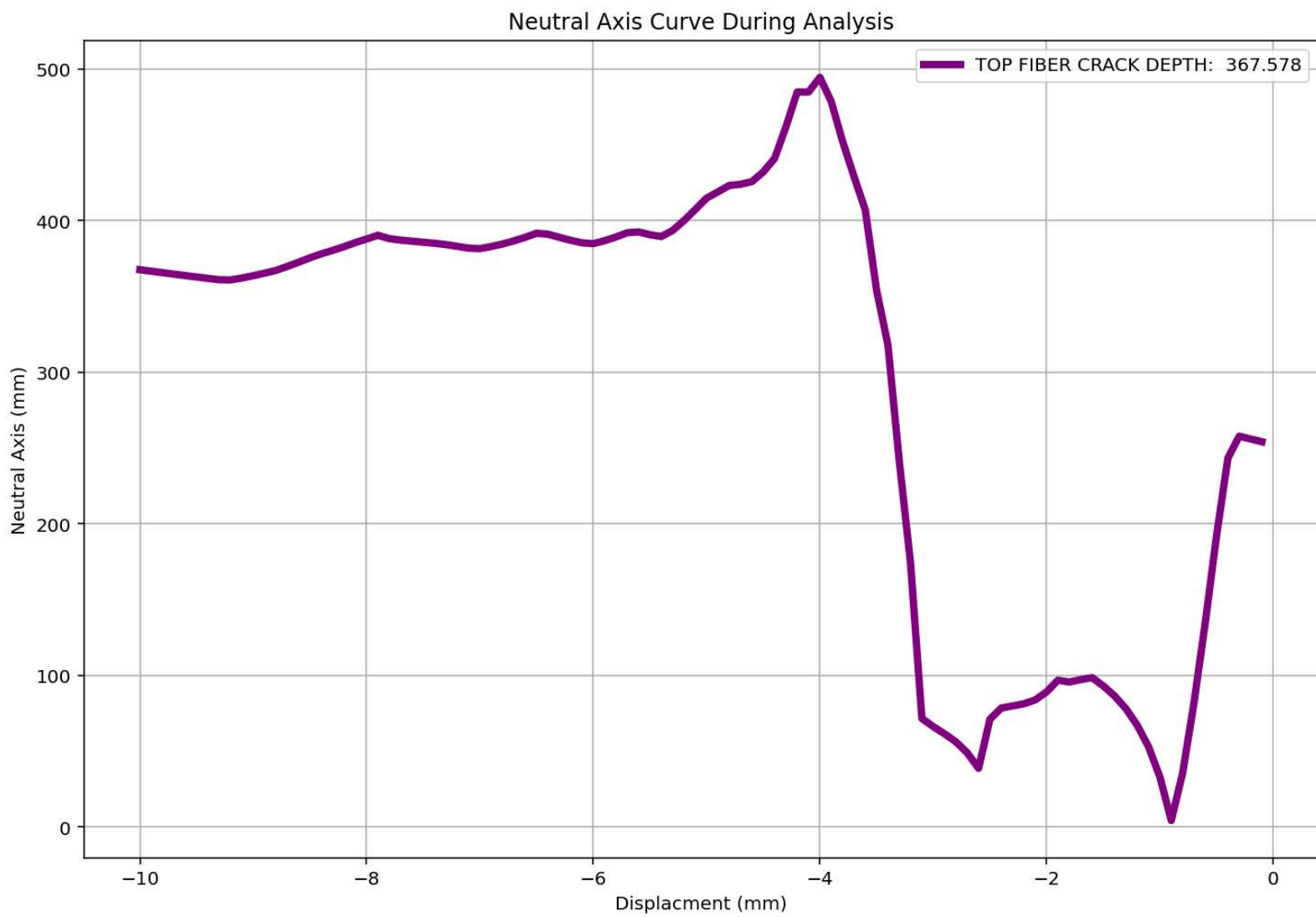


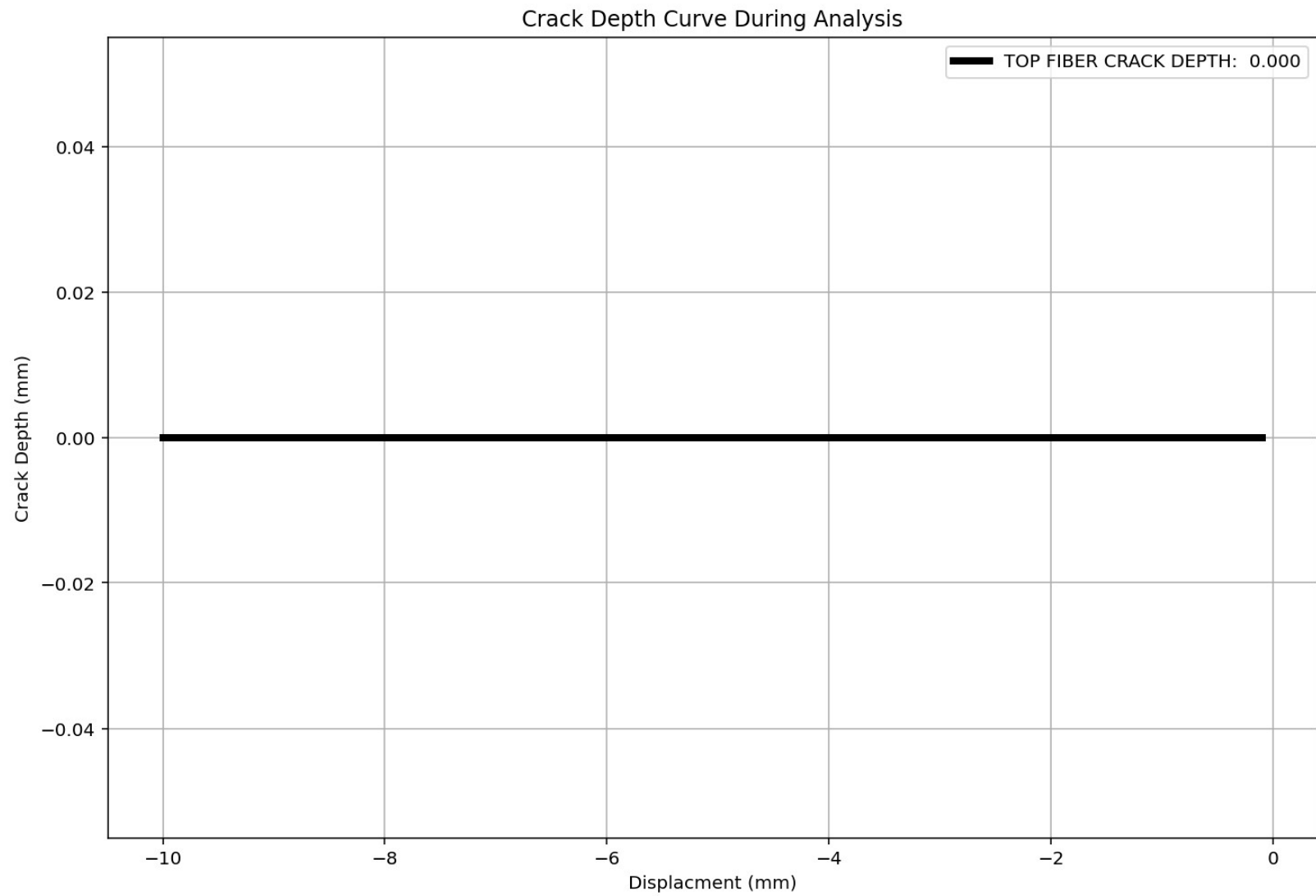


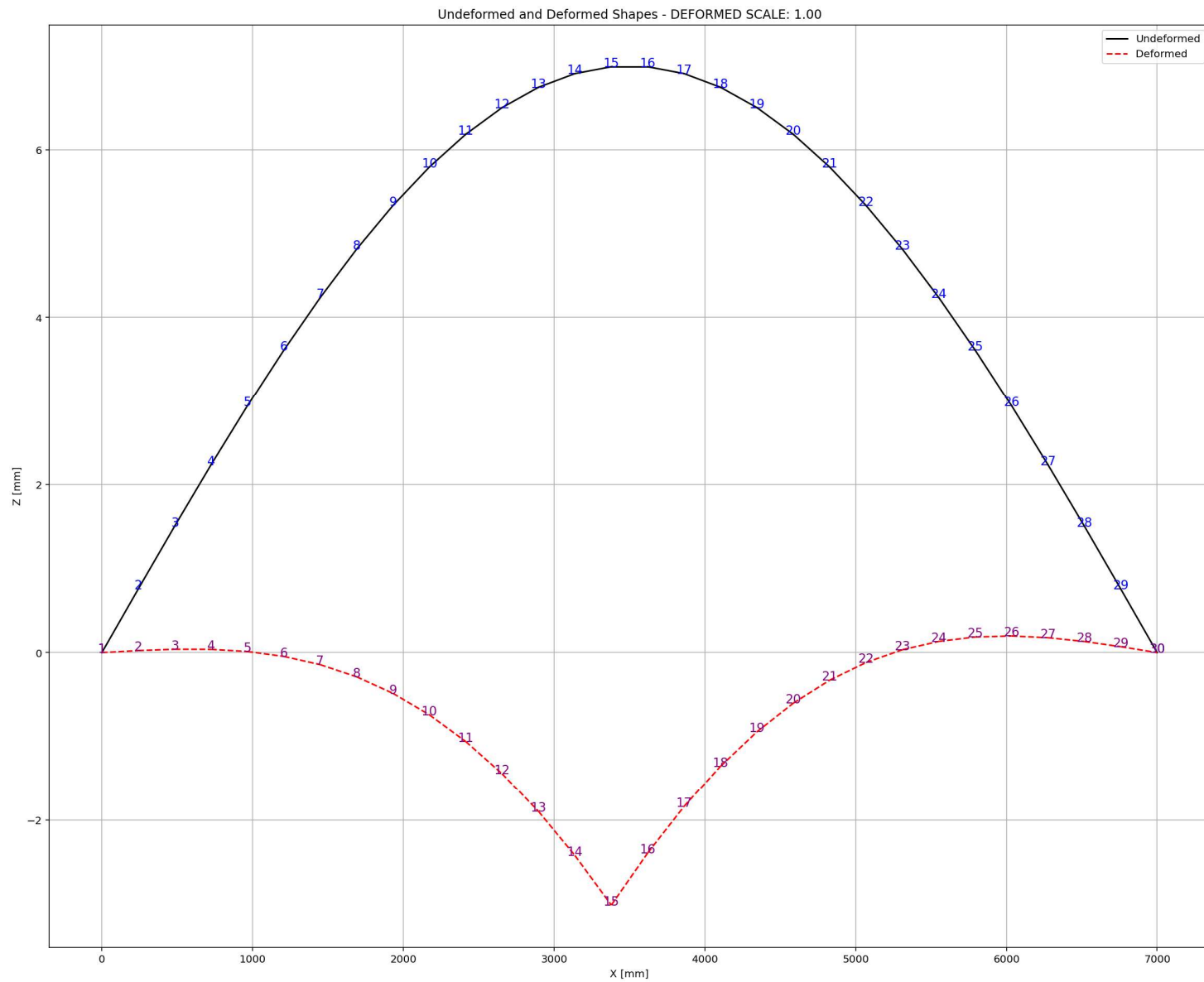
Period of Structure During Pushover Analysis













# **CYCLIC DISPLACEMENT ANALYSIS**

Spyder (Python 3.12)

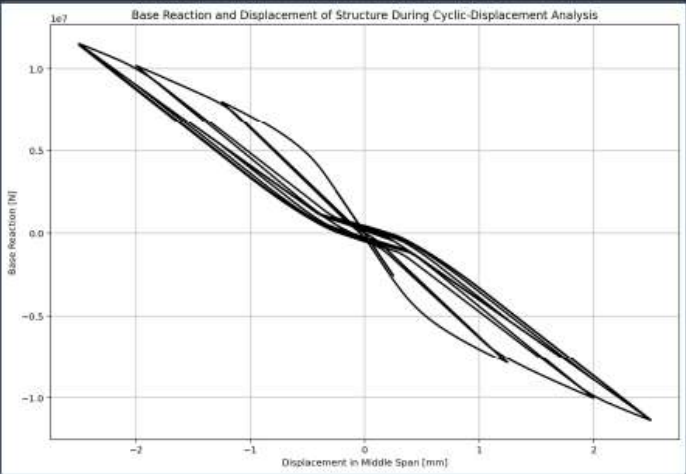
File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES\_FILES\+EIGHT\_ANALY...\_CONCRETE\_SLAB\_SUPERSTRUCTURE\_SIMPLY\_SUPPORTED.py

P(t)\_BRIDGE\_CONCRE...IMPLY\_SUPPORTED.py X

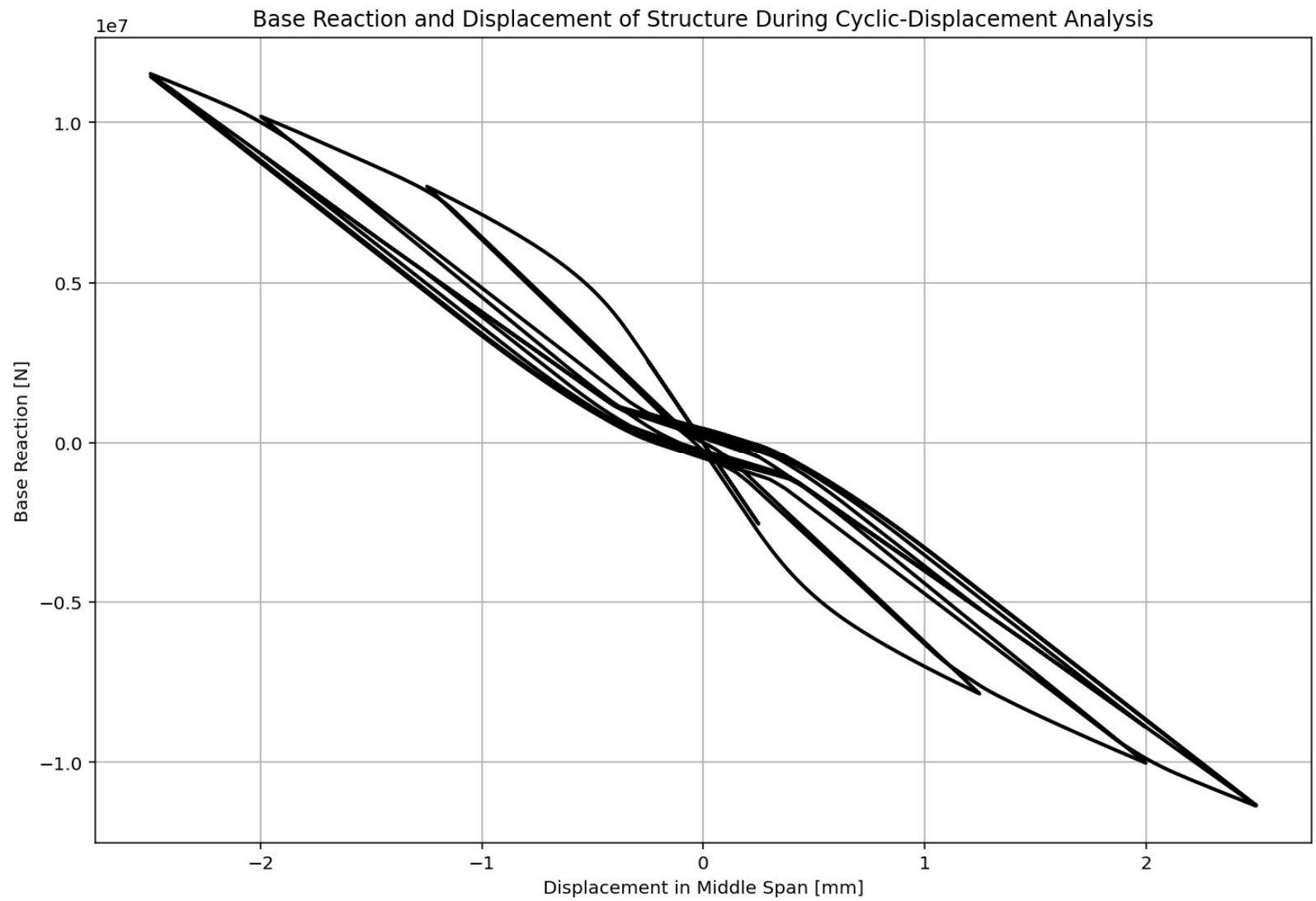
```
503 if ANAL_TYPE == 'CYCLIC_DISPLACEMENT': # STATIC TIME-HISTORY ANALYSIS
504     IDctrlDOF = 2 # 1: Horizontal Displacement - 2: Vertical Displacement
505     DMAX = -2.50 # [mm] Max. Vertical Displacement
506     n_points = 10000
507     # 4. CYCLIC DISPLACEMENT PROTOCOL
508     # Key strain points (same logic as your protocol)
509     key_disp = np.array([
510         0.0,
511         0.1*DMAX, -0.1*DMAX,
512         0.5*DMAX, -0.5*DMAX,
513         0.8*DMAX, -0.8*DMAX,
514         DMAX, -DMAX,
515         0.1*DMAX, -0.1*DMAX,
516         0.5*DMAX, -0.5*DMAX,
517         0.8*DMAX, -0.8*DMAX,
518         DMAX, -DMAX,
519         0.0
520     ])
521
522     # Generate 10000-point displacement protocol
523     disp_protocol = np.interp(
524         np.linspace(0, len(key_disp) - 1, n_points),
525         np.arange(len(key_disp)),
526         key_disp
527     )
528
529     ops.analysis('Static')
530     STEP = 0.0
531     for target_disp in disp_protocol:
532         current_disp = ops.nodeDisp(center_node, 2) # DISPALCEMENT APPLIED IN MIDDLE N
533         dU = target_disp - current_disp
534         ops.integrator('DisplacementControl', center_node, IDctrlDOF, dU)
535         OK = ops.analyze(1)
536         S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
```

Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis

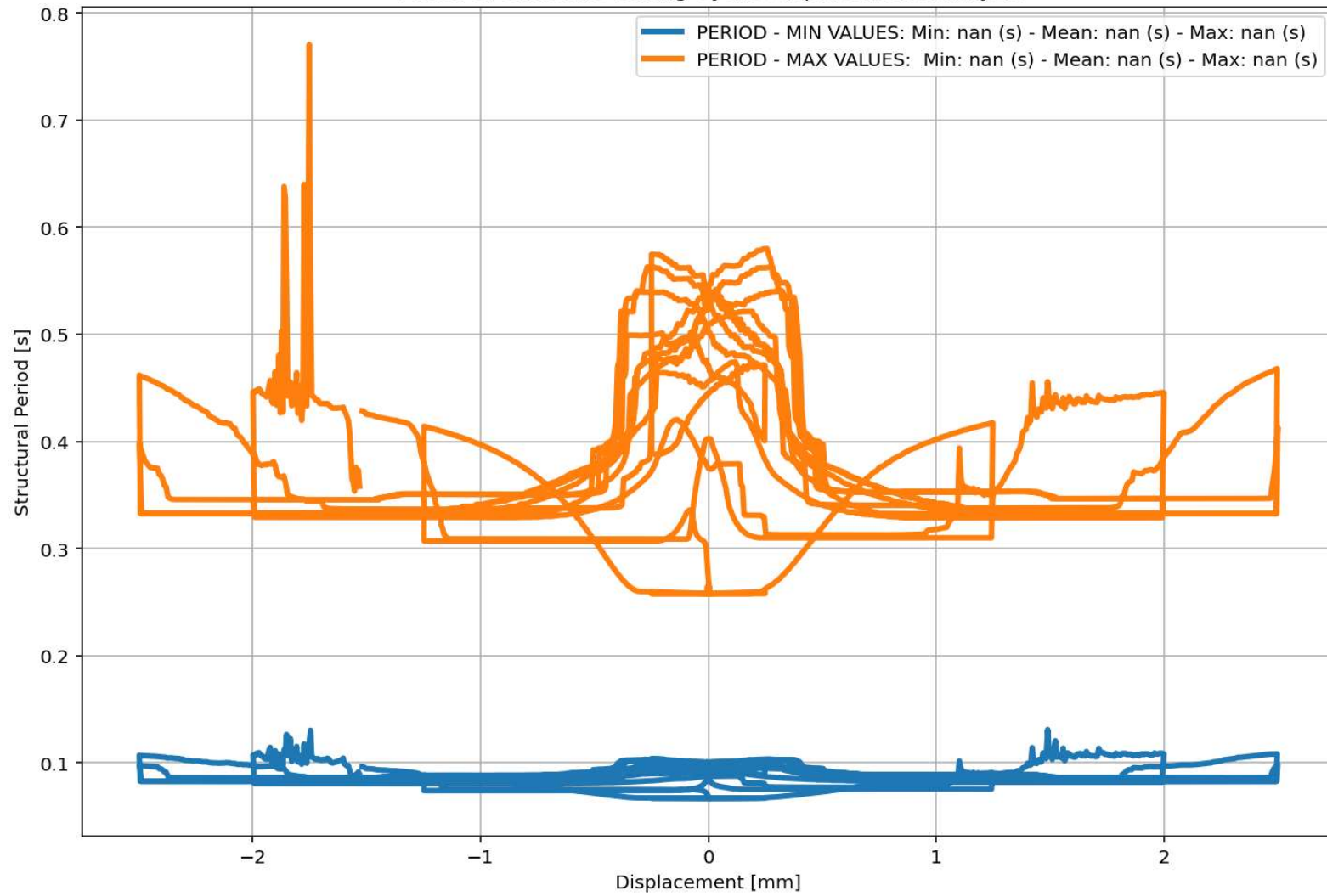


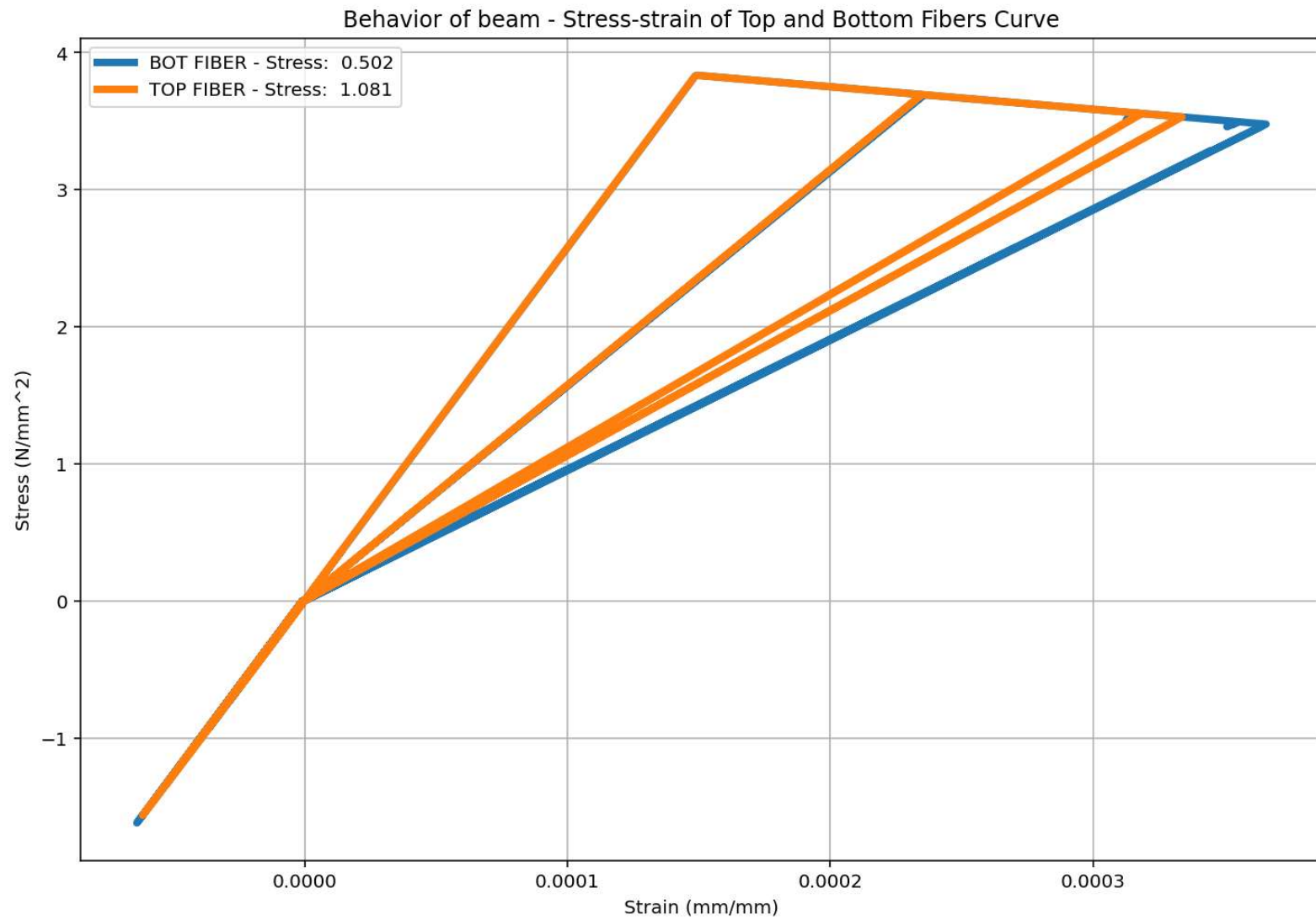
IPython Console Files Help Variable Explorer Debugger Plots History

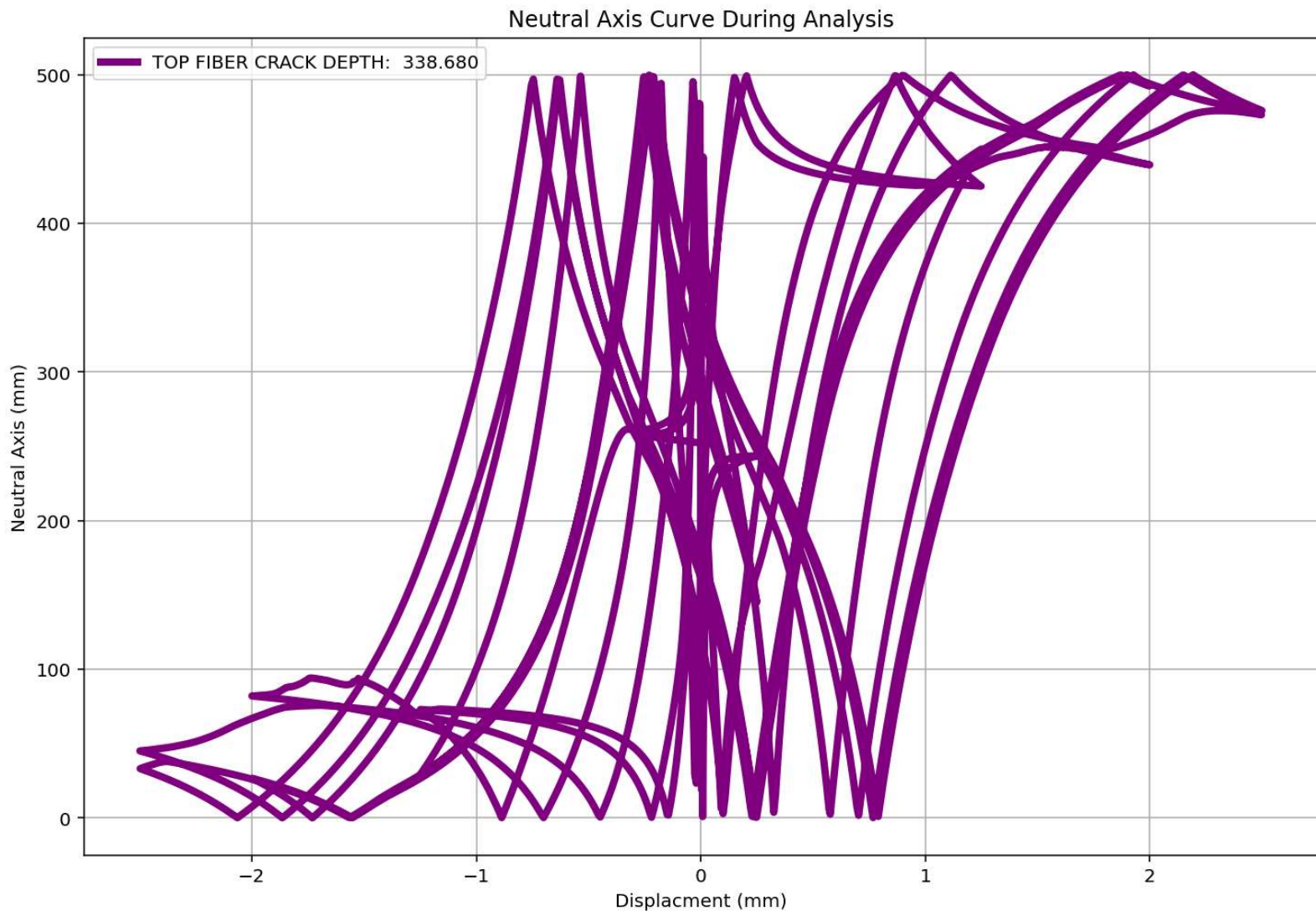
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 449, Col 18 UTF-8 CRLF RW Mem 41%



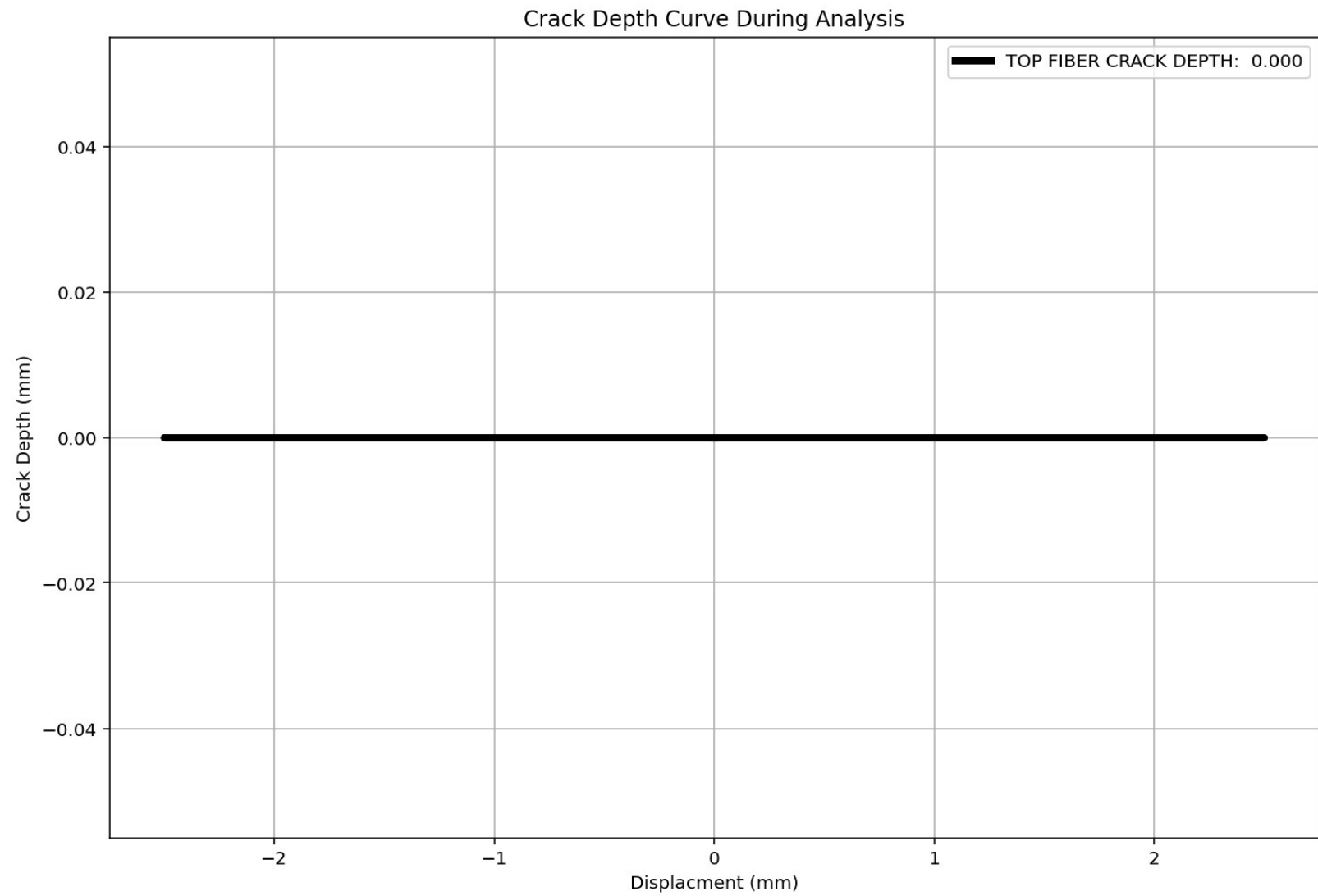
Period of Structure During Cyclic Displacement Analysis



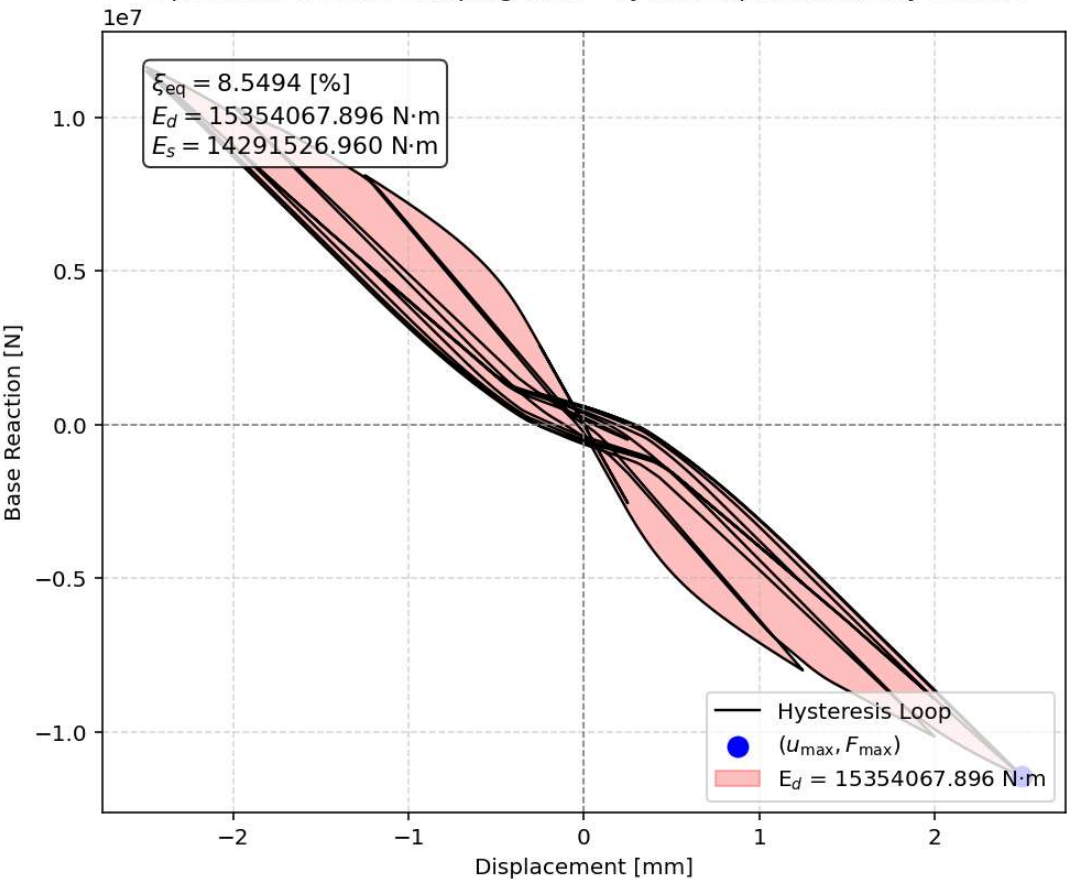


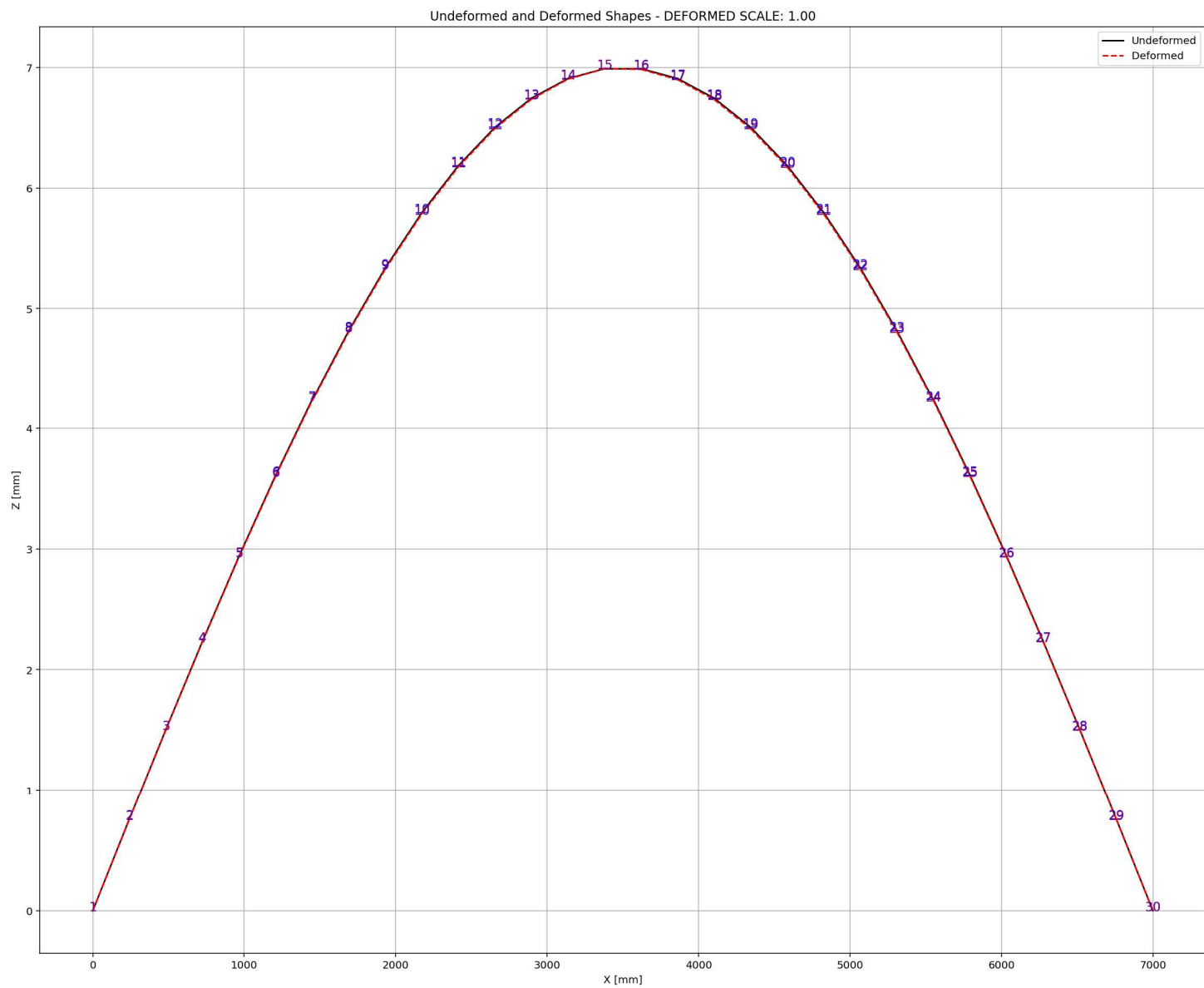




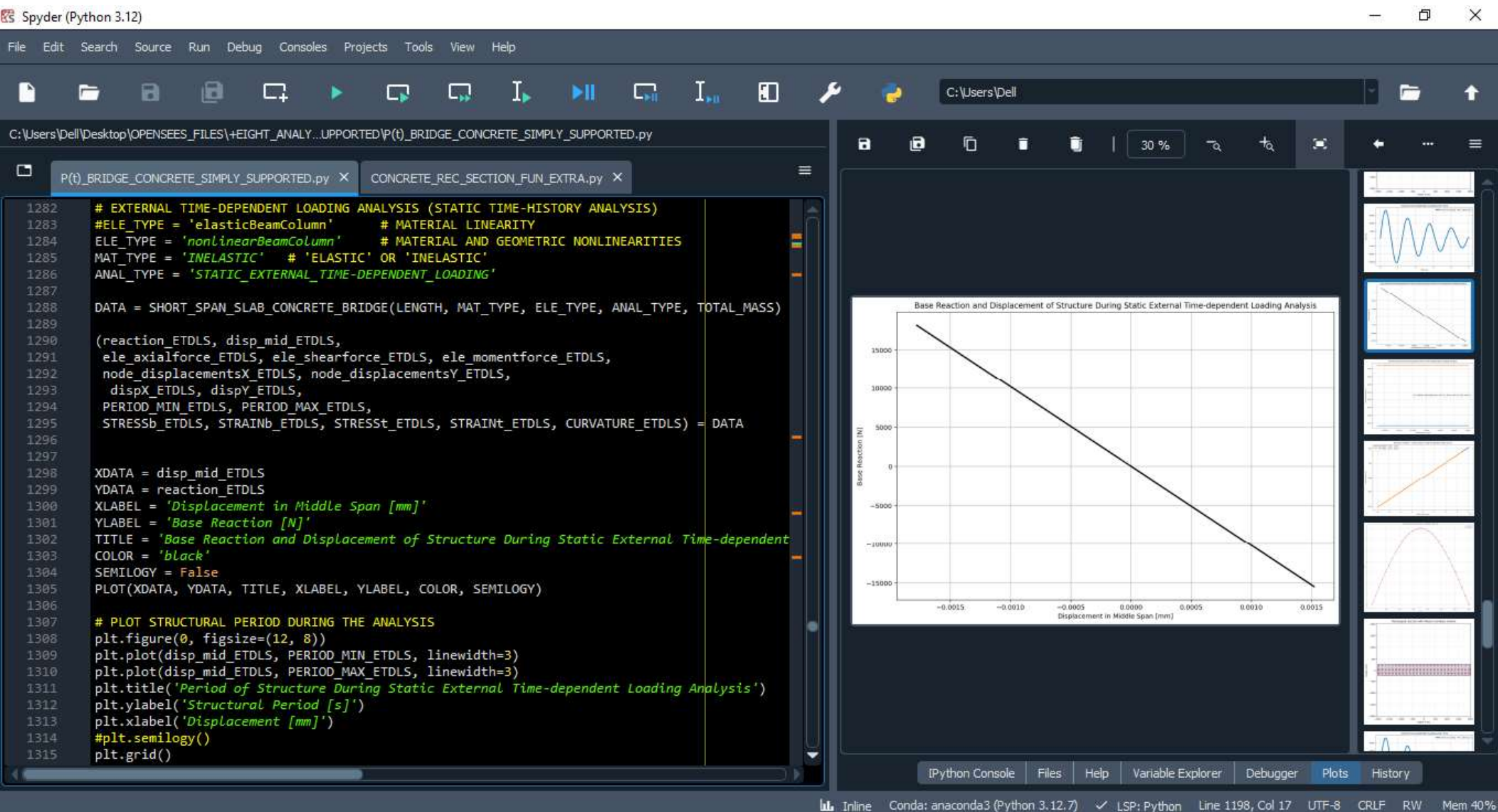


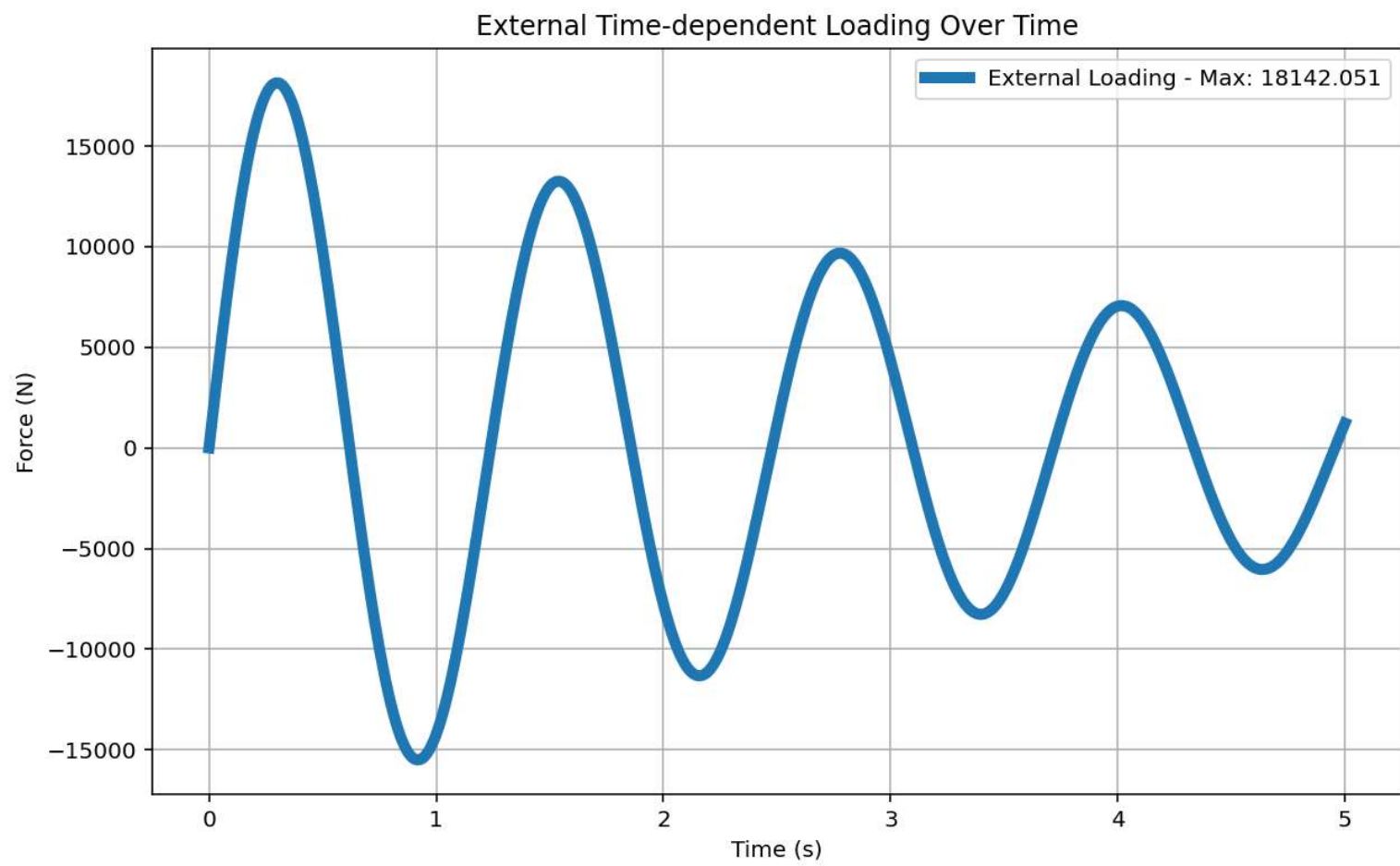
Equivalent viscous damping ratio - Cyclic Displacement Hysteresis





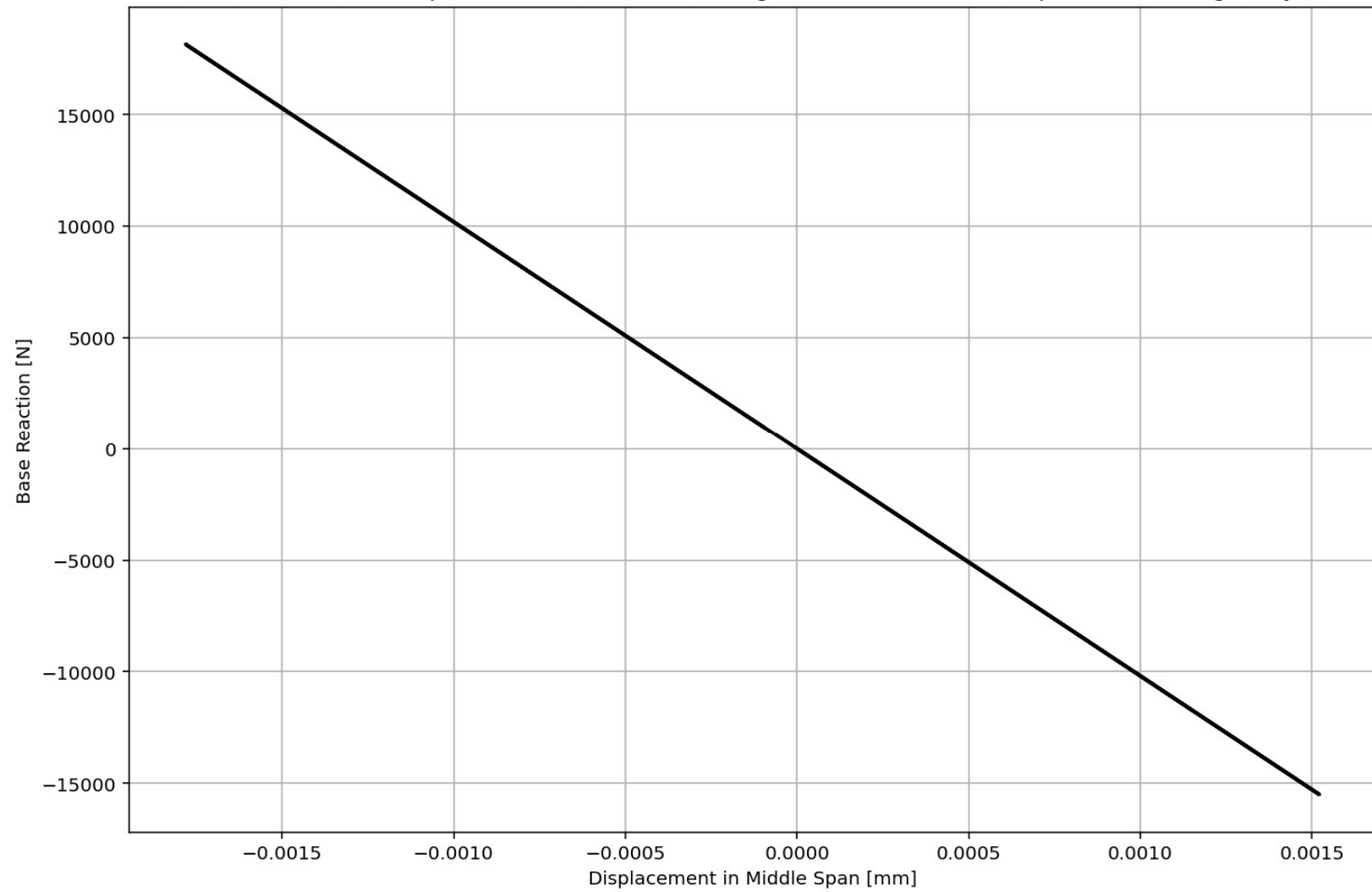
# **STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS**



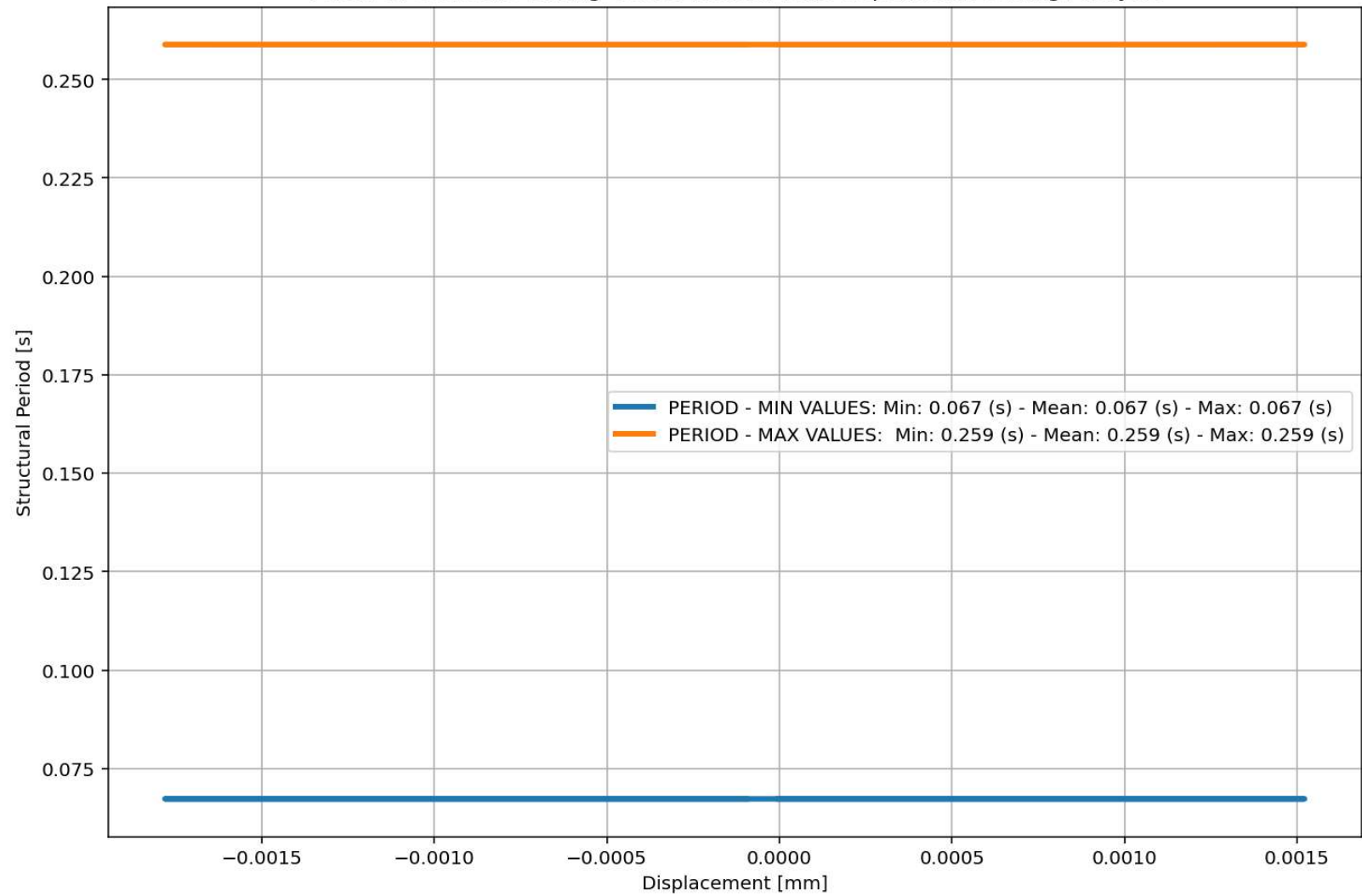


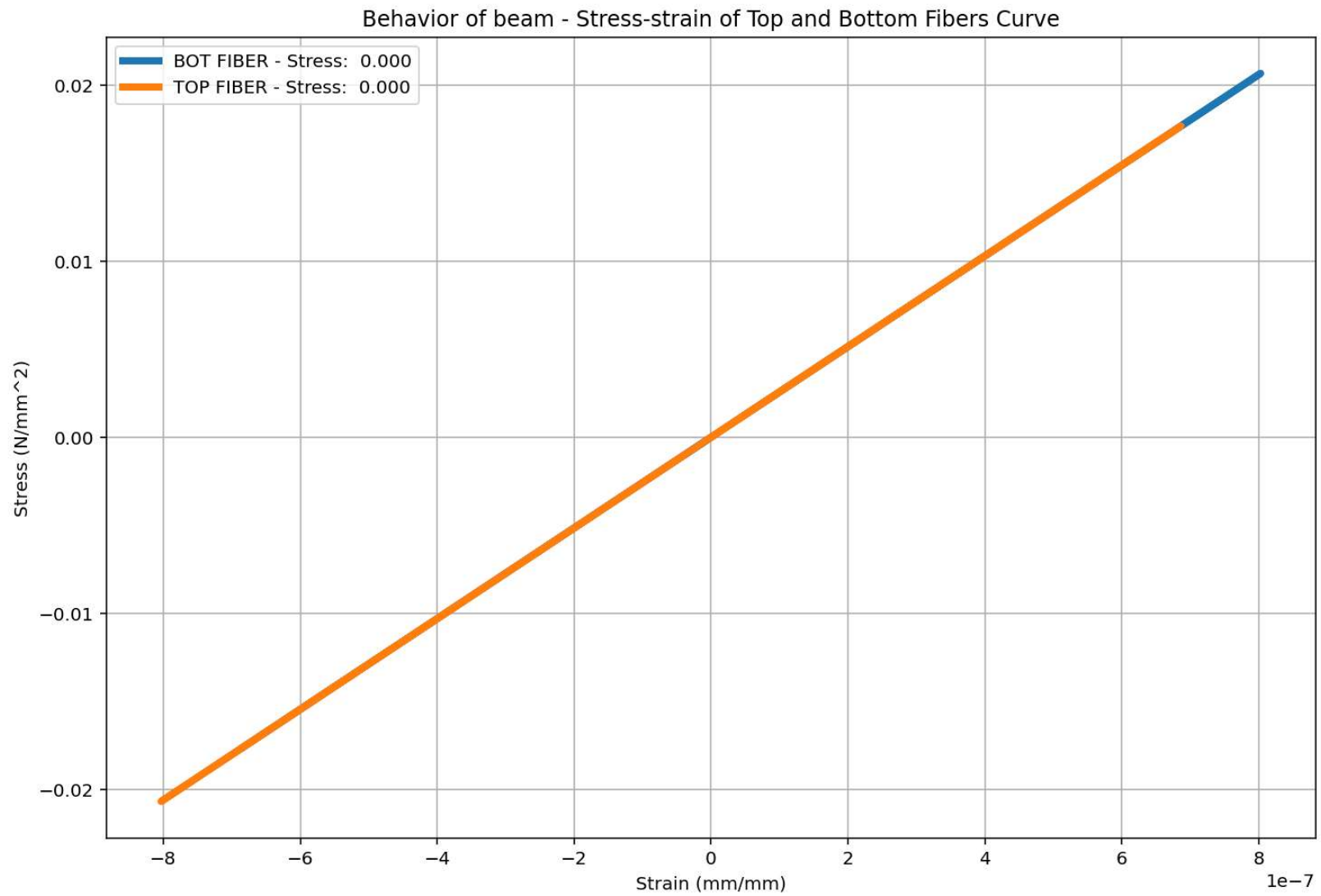


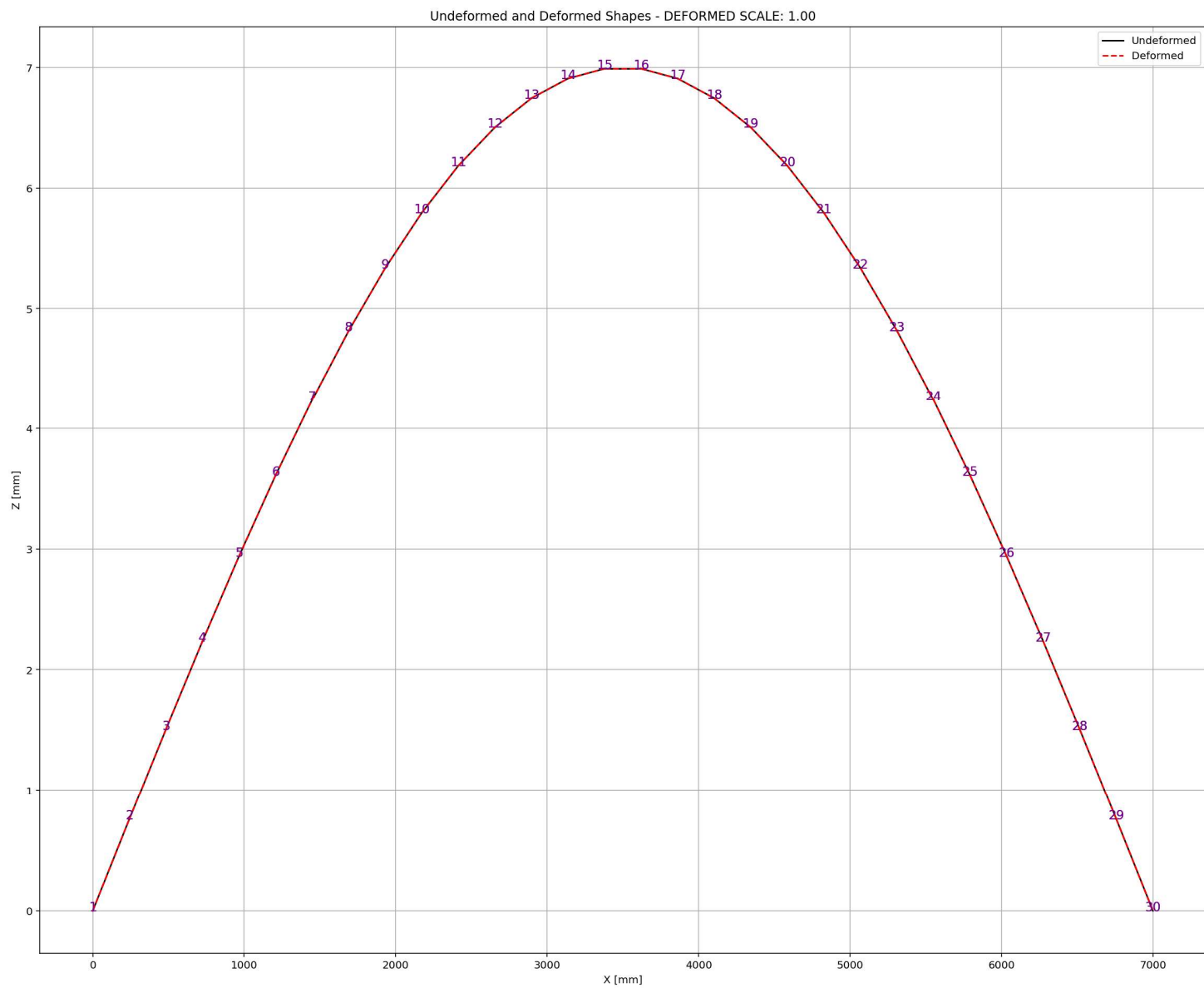
Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis



Period of Structure During Static External Time-dependent Loading Analysis







# **DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS**

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES\_FILES\+EIGHT\_ANALY...UPORTED\P(t)\_BRIDGE\_CONCRETE\_SIMPLY\_SUPPORTED.py

P(t)\_BRIDGE\_CONCRETE\_SIMPLY\_SUPPORTED.py X CONCRETE\_REC\_SECTION\_FUN\_EXTRA.py X

```
1334 # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1335 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1336 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1337 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1338 ANAL_TYPE = 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1339
1340 DATA = SHORT_SPAN_SLAB_CONCRETE_BRIDGE(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1341
1342 (time_ETDLD, reaction_ETDLD, disp_mid_ETDLD,
1343 ele_axialforce_ETDLD, ele_shearforce_ETDLD, ele_momentforce_ETDLD,
1344 node_displacementsX_ETDLD, node_displacementsY_ETDLD,
1345 dispX_ETDLD, dispY_ETDLD,
1346 veloX_ETDLD, veloY_ETDLD,
1347 accX_ETDLD, accY_ETDLD,
1348 stiffness_ETDLD, PERIOD_ETDLD, damping_ratio_ETDLD,
1349 PERIOD_MIN_ETDLD, PERIOD_MAX_ETDLD,
1350 STRESSb_ETDLD, STRAINb_ETDLD, STRESSt_ETDLD, STRAINT_ETDLD, CURVATURE_ETDLD) = DATA
1351
1352
1353 XDATA = disp_mid_ETDLD
1354 YDATA = reaction_ETDLD
1355 XLABEL = 'Displacement in Middle Span [mm]'
1356 YLABEL = 'Base Reaction [N]'
1357 TITLE = 'Base Reaction and Displacement of Structure During Dynamic External Time-dependen
1358 COLOR = 'black'
1359 SEMIOLOGY = False
1360 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
1361
1362 PLOT_TIME_HISTORY(time_ETDLD, reaction_ETDLD, disp_mid_ETDLD,
1363 dispX_ETDLD, dispY_ETDLD,
1364 veloX_ETDLD, veloY_ETDLD,
1365 accX_ETDLD, accY_ETDLD)
1366
1367 # PLOT ELEMENTS AXIAL FORCE
```

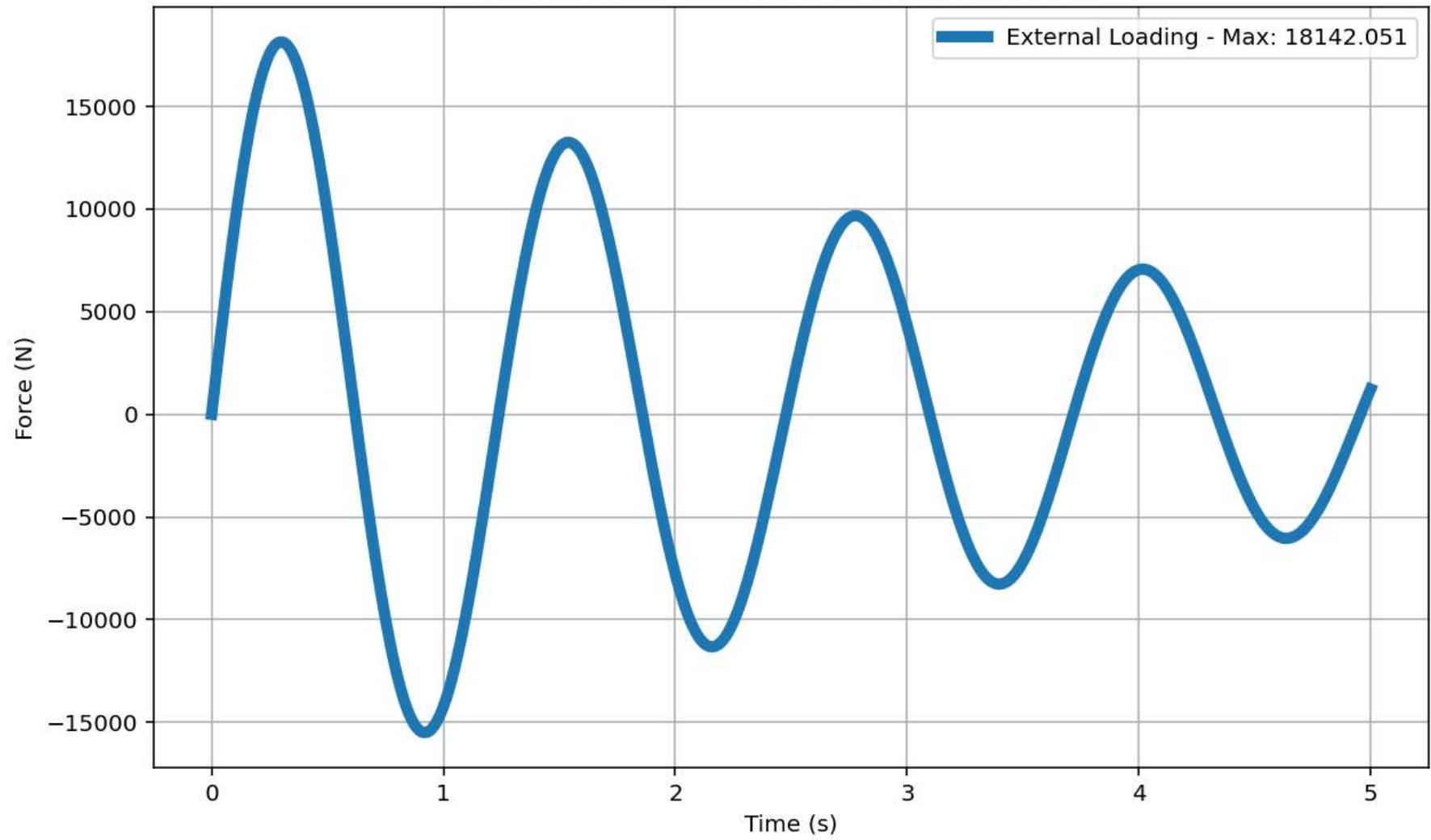
Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis

30 %

[Python Console] Files Help Variable Explorer Debugger Plots History

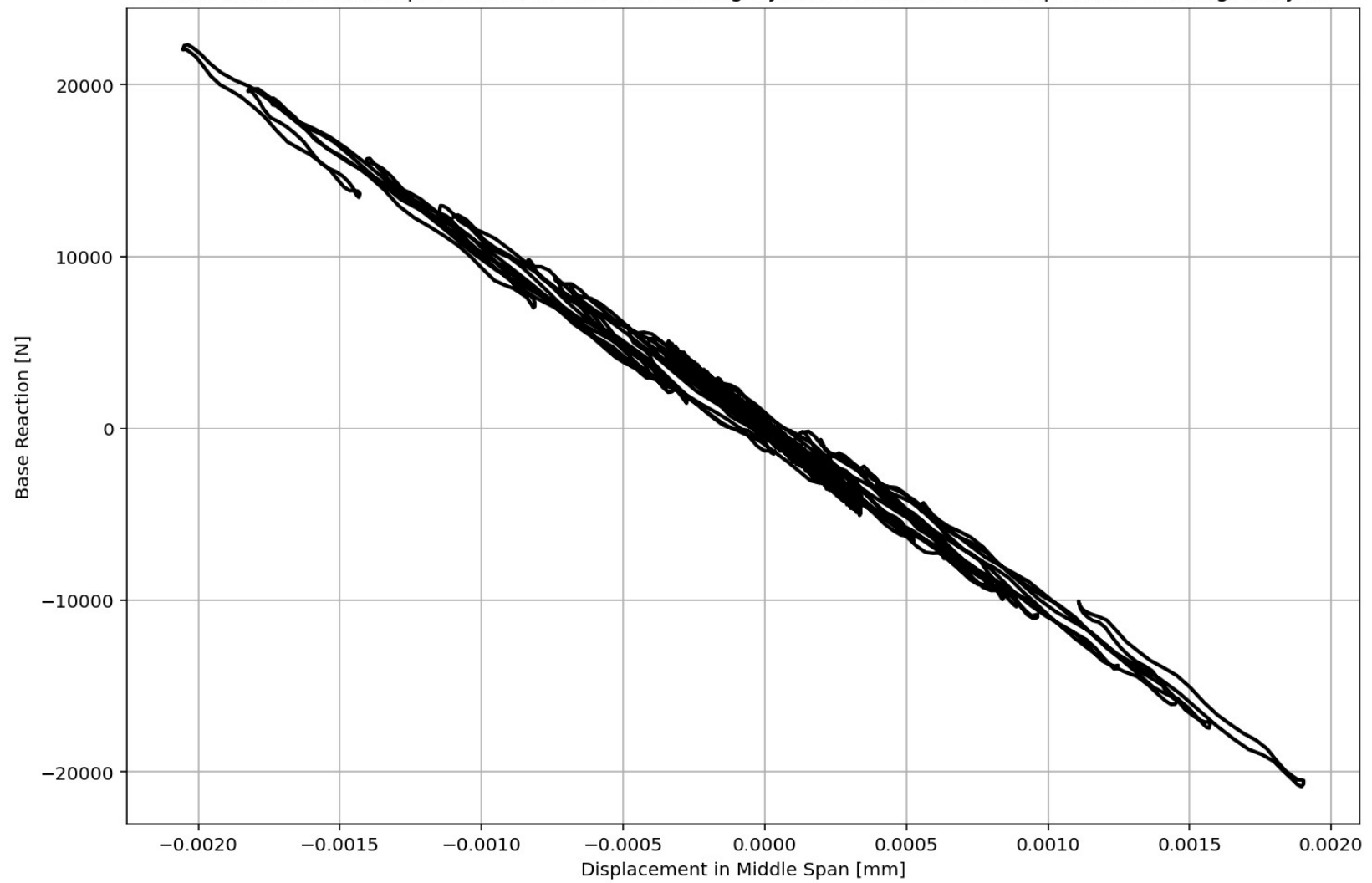
Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 1198, Col 17 UTF-8 CRLF RW Mem 41%

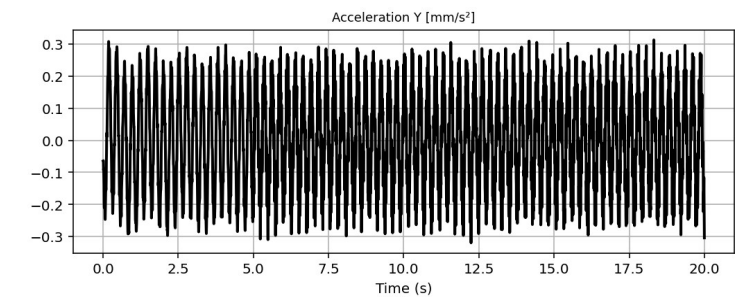
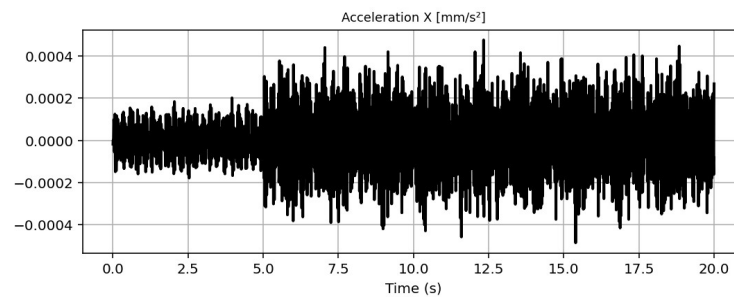
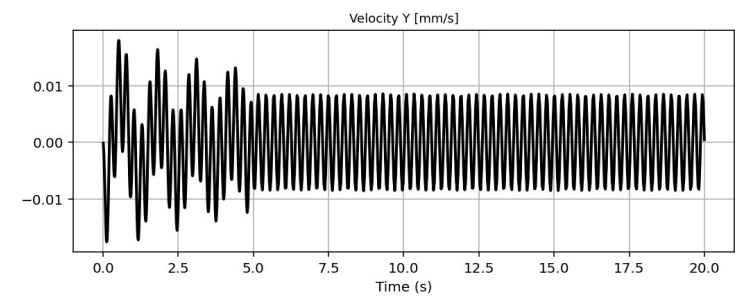
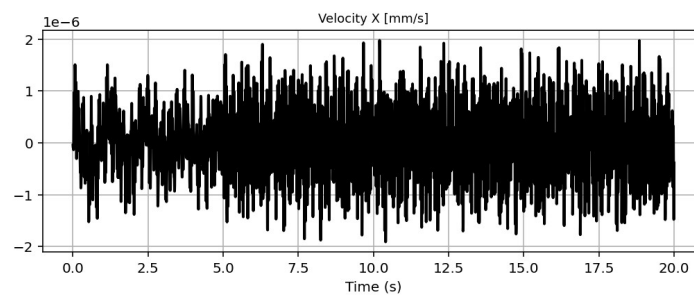
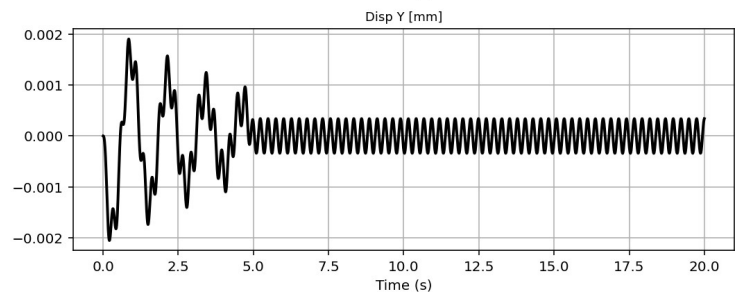
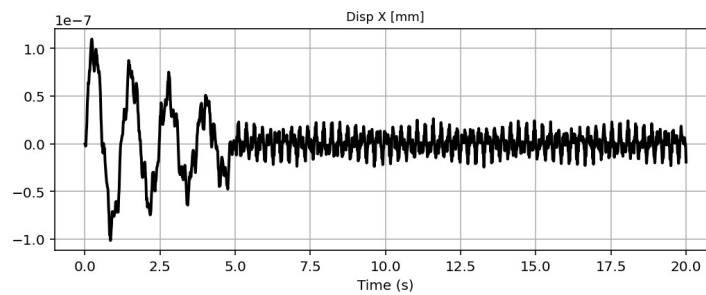
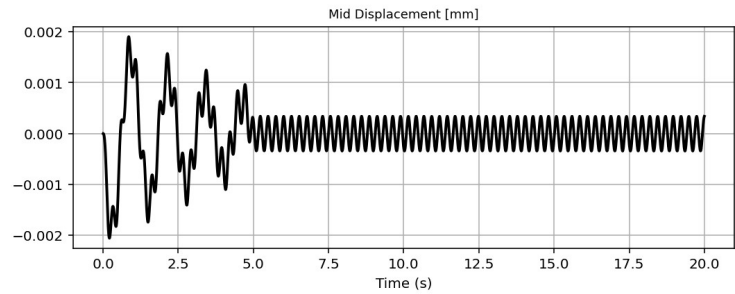
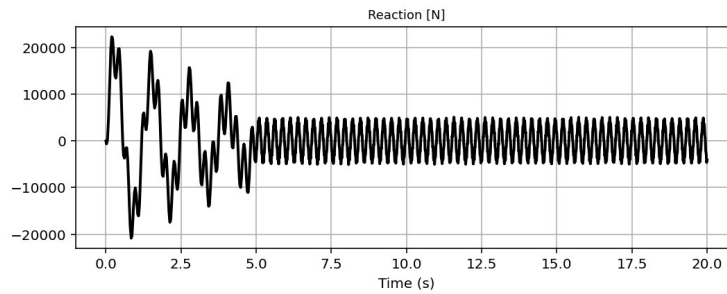
External Time-dependent Loading Over Time



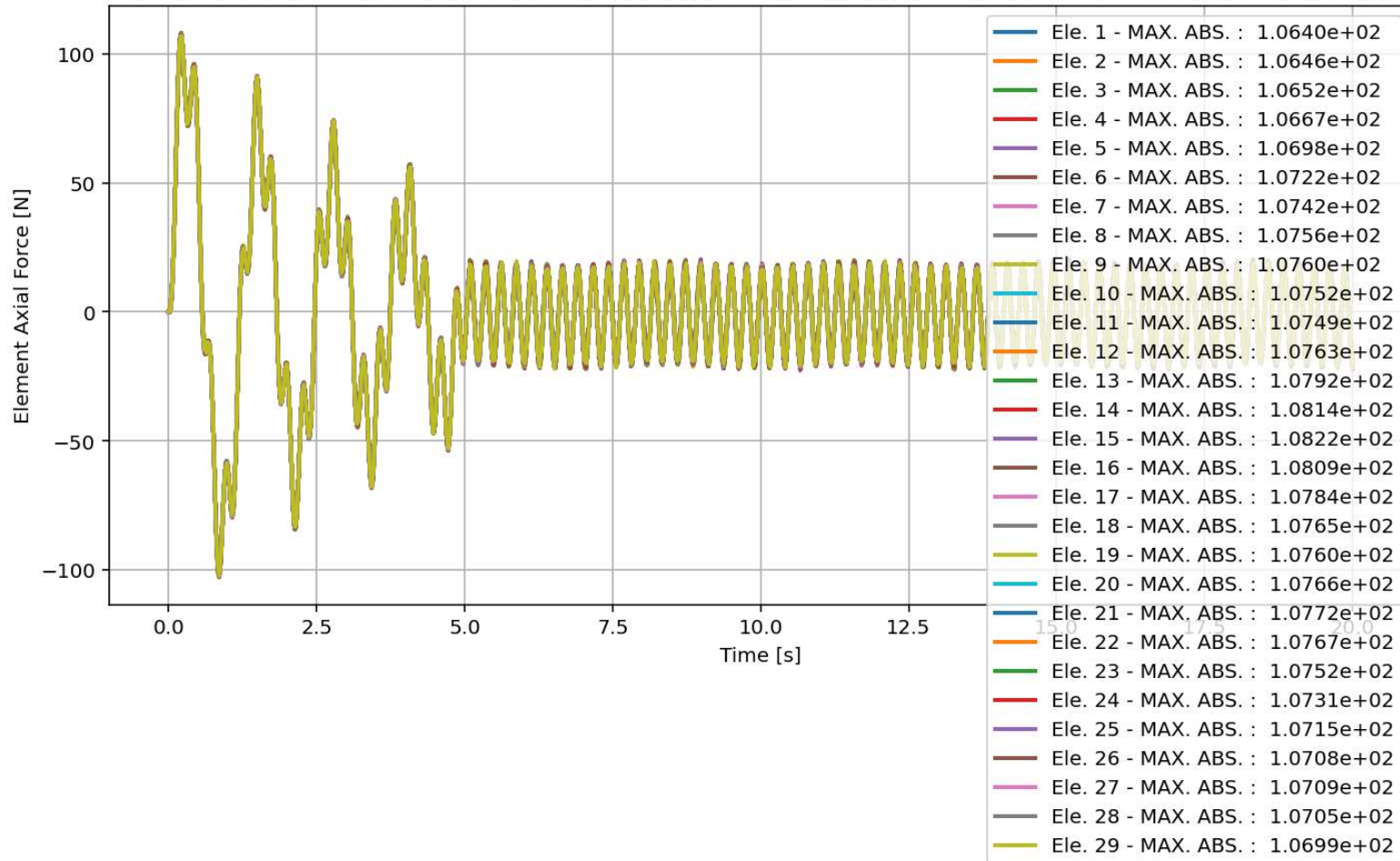


Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis

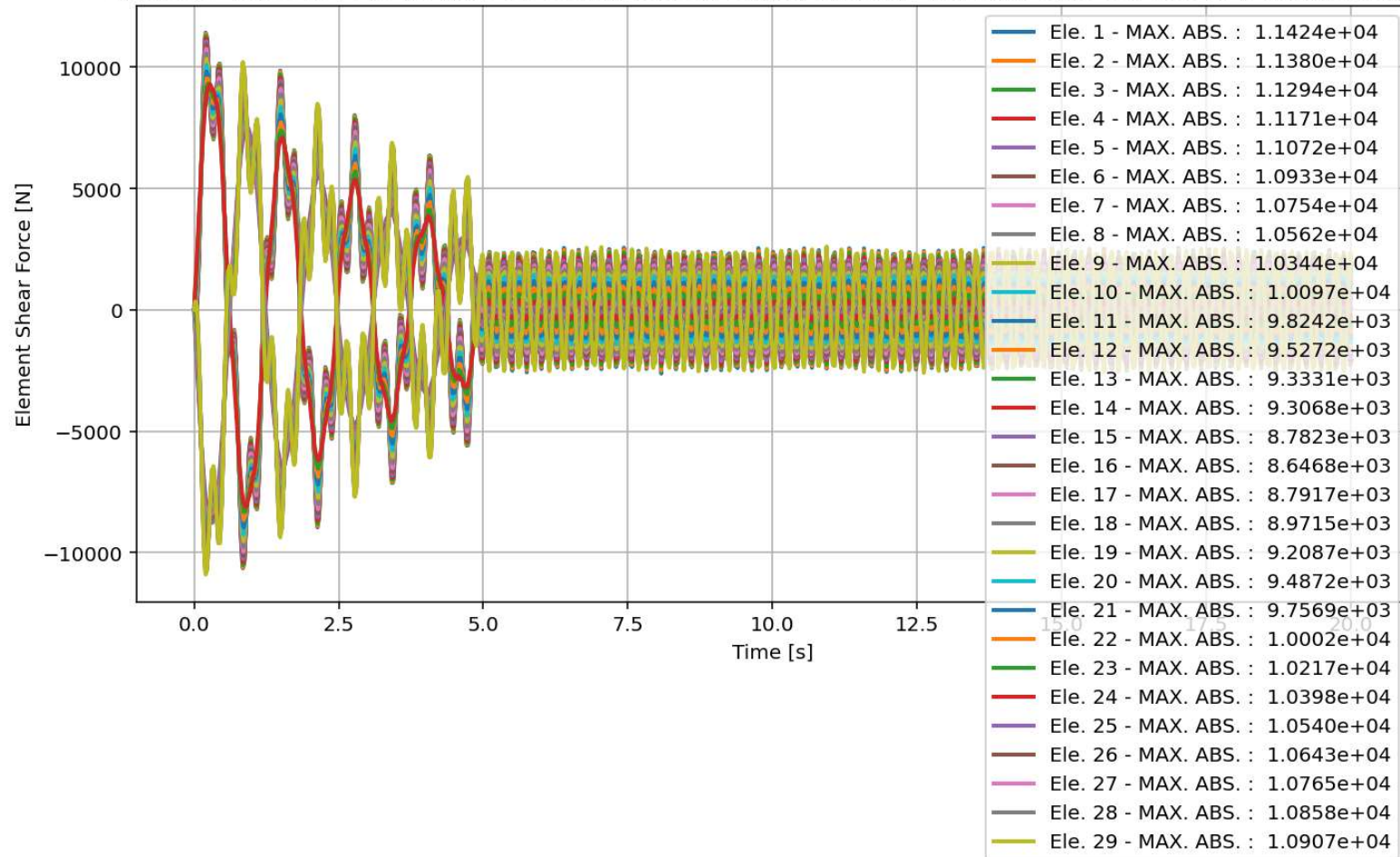




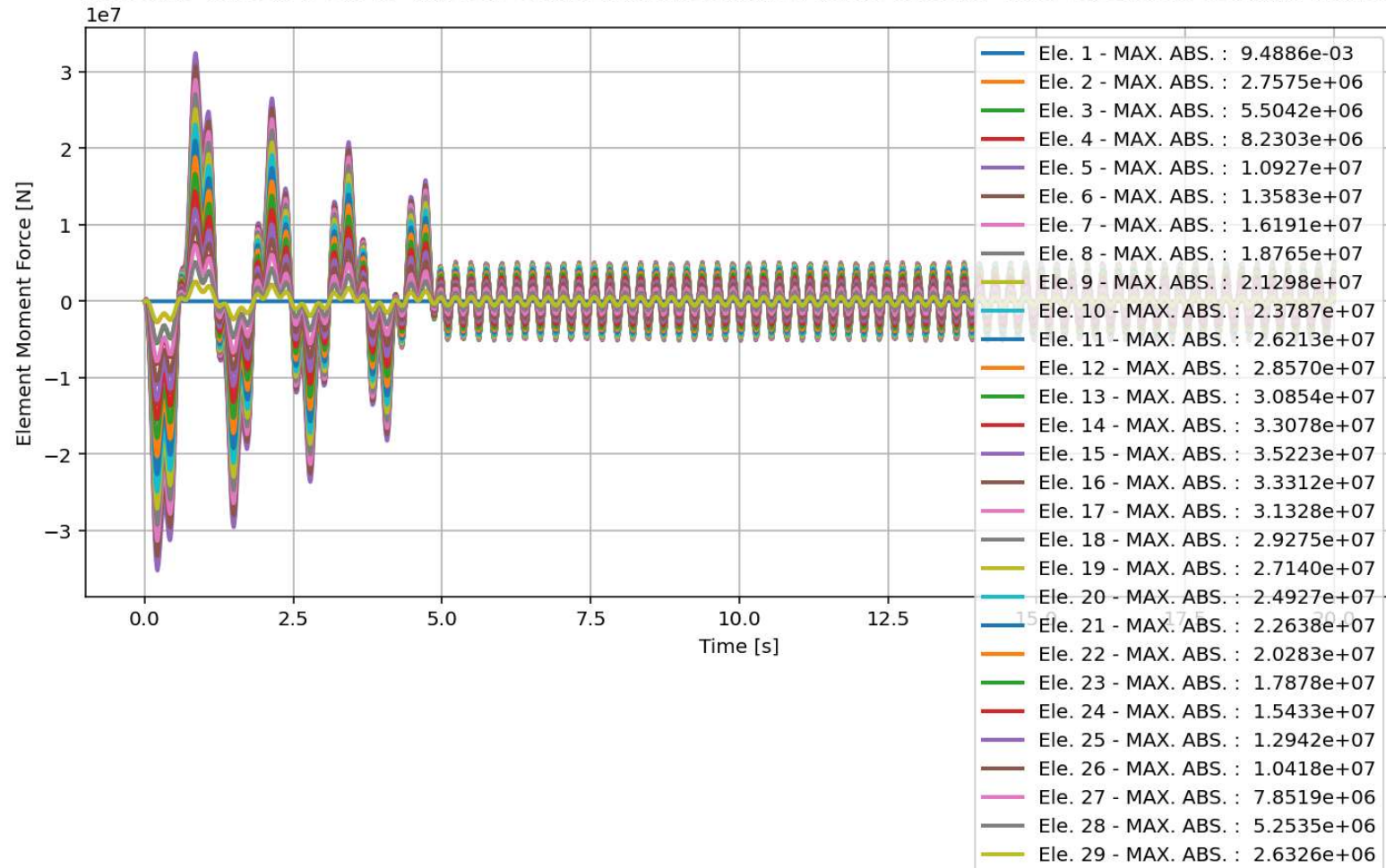
elements force in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis



elements force in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis

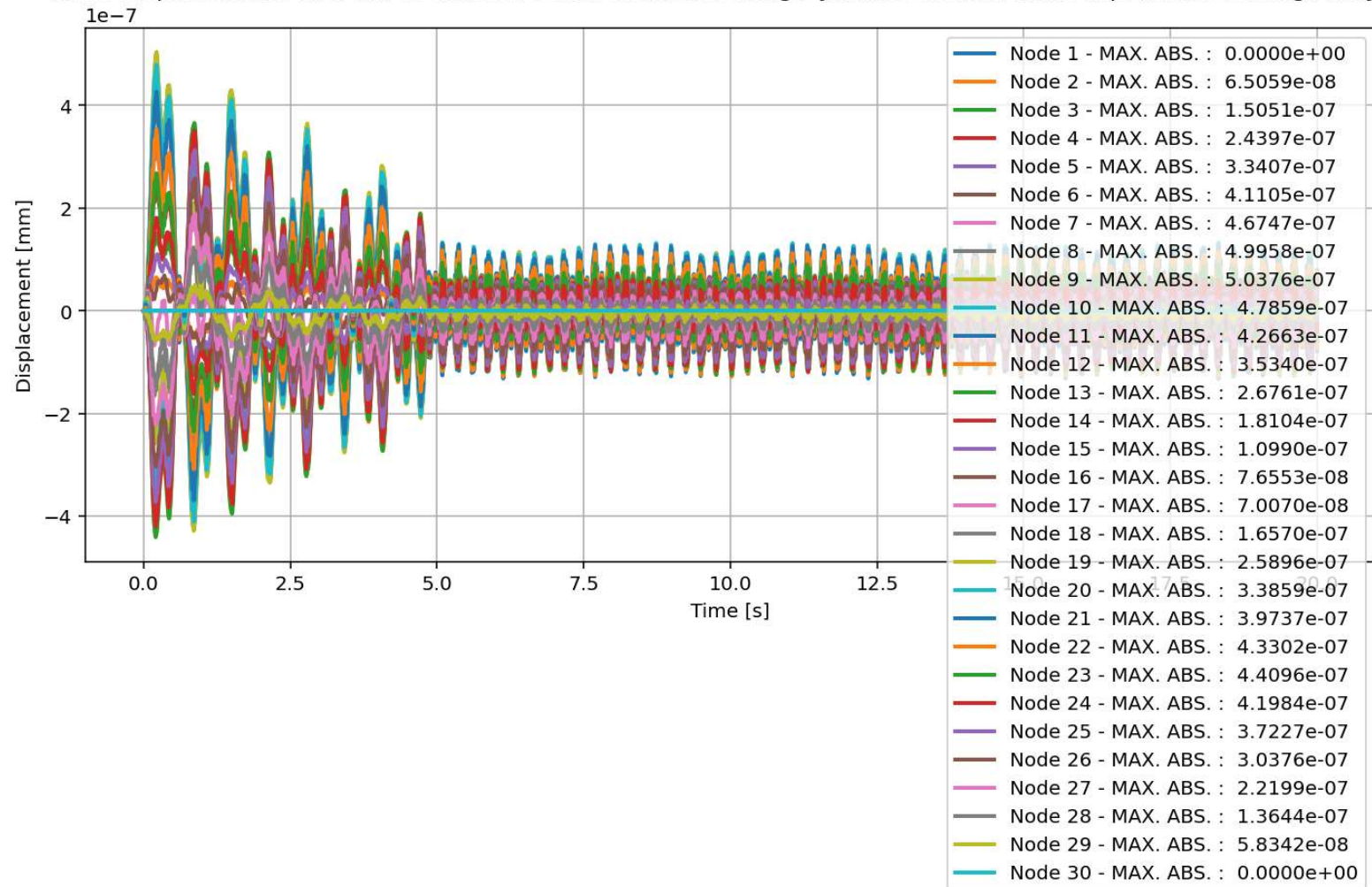


elements force in Z Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis

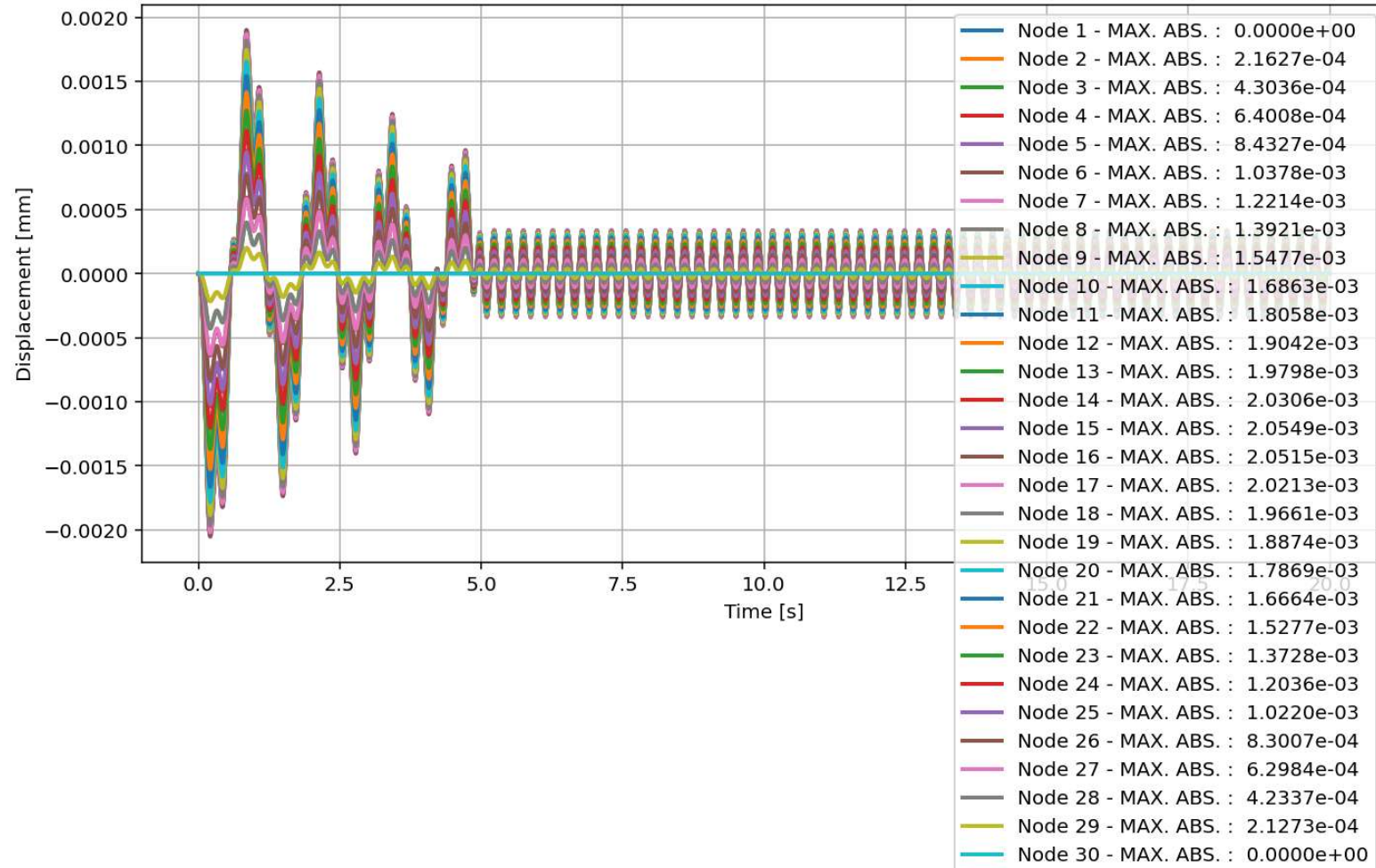




Node Displacements in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis

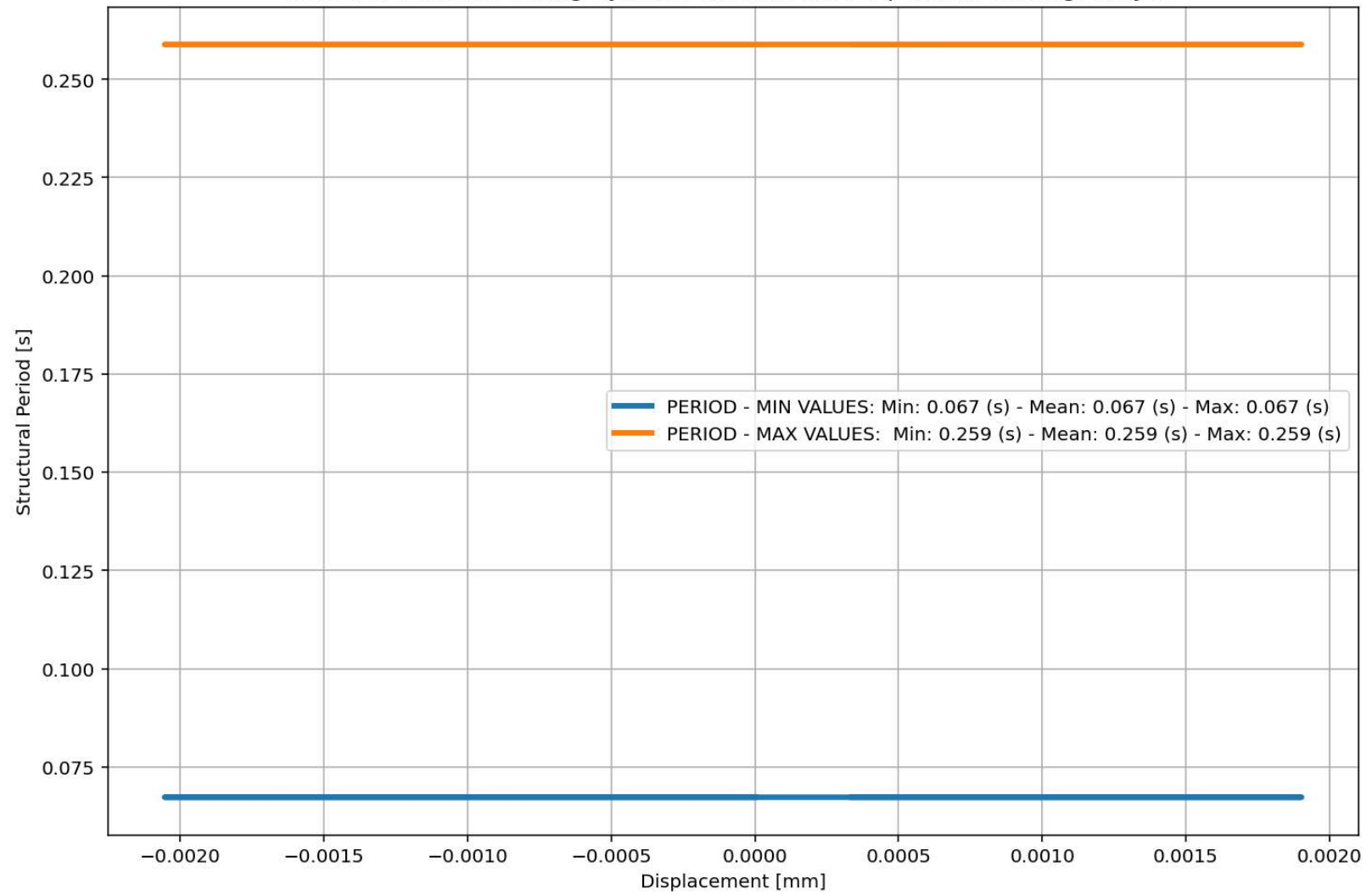


Node Displacements in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis

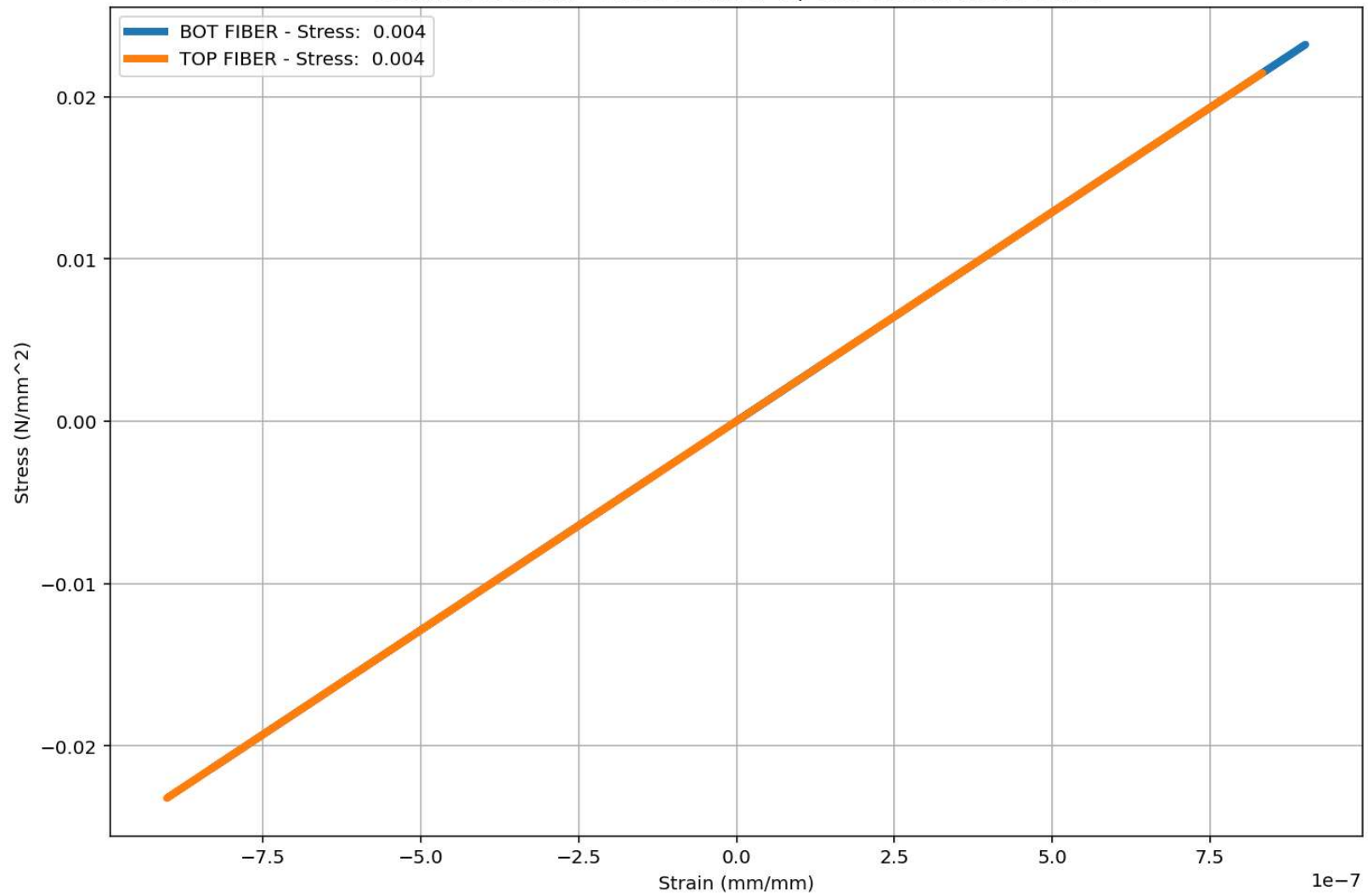


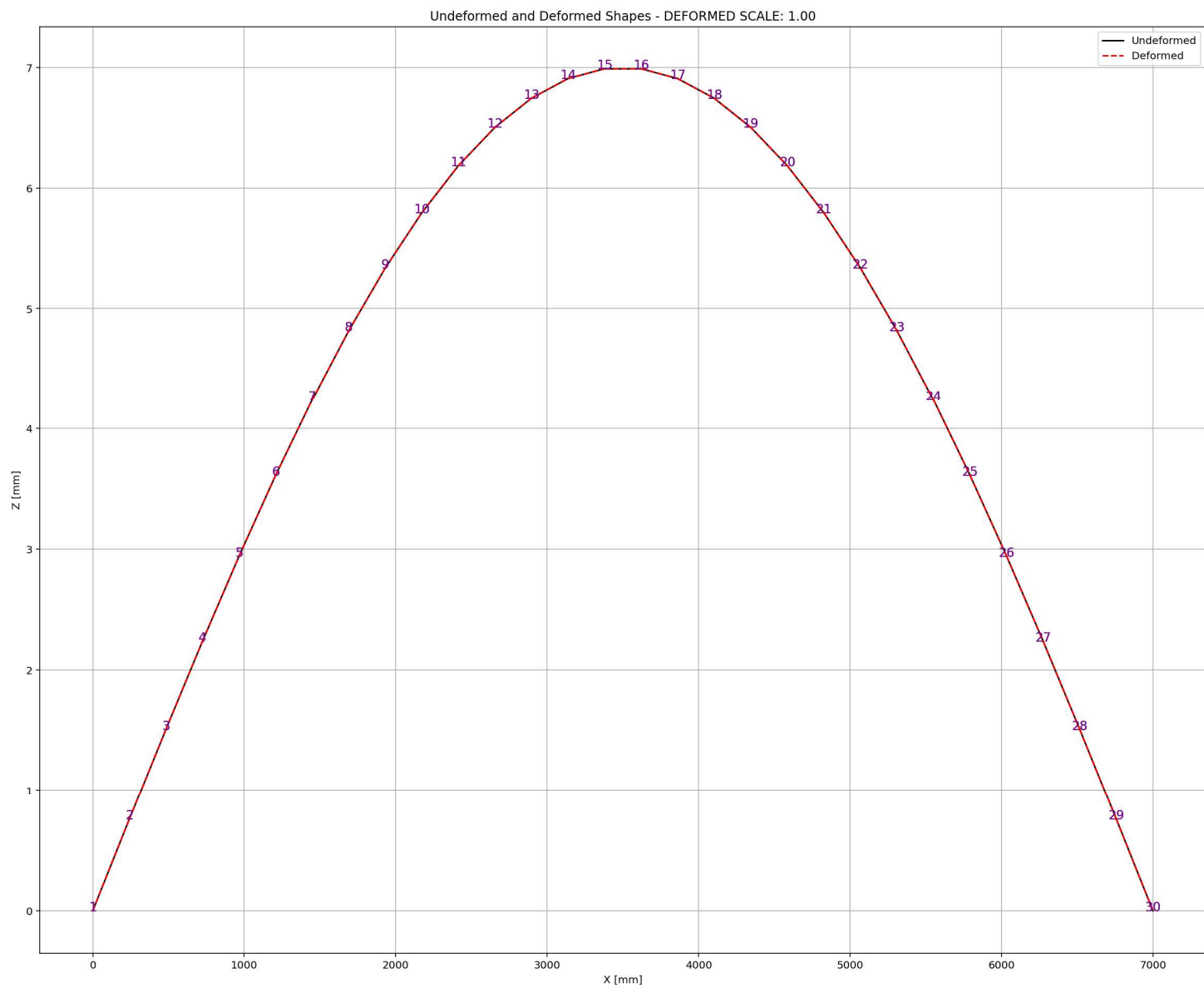


Period of Structure During Dynamic External Time-dependent Loading Analysis



Behavior of beam - Stress-strain of Top and Bottom Fibers Curve





# **FREE-VIBRATION ANALYSIS**

Spyder (Python 3.12)

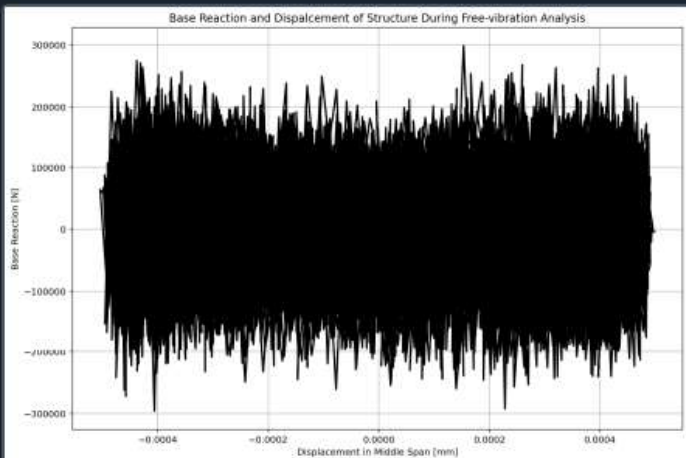
File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES\_FILES\+HEIGHT\_ANALY...\_CONCRETE\_SLAB\_SUPERSTRUCTURE\_SIMPLY\_SUPPORTEDED.py

P(t)\_BRIDGE\_CONCRE...IMPLY\_SUPPORTEDED.py X CONCRETE\_I\_SECTION...LANGE\_FUN\_EXTRA.py X untitled0.py\* X

```
1445 # FREE-VIBRATION ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1446 #ELE_TYPE = 'elasticBeamColumn' # MATERIAL LINEARITY
1447 ELE_TYPE = 'nonlinearBeamColumn' # MATERIAL AND GEOMETRIC NONLINEARITIES
1448 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1449 ANAL_TYPE = 'FREE-VIBRATION'
1450 DATA = SHORT_SPAN_SLAB_CONCRETE_BRIDGE(LENGTH, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MASS)
1451
1452 (time_FV, reaction_FV, disp_mid_FV,
1453 ele_axialforce_FV, ele_shearforce_FV, ele_momentforce_FV,
1454 node_displacementsX_FV, node_displacementsY_FV,
1455 dispX_FV, dispY_FV,
1456 veloX_FV, veloY_FV,
1457 accX_FV, accY_FV,
1458 stiffness_FV, PERIOD_FV, damping_ratio_FV,
1459 PERIOD_MIN_FV, PERIOD_MAX_FV,
1460 STRESSb_FV, STRAINb_FV, STRESSt_FV, STRAINt_FV, CURVATURE_FV) = DATA
1461
1462 XDATA = disp_mid_FV
1463 YDATA = reaction_FV
1464 XLABEL = 'Displacement in Middle Span [mm]'
1465 YLABEL = 'Base Reaction [N]'
1466 TITLE = 'Base Reaction and Displacment of Structure During Free-vibration Analysis'
1467 COLOR = 'black'
1468 SEMILOGY = False
1469 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1470
1471 PLOT_TIME_HISTORY(time_FV, reaction_FV, disp_mid_FV,
1472 dispX_FV, dispY_FV,
1473 veloX_FV, veloY_FV,
1474 accX_FV, accY_FV)
1475
1476 # PLOT ELEMENTS AXIAL FORCE
1477 YLABEL = 'Element Axial Force [N]'
1478 TITLE = "elements force in X Dir. vs Time for Beam Element During Free-vibration Analysis"
```

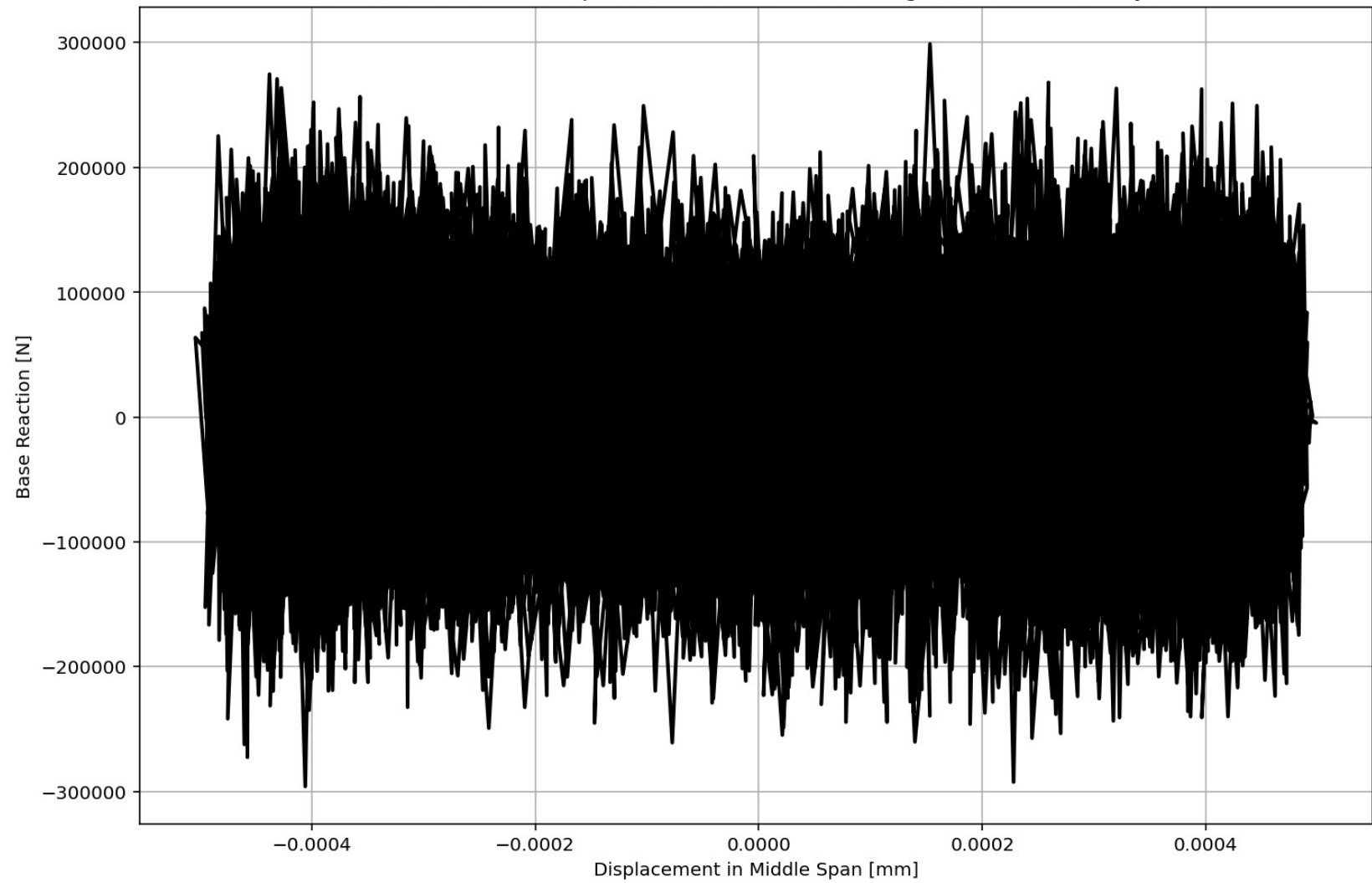
Base Reaction and Displacment of Structure During Free-vibration Analysis

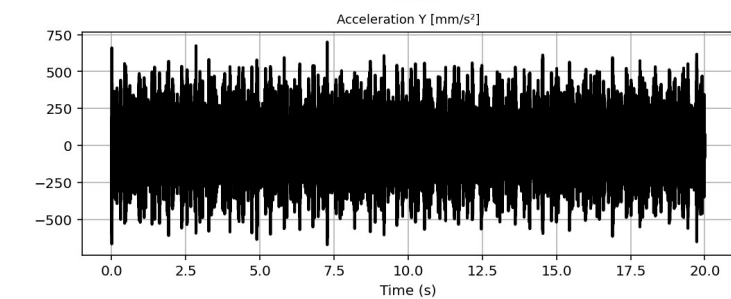
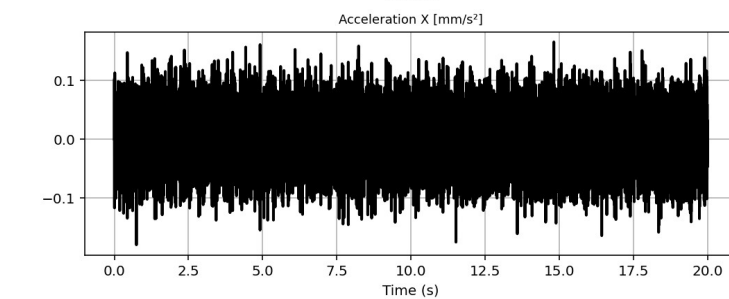
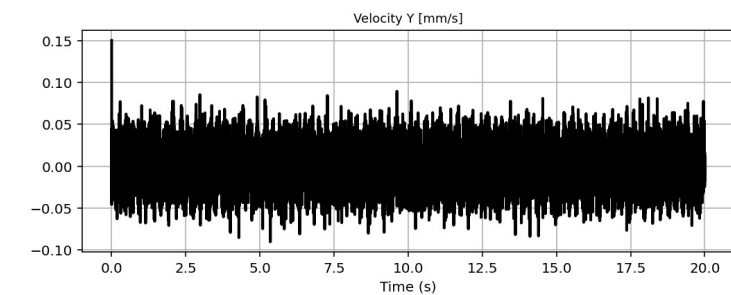
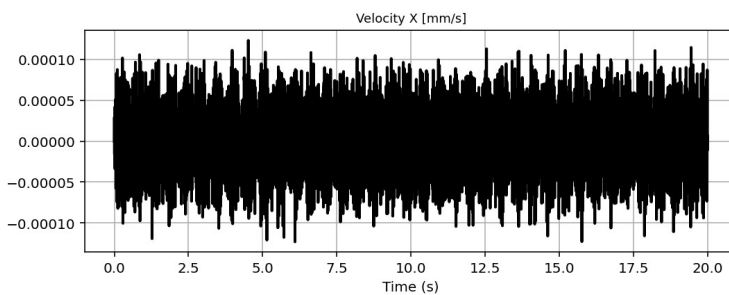
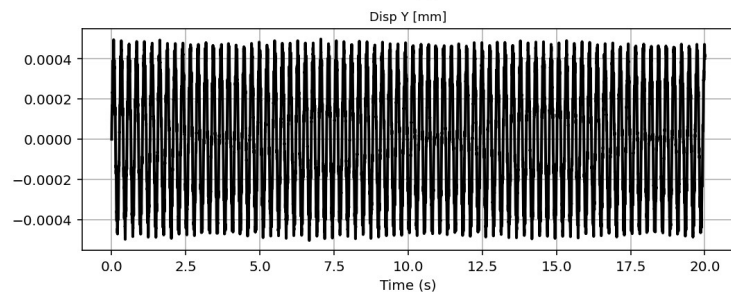
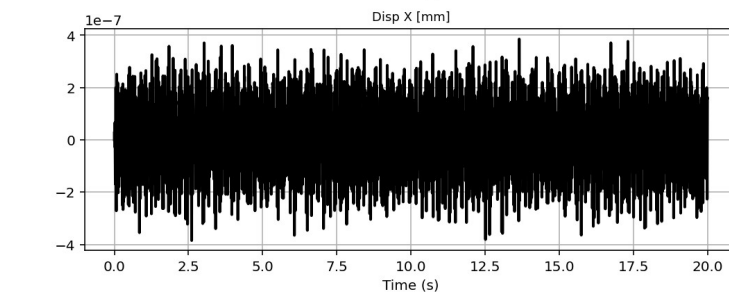
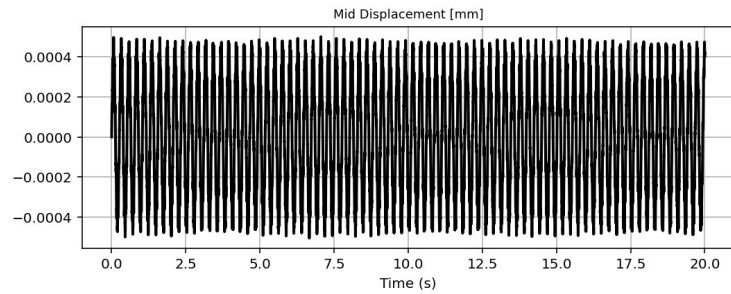
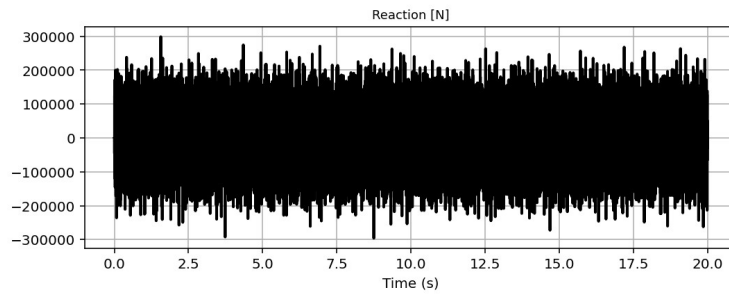


Python Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 172, Col 35 UTF-8 CRLF RW Mem 47%

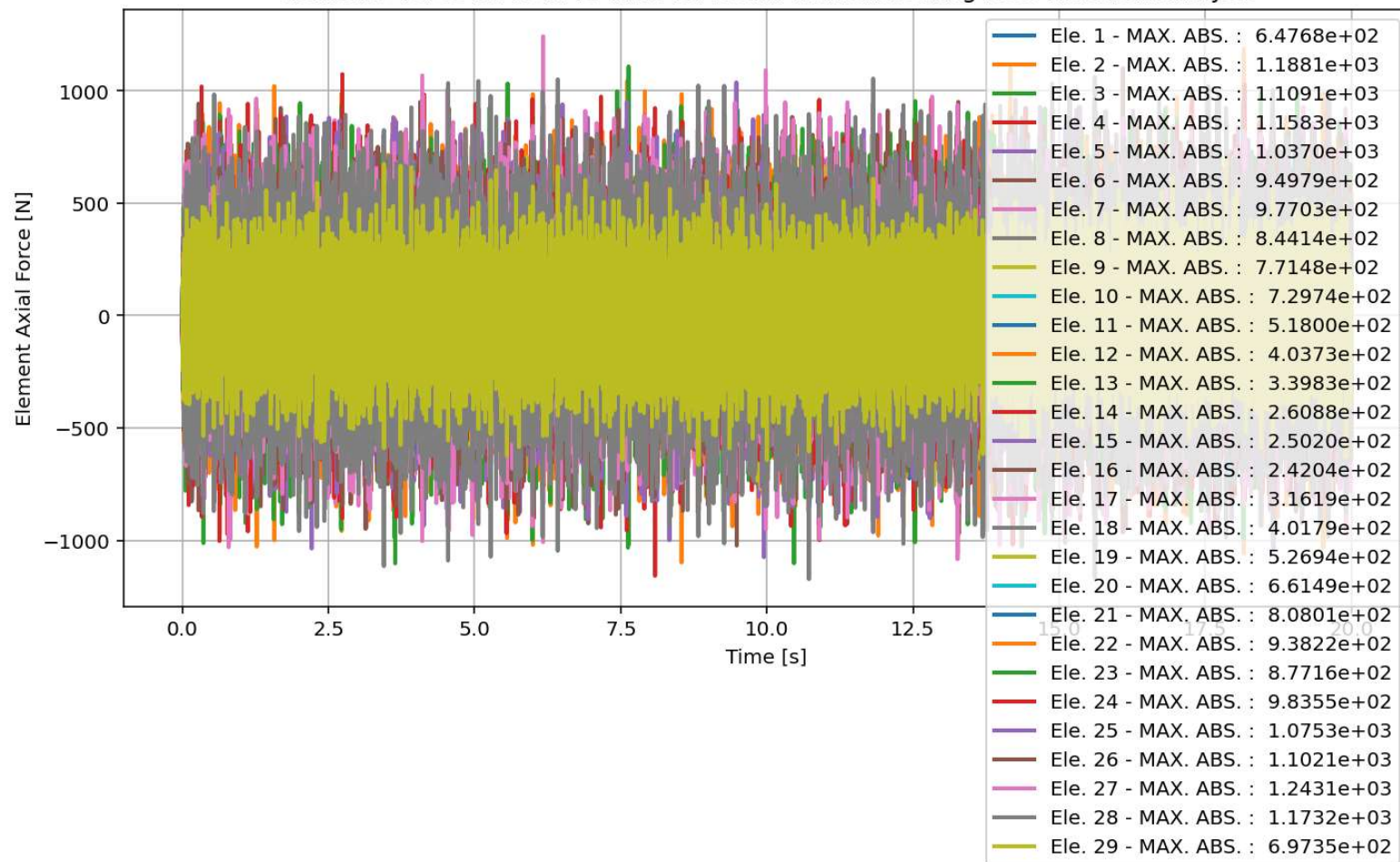
Base Reaction and Displacement of Structure During Free-vibration Analysis



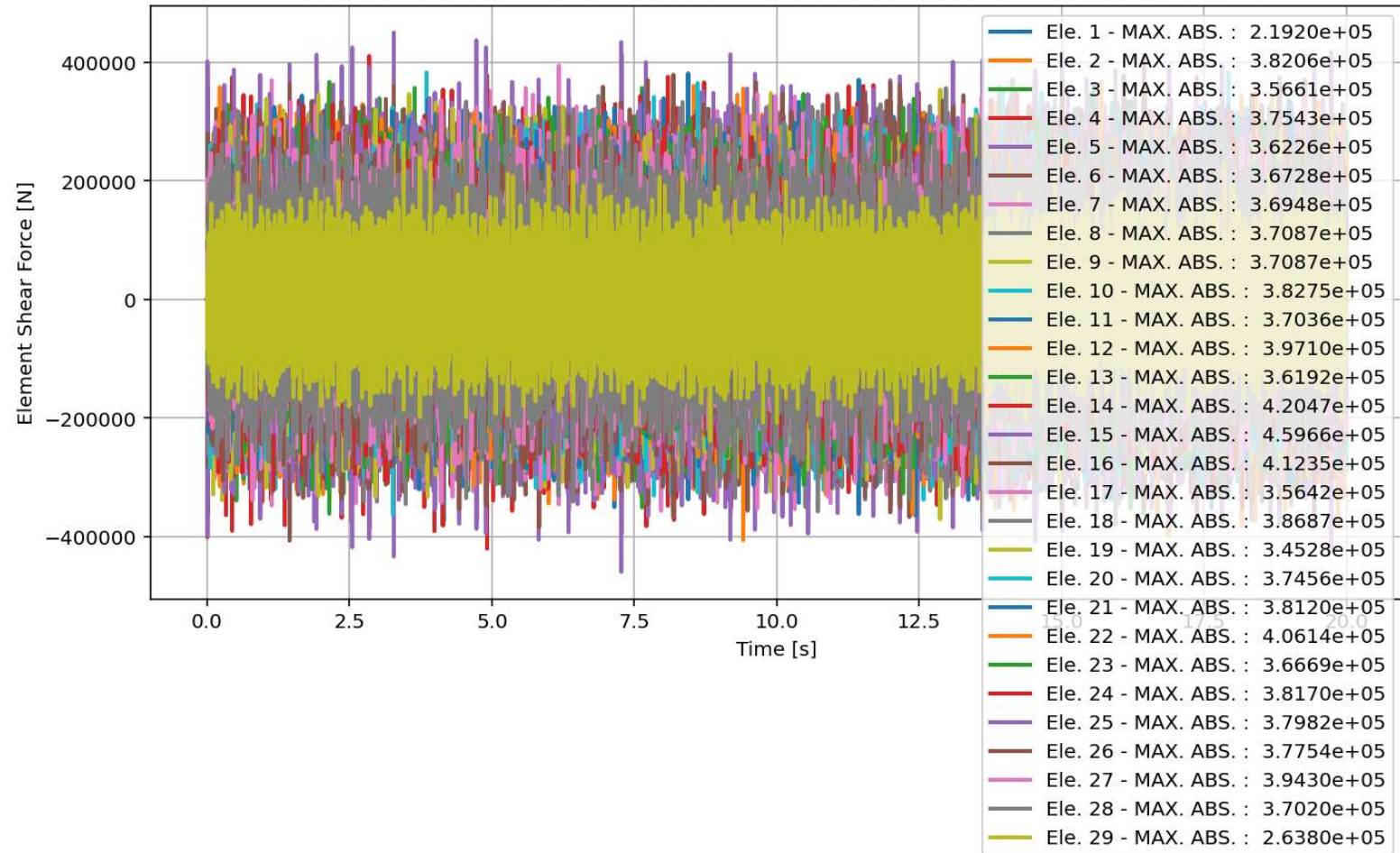


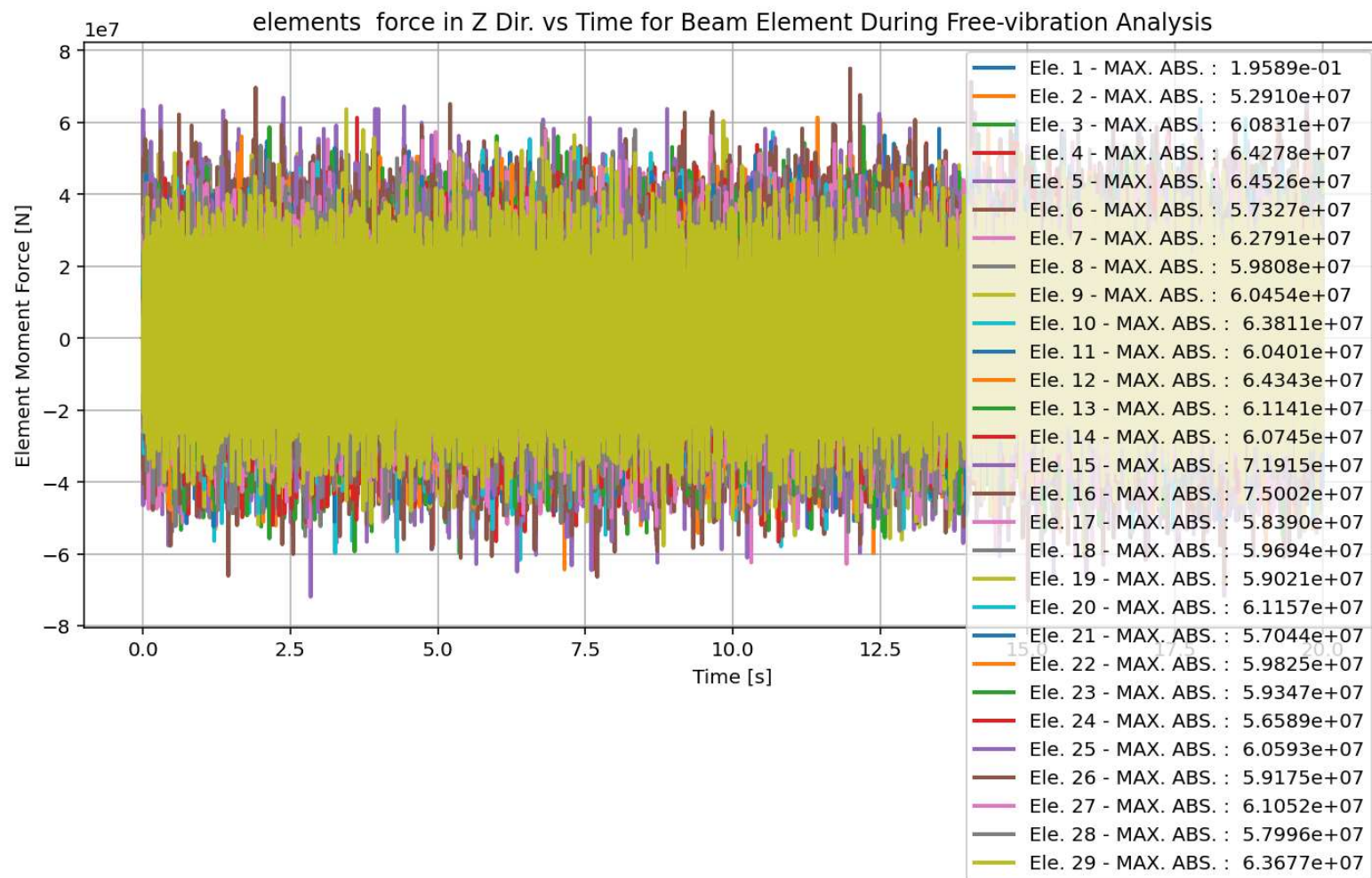


elements force in X Dir. vs Time for Beam Element During Free-vibration Analysis

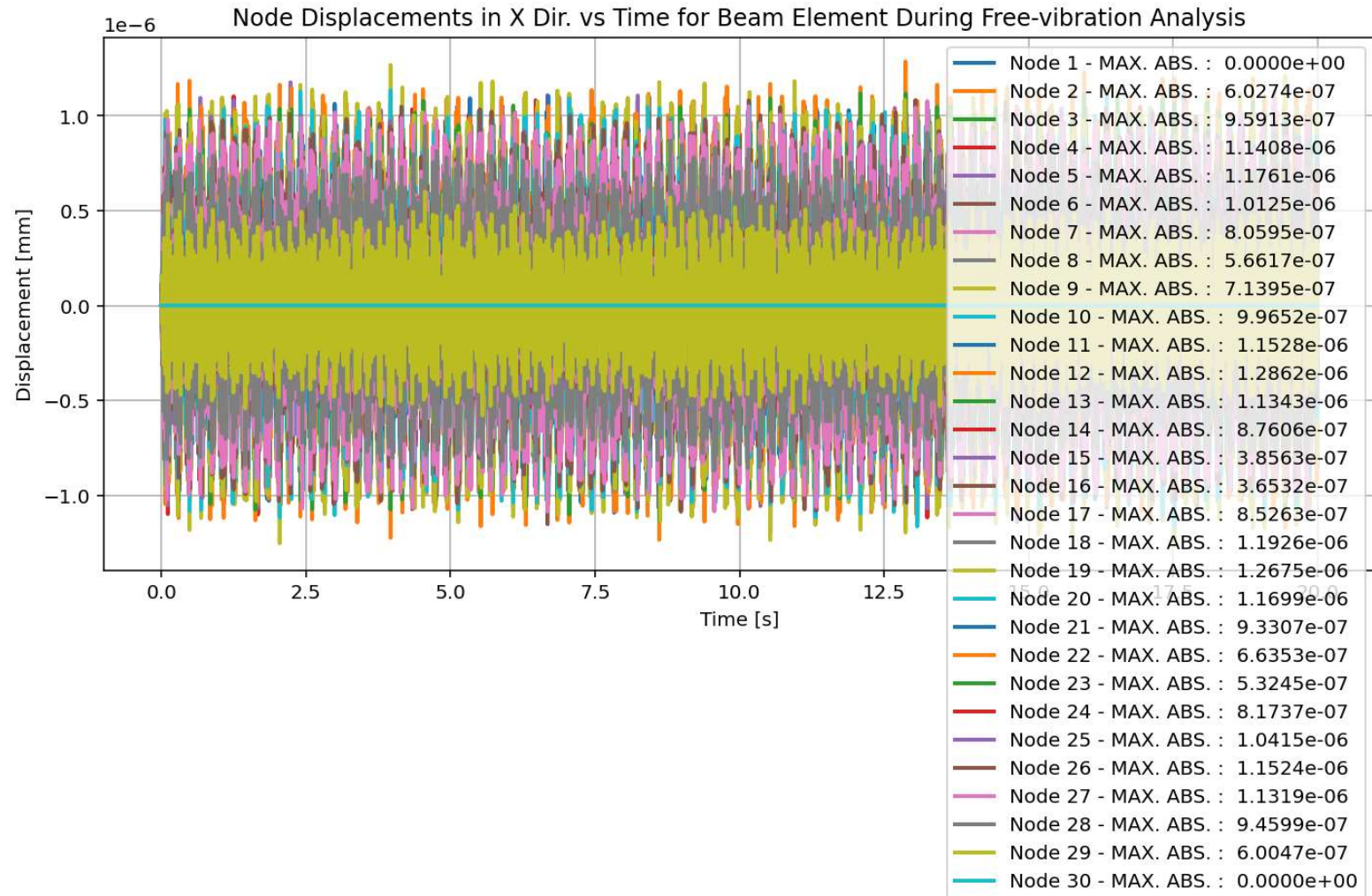


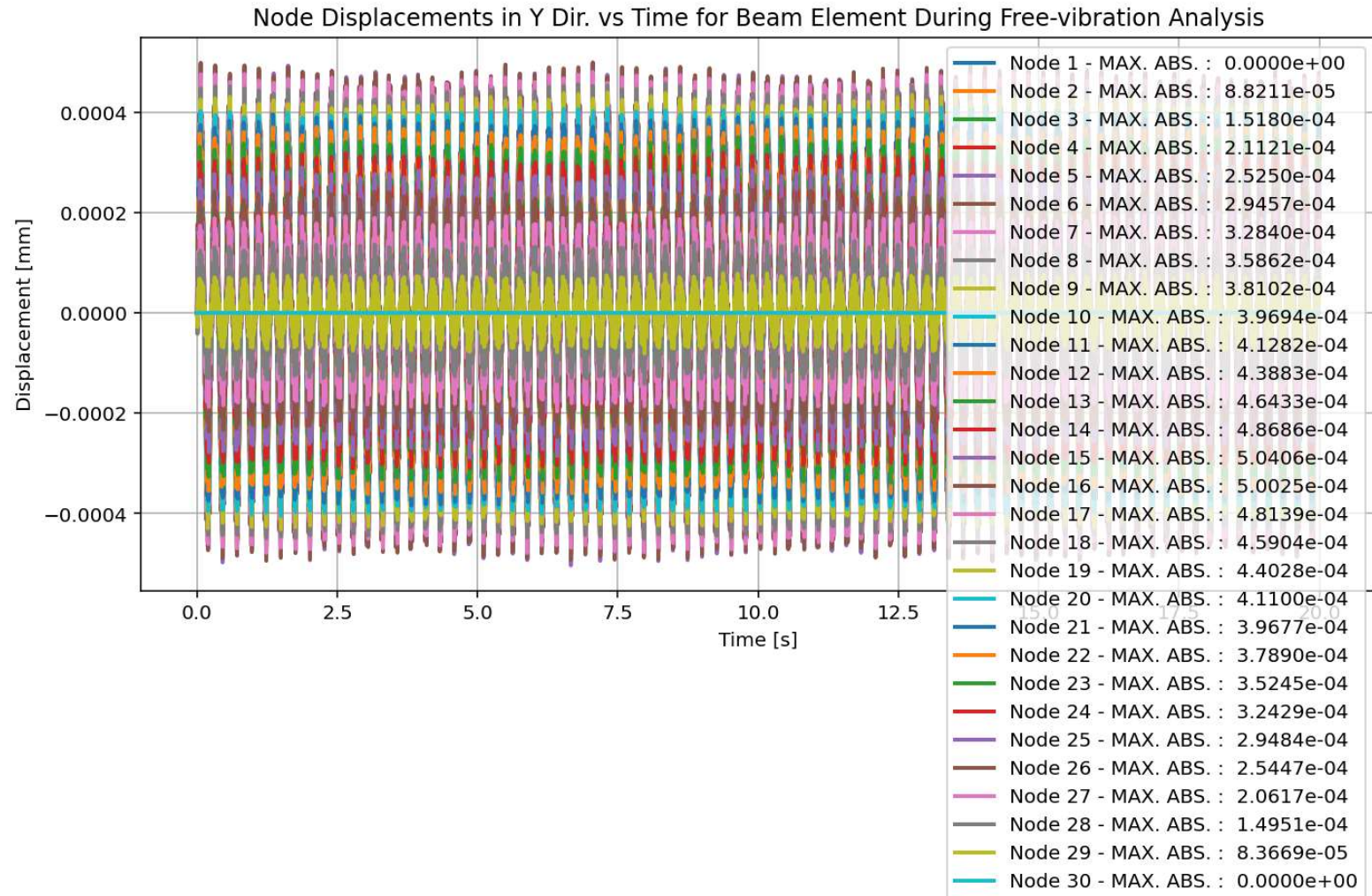
elements force in Y Dir. vs Time for Beam Element During Free-vibration Analysis



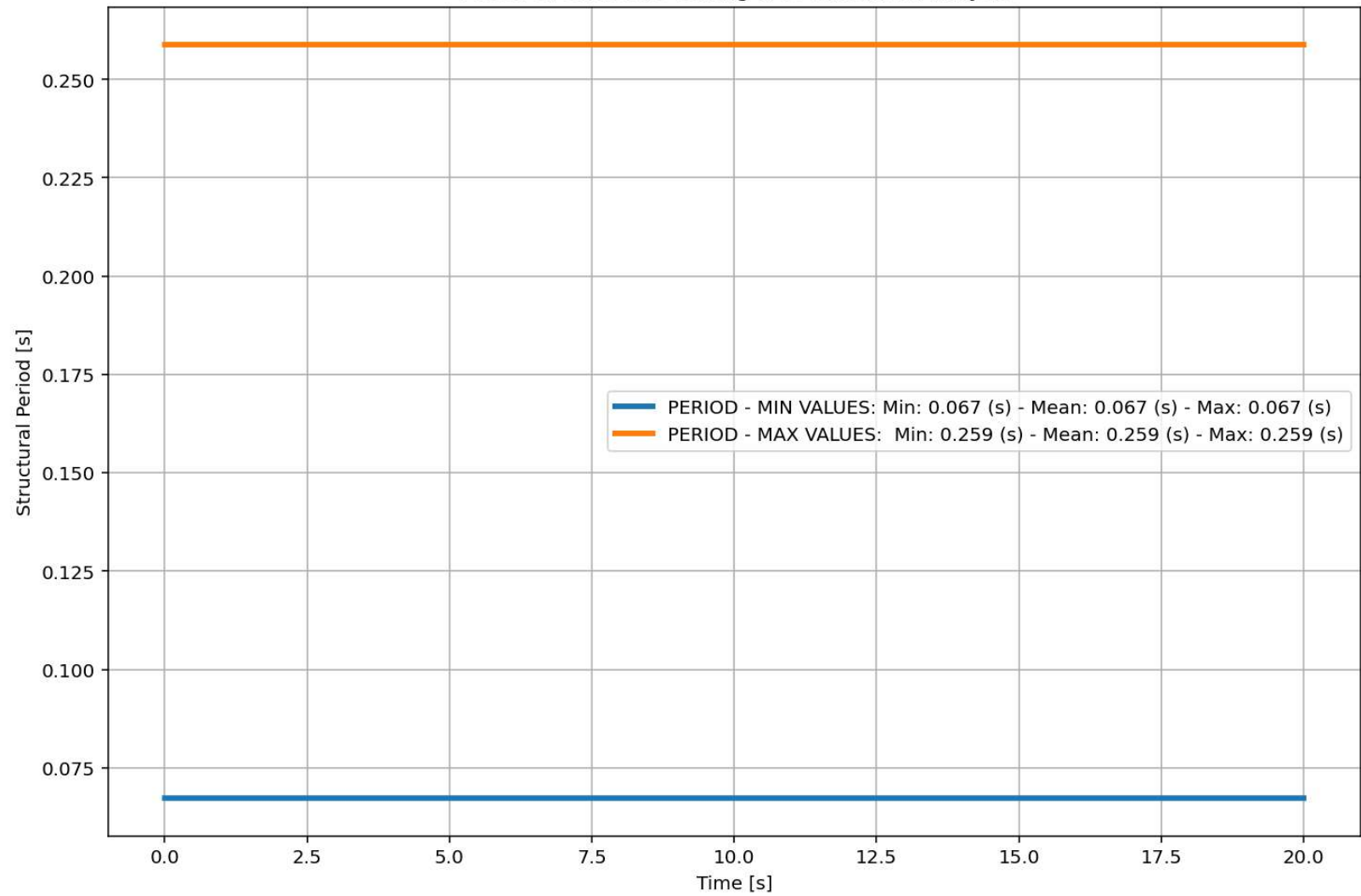


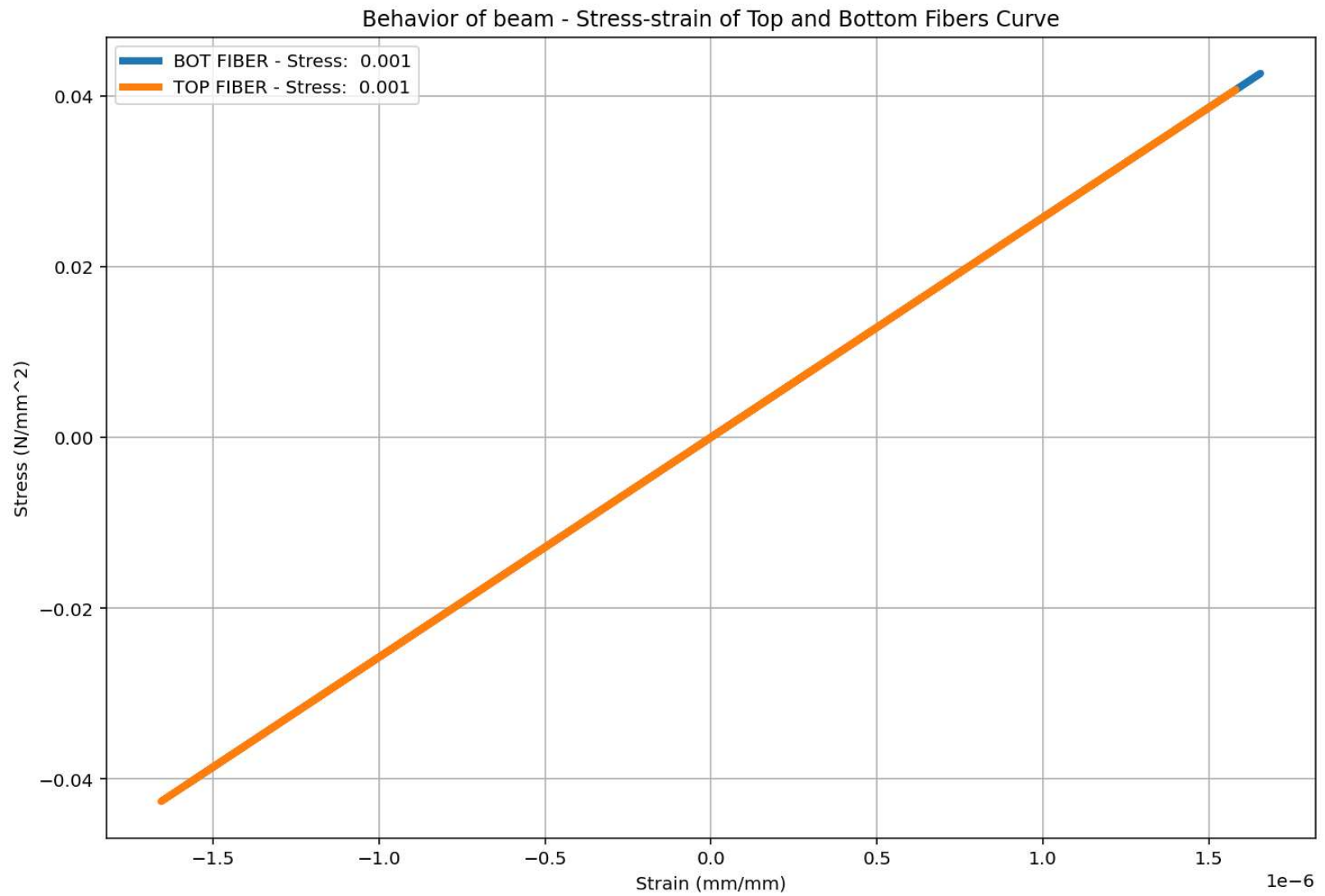




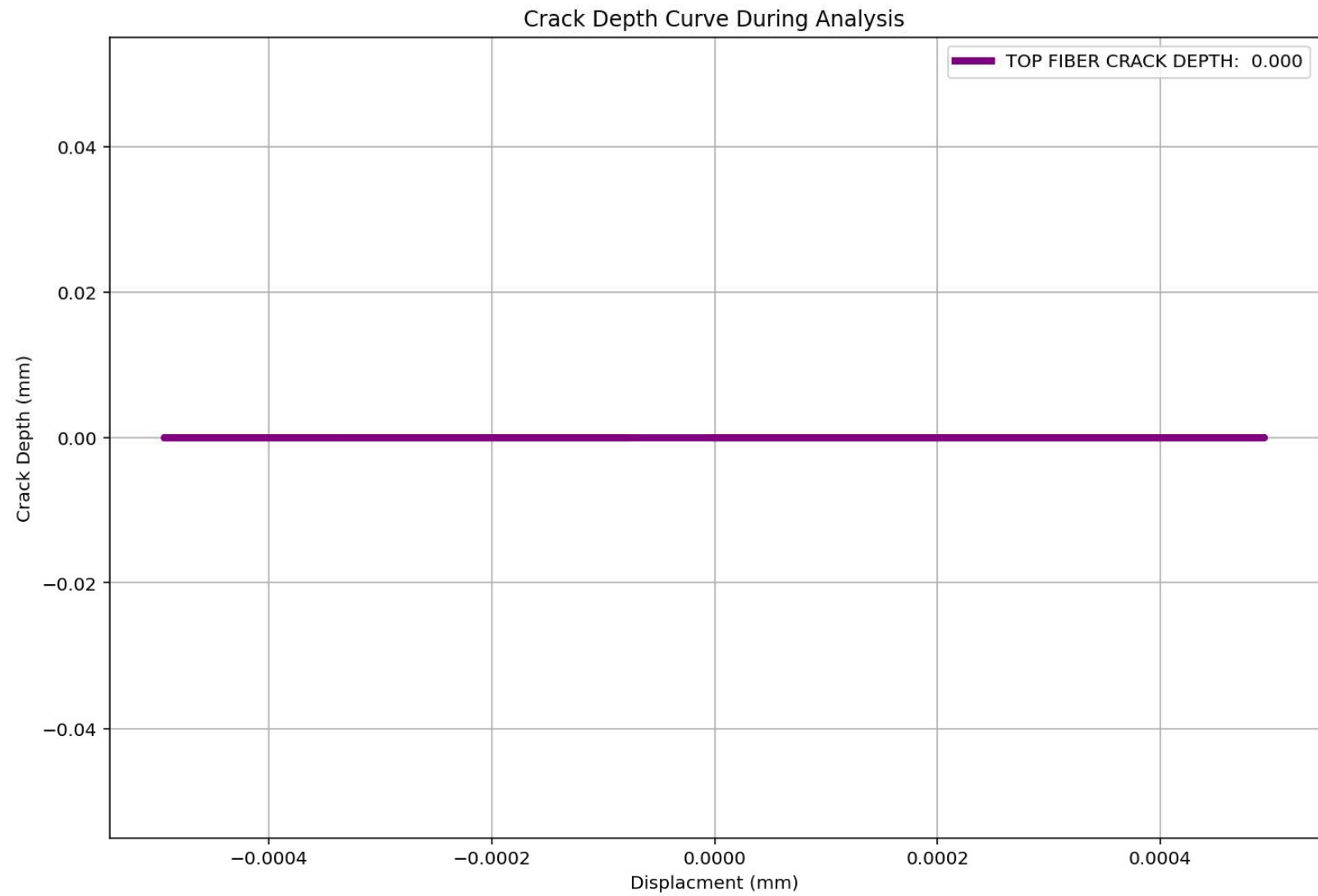


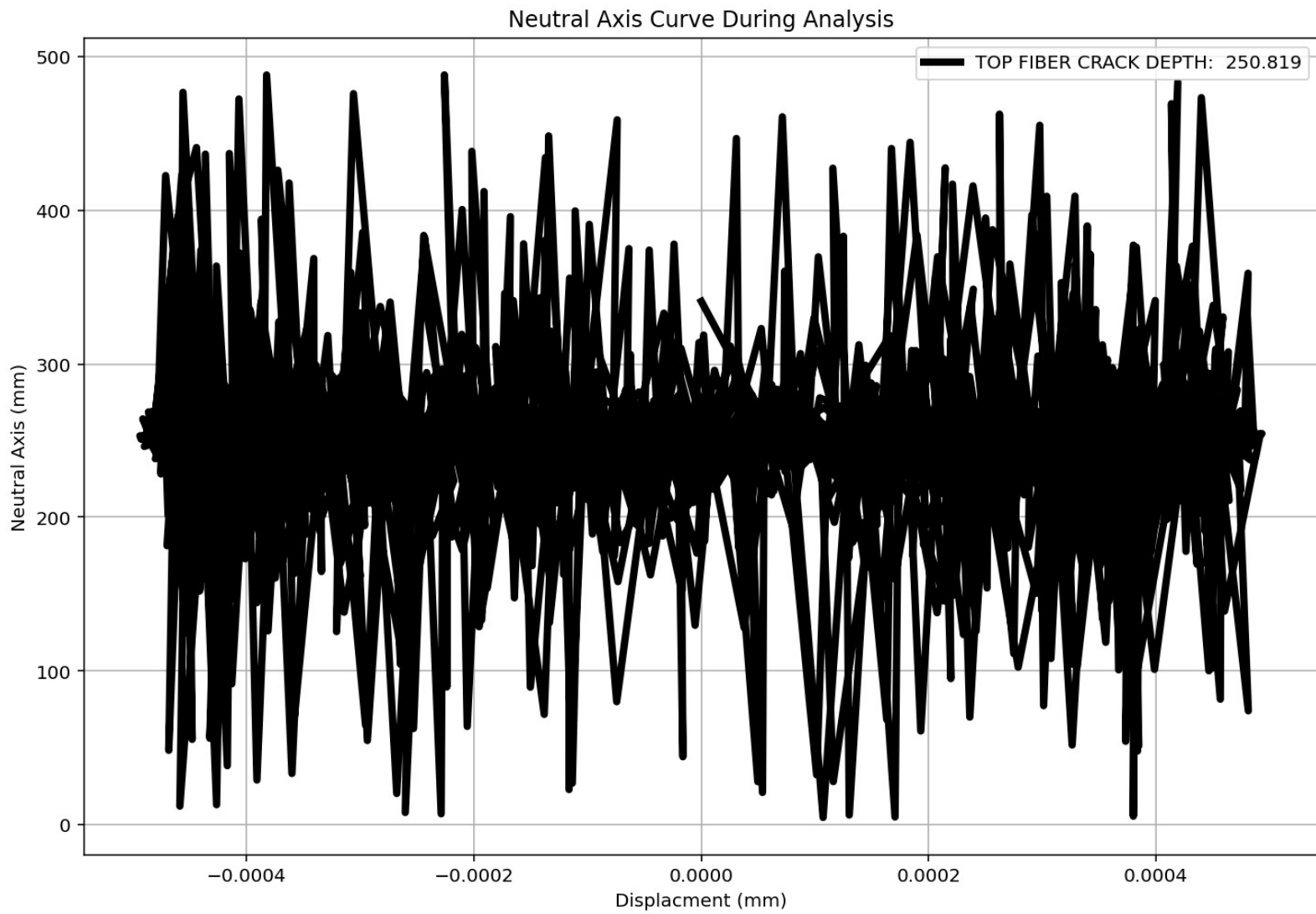
Period of Structure During Free-vibration Analysis

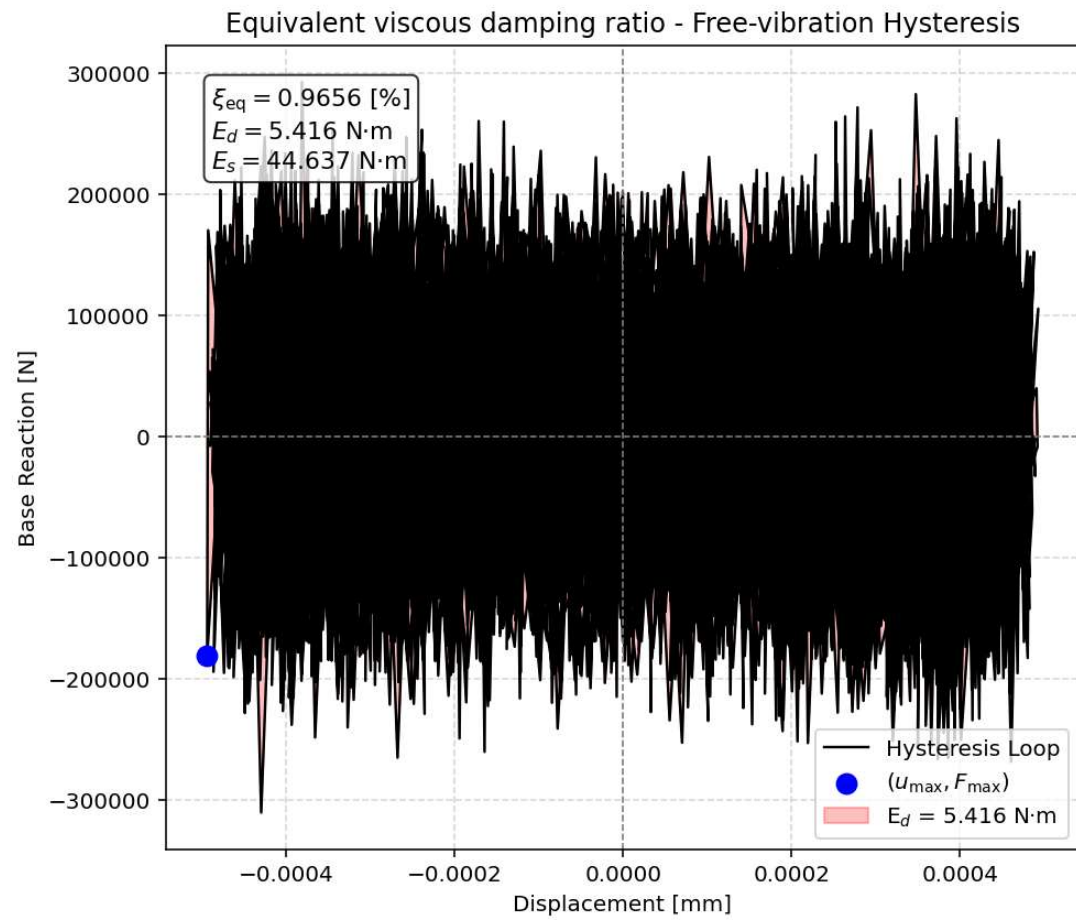


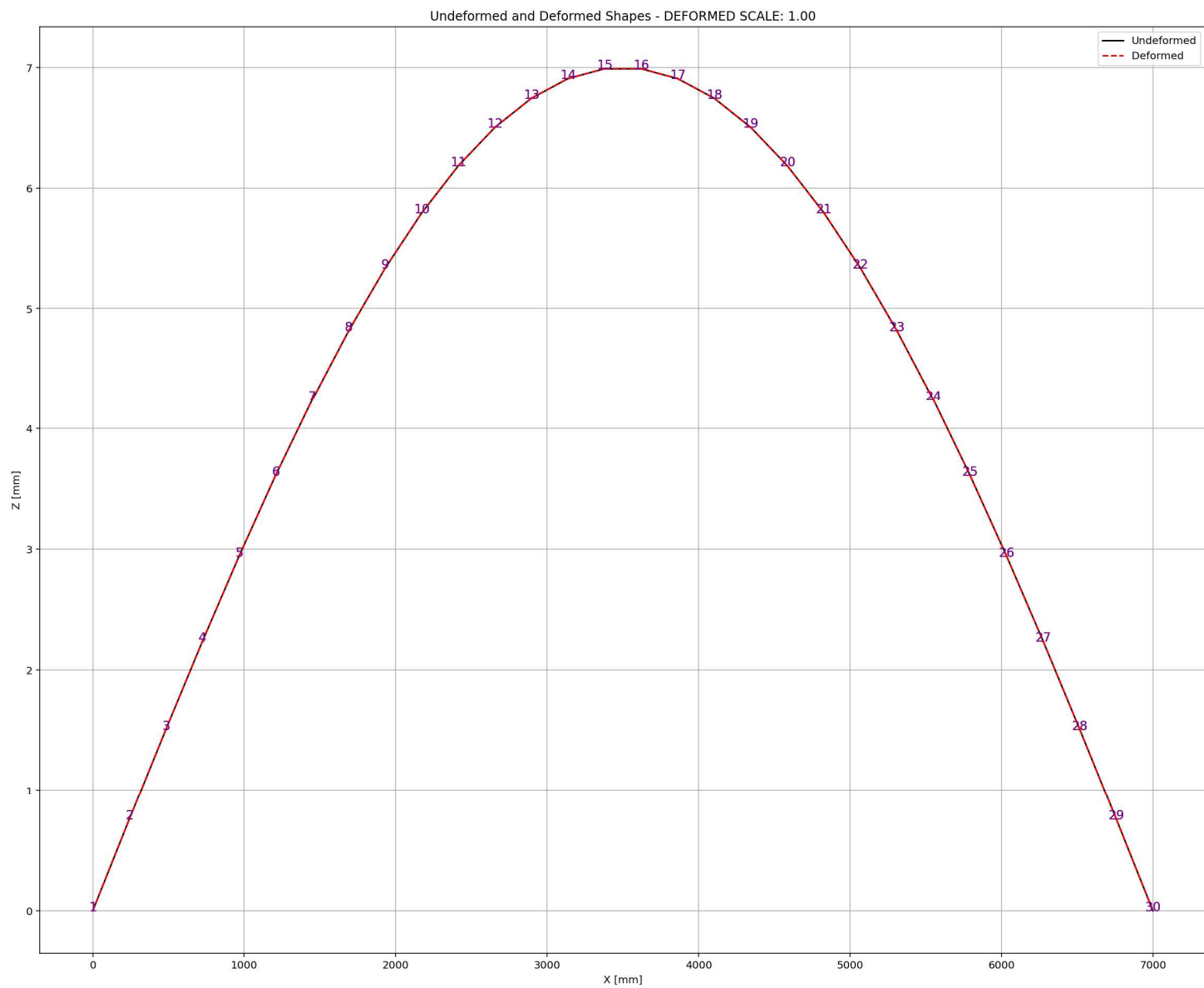






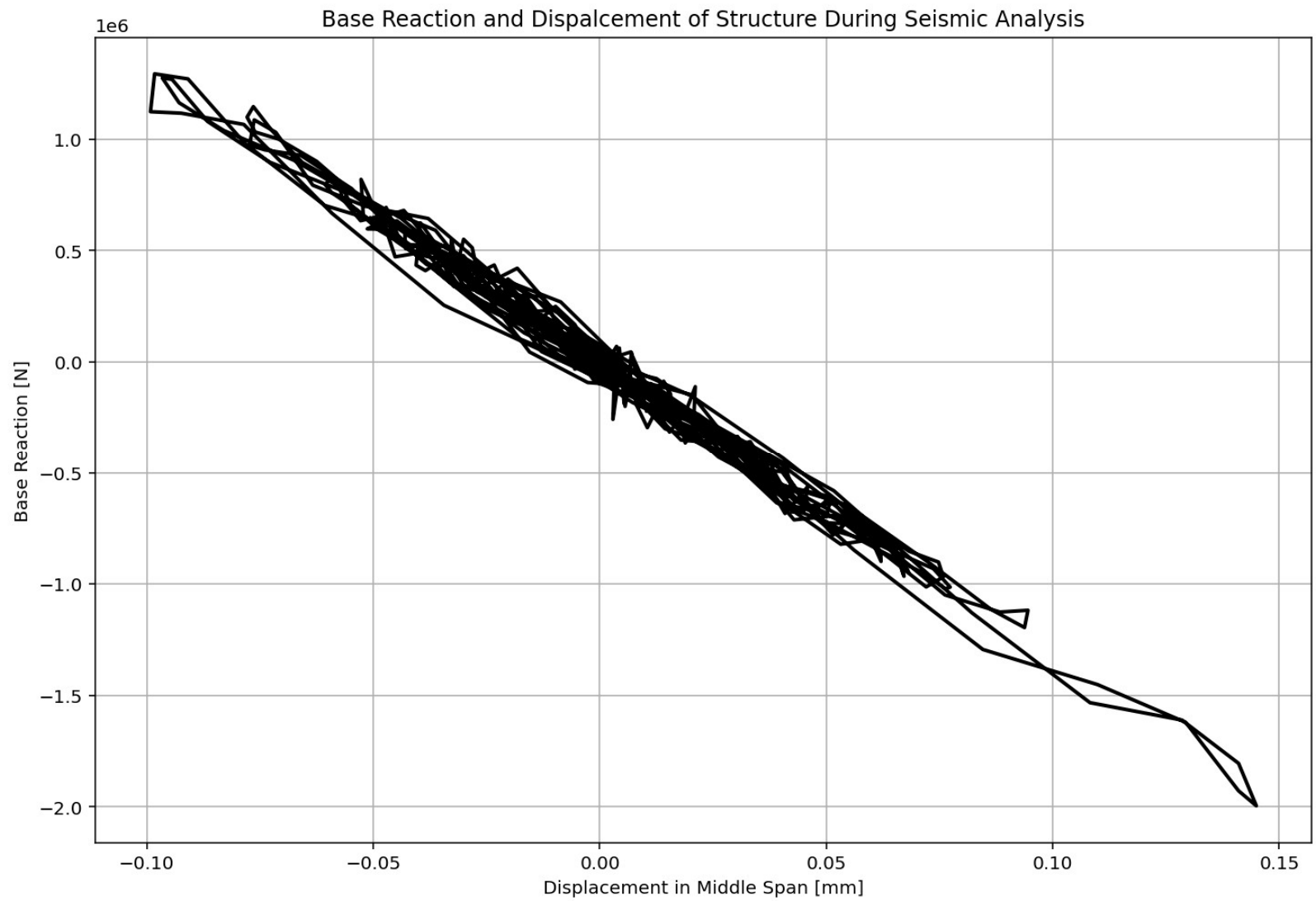




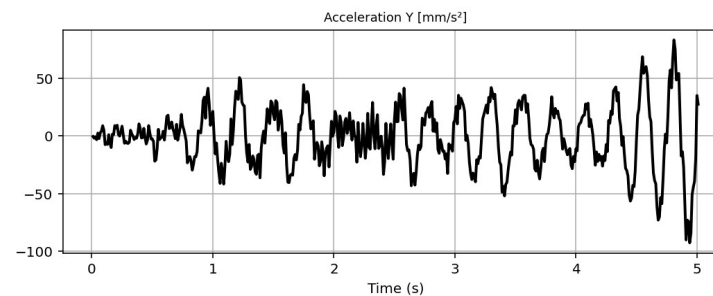
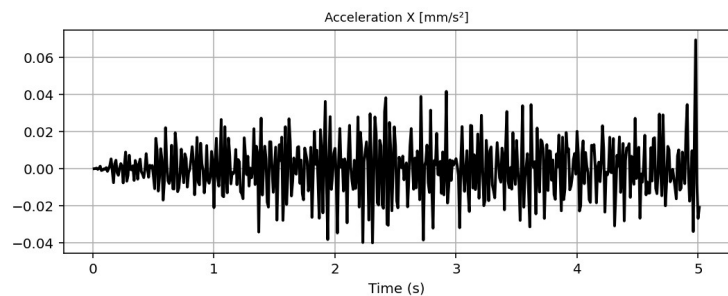
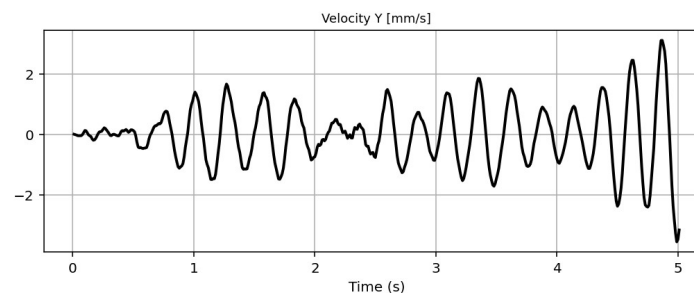
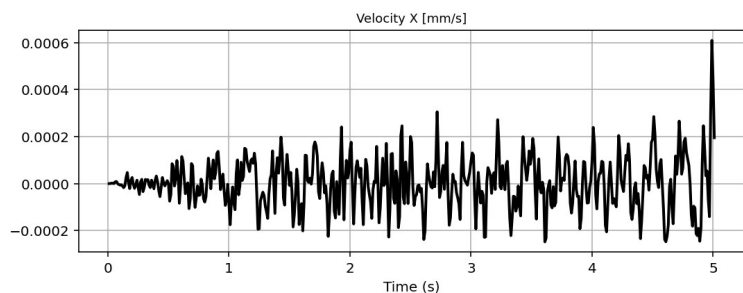
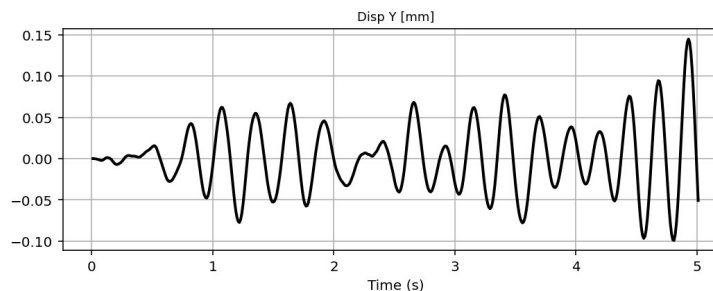
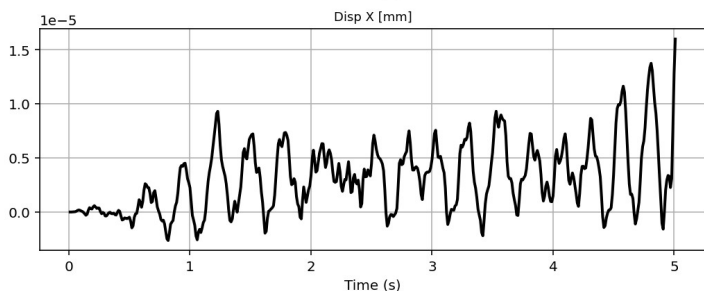
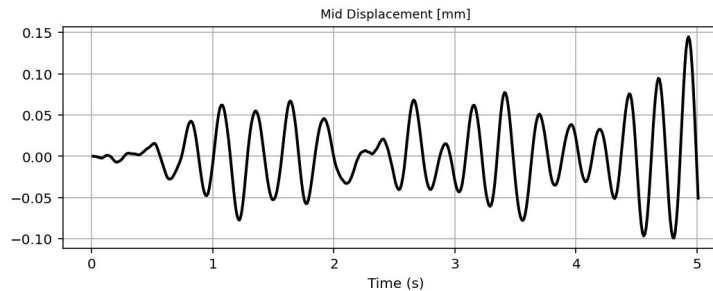
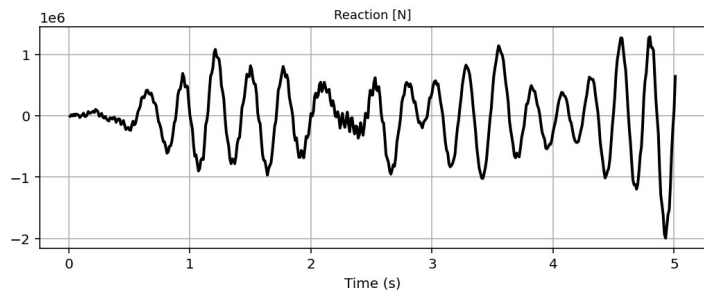


# SEISMIC ANALYSIS

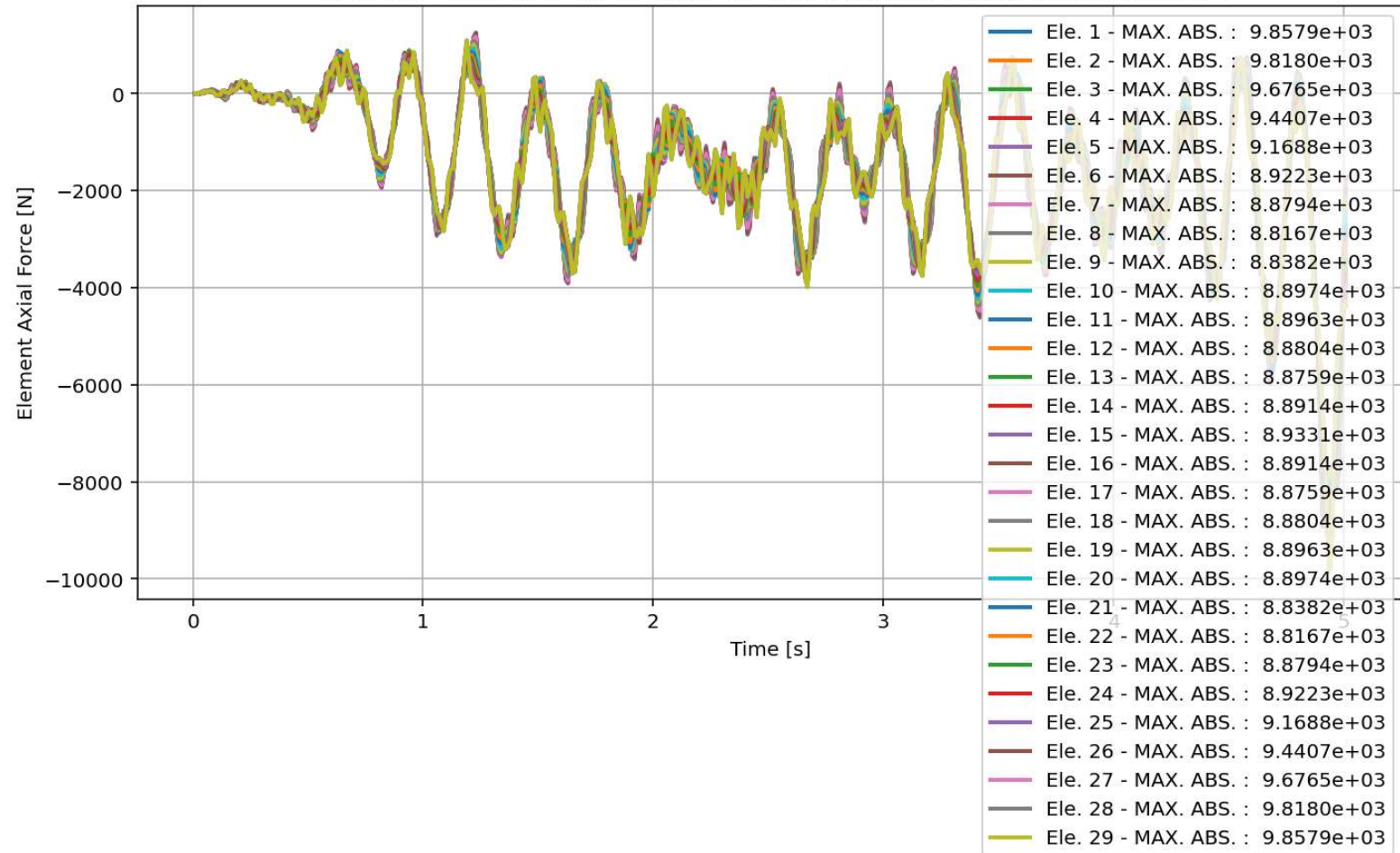


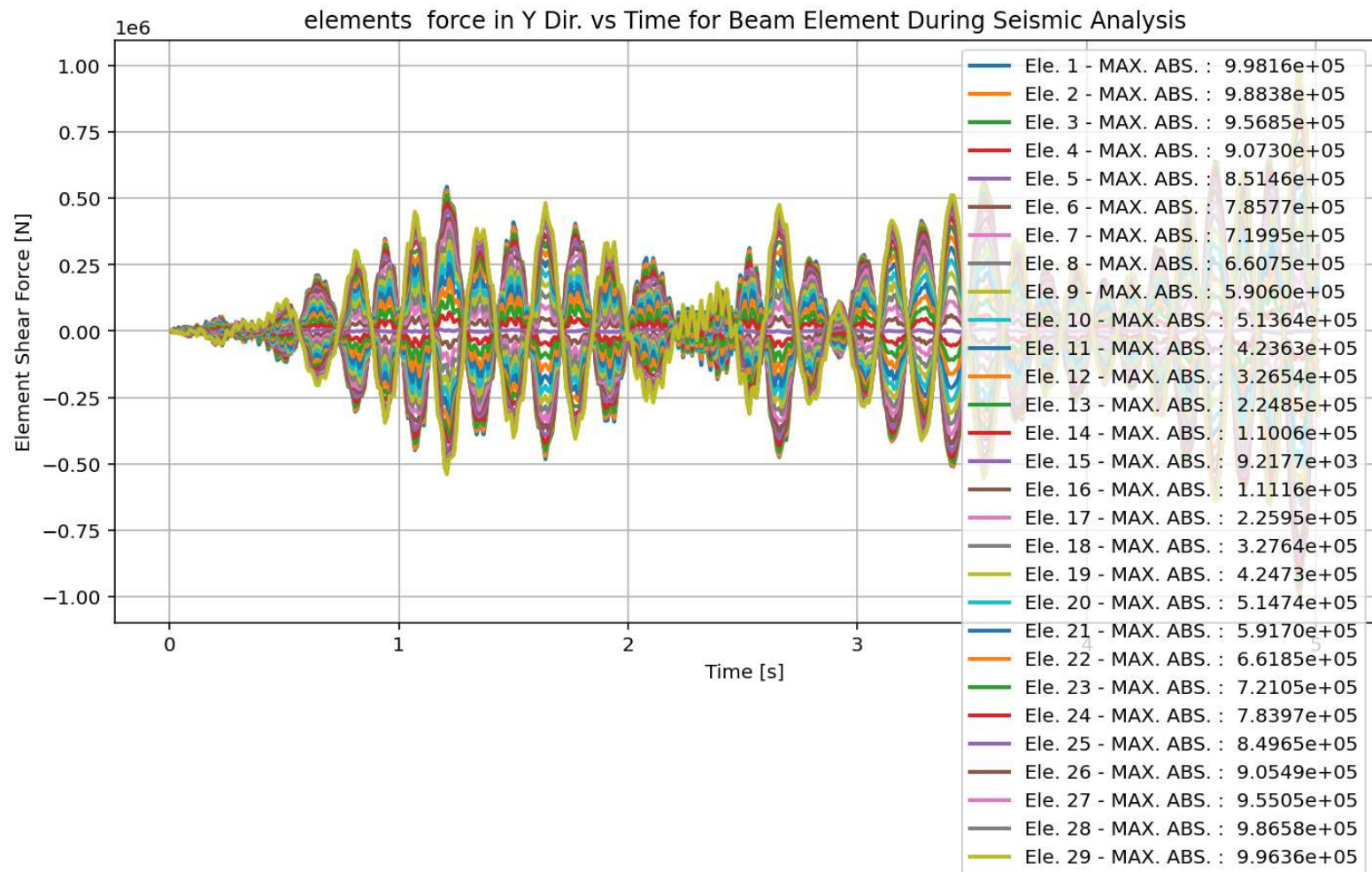


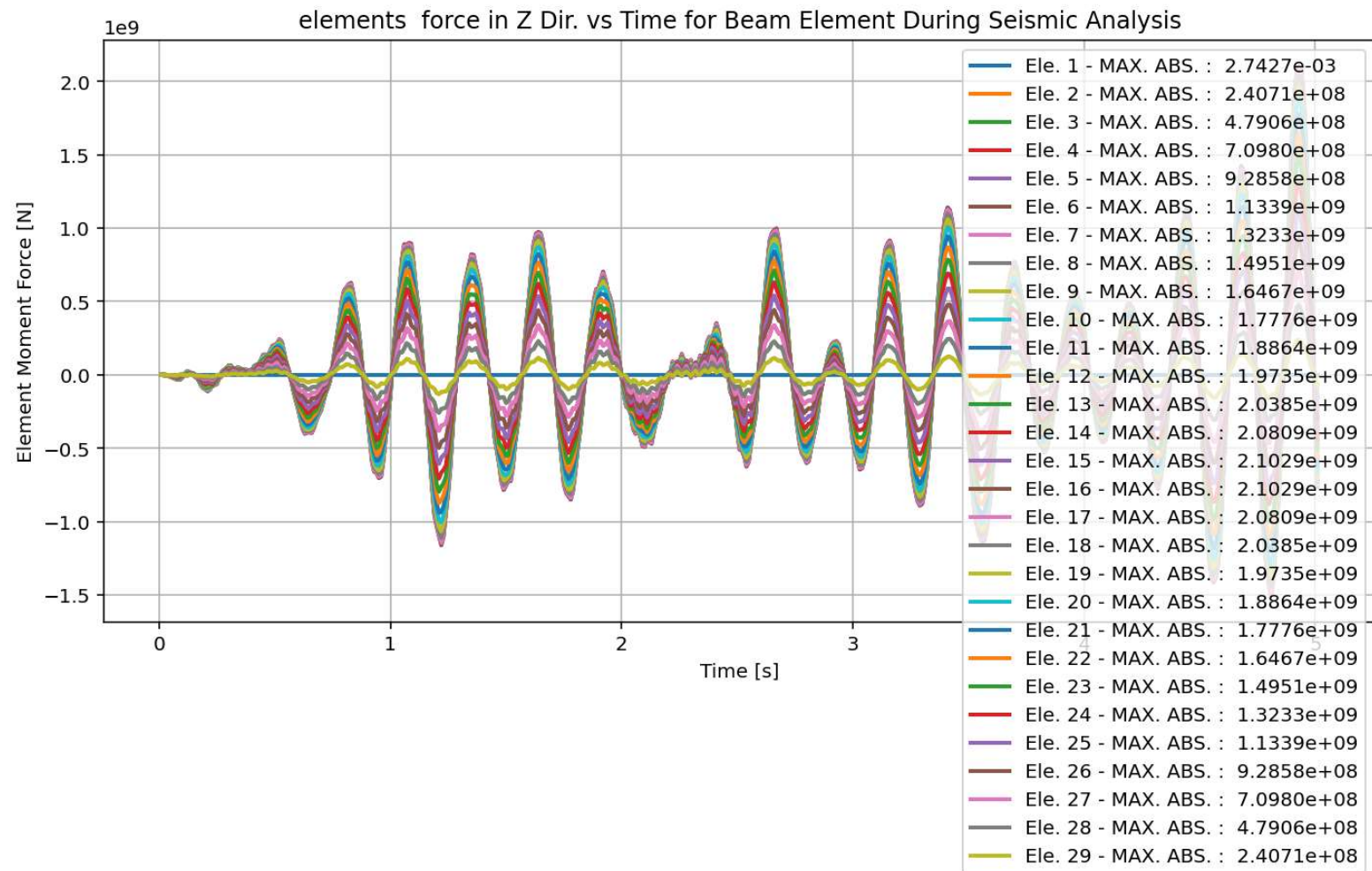




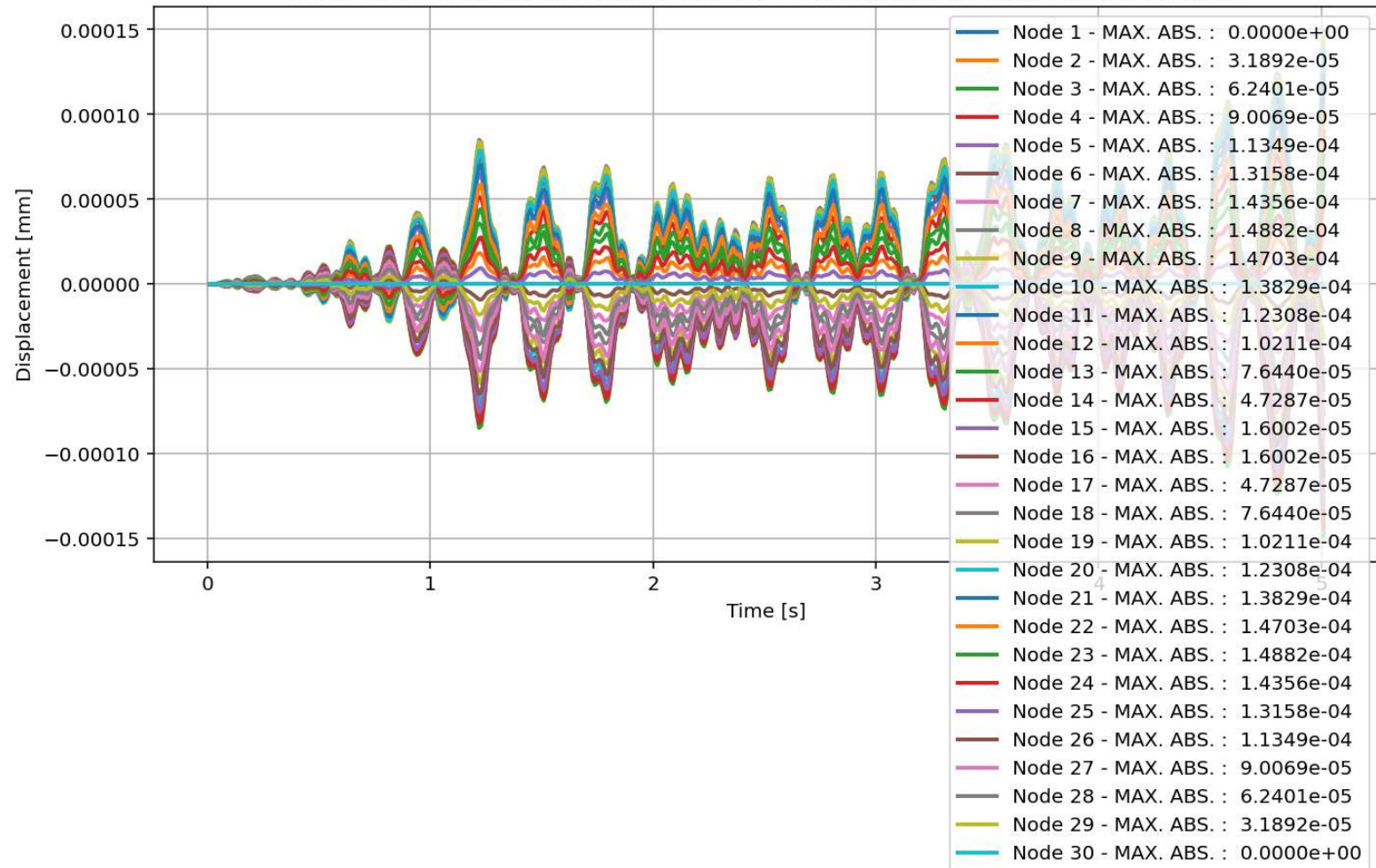
elements force in X Dir. vs Time for Beam Element During Seismic Analysis





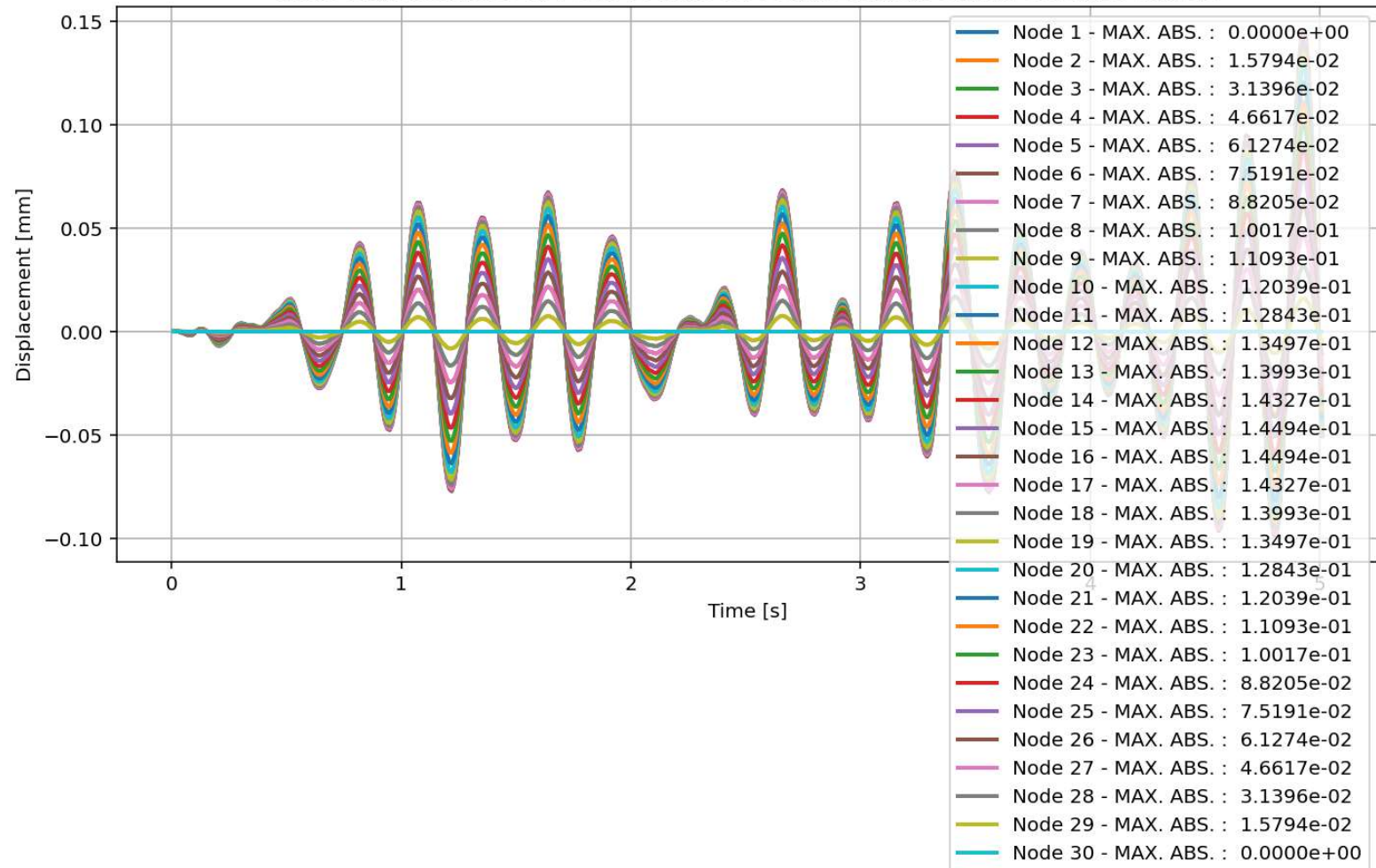


Node Displacements in X Dir. vs Time for Beam Element During Seismic Analysis

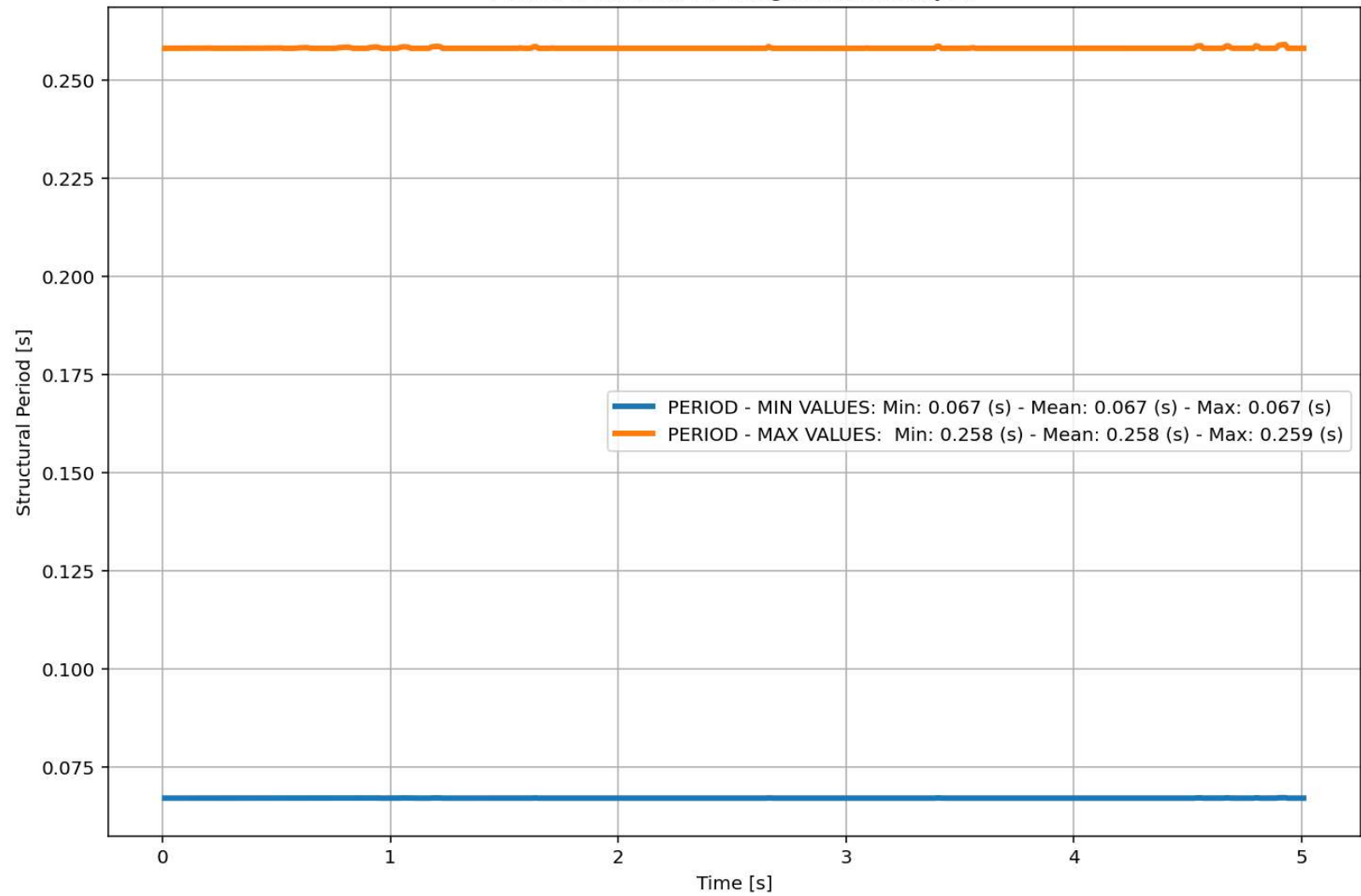




Node Displacements in Y Dir. vs Time for Beam Element During Seismic Analysis

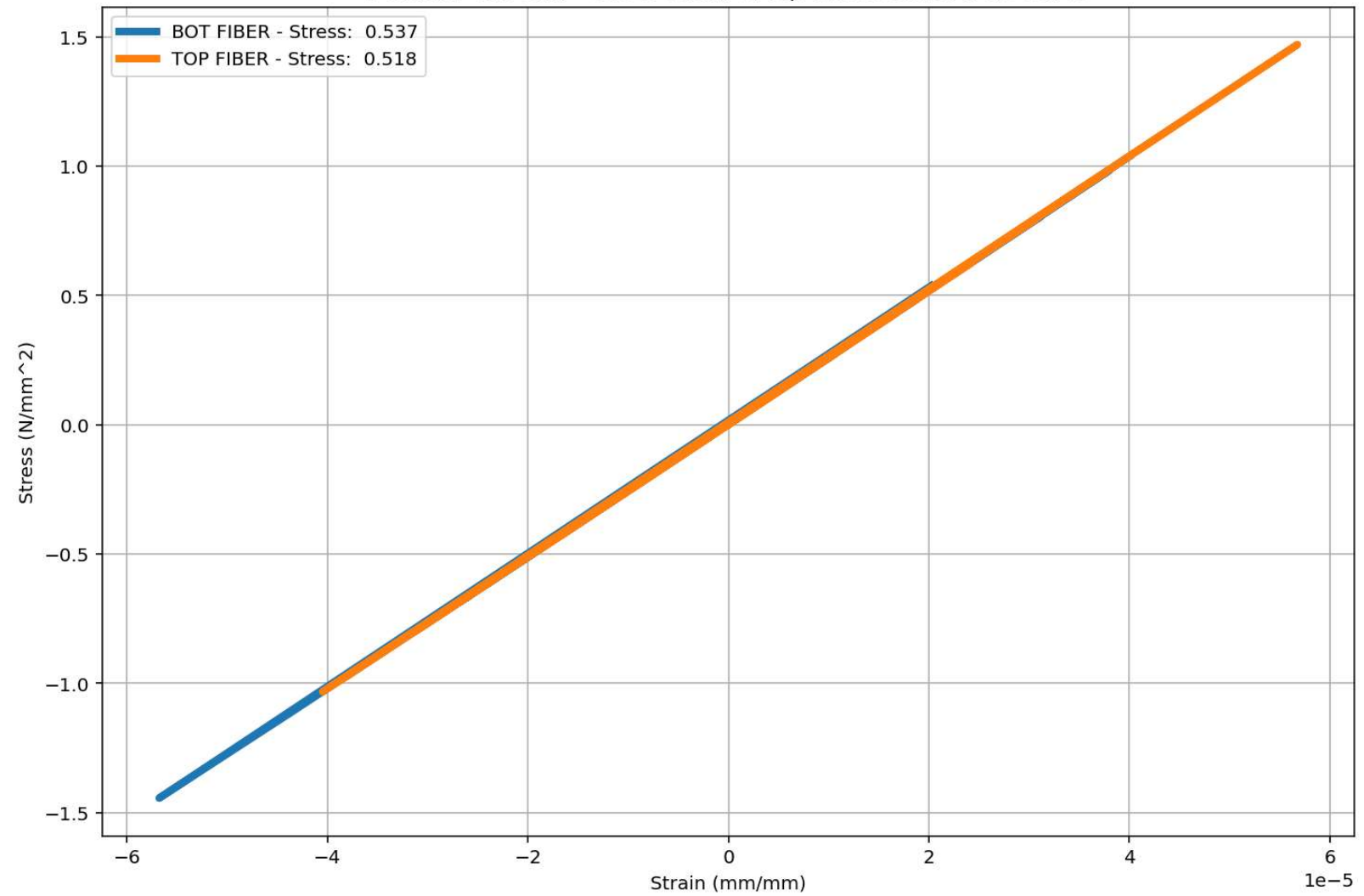


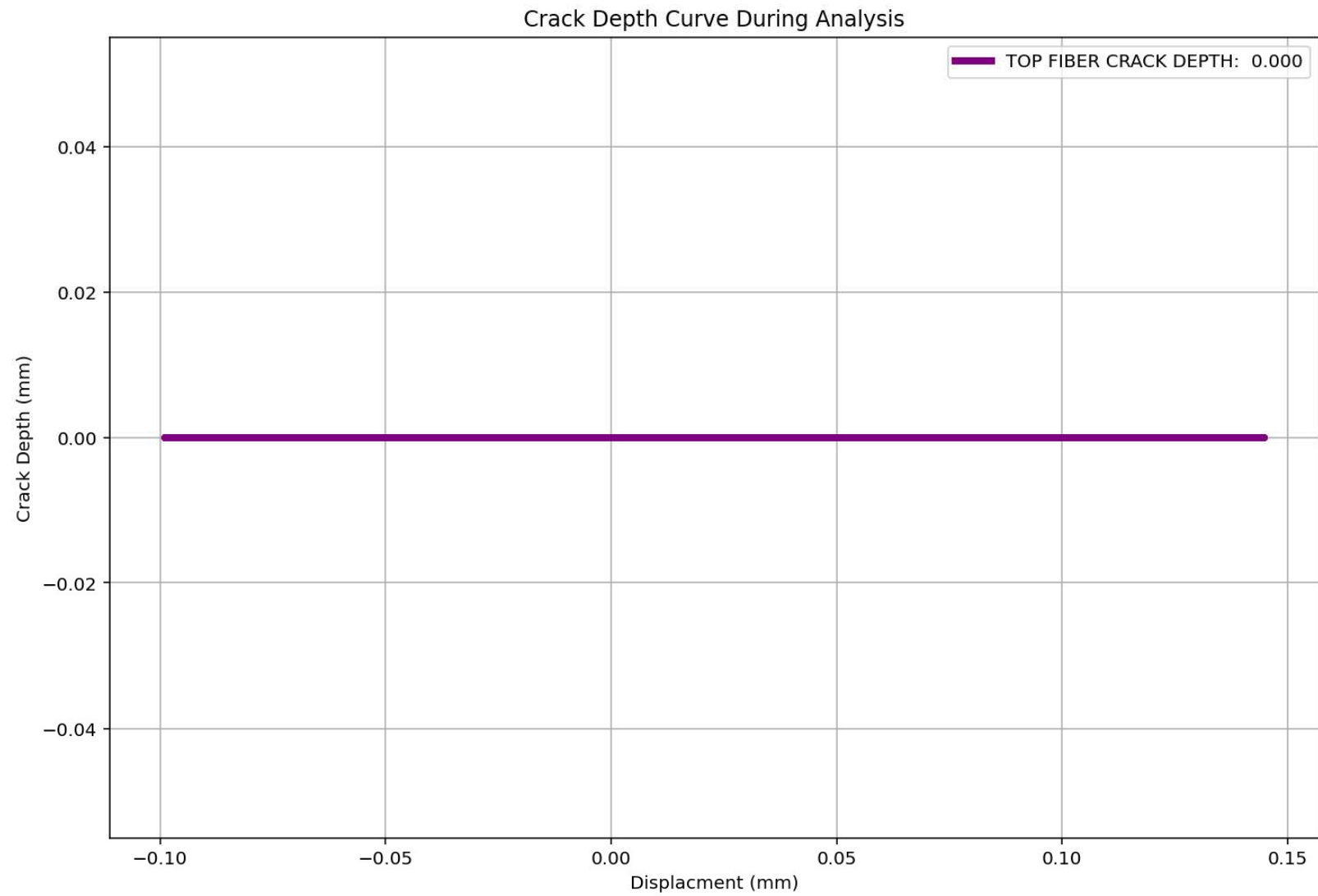
Period of Structure During Seismic Analysis

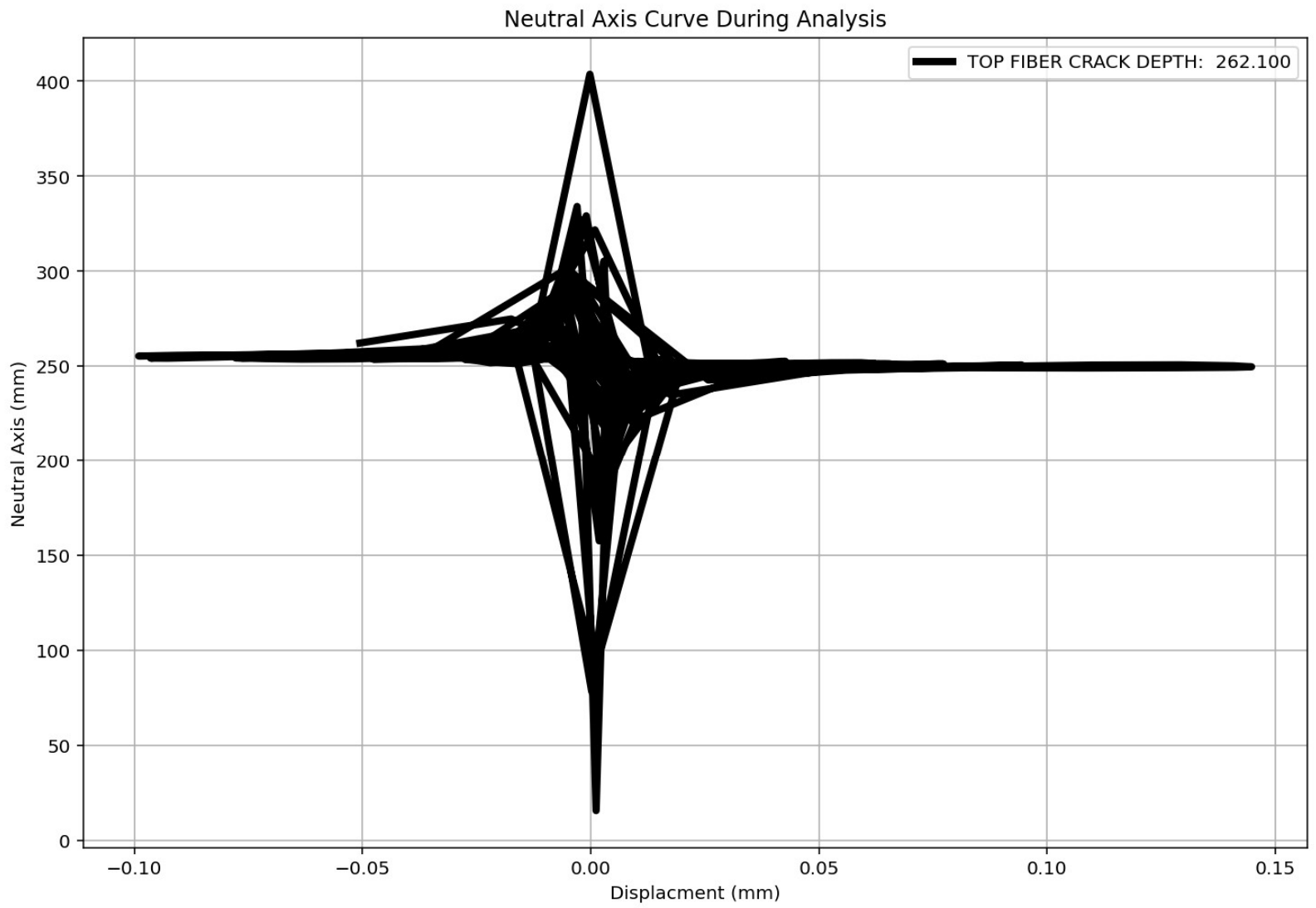


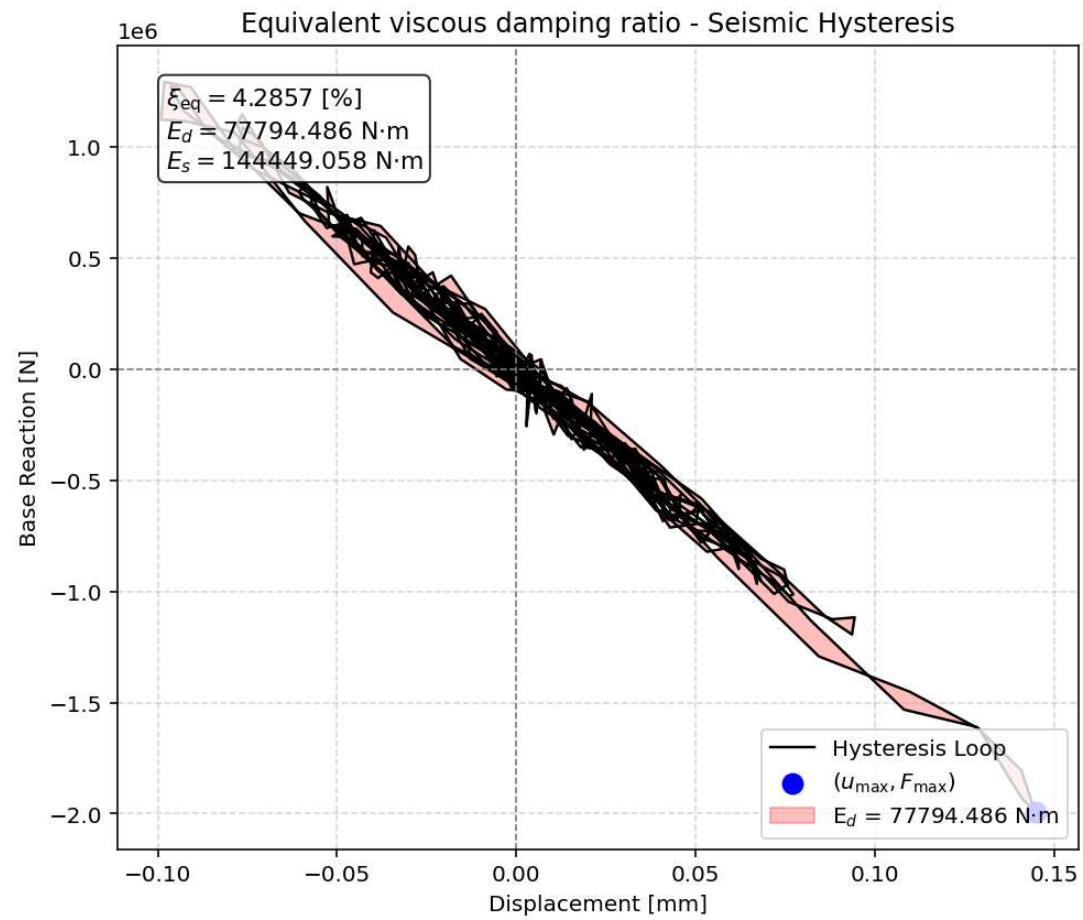


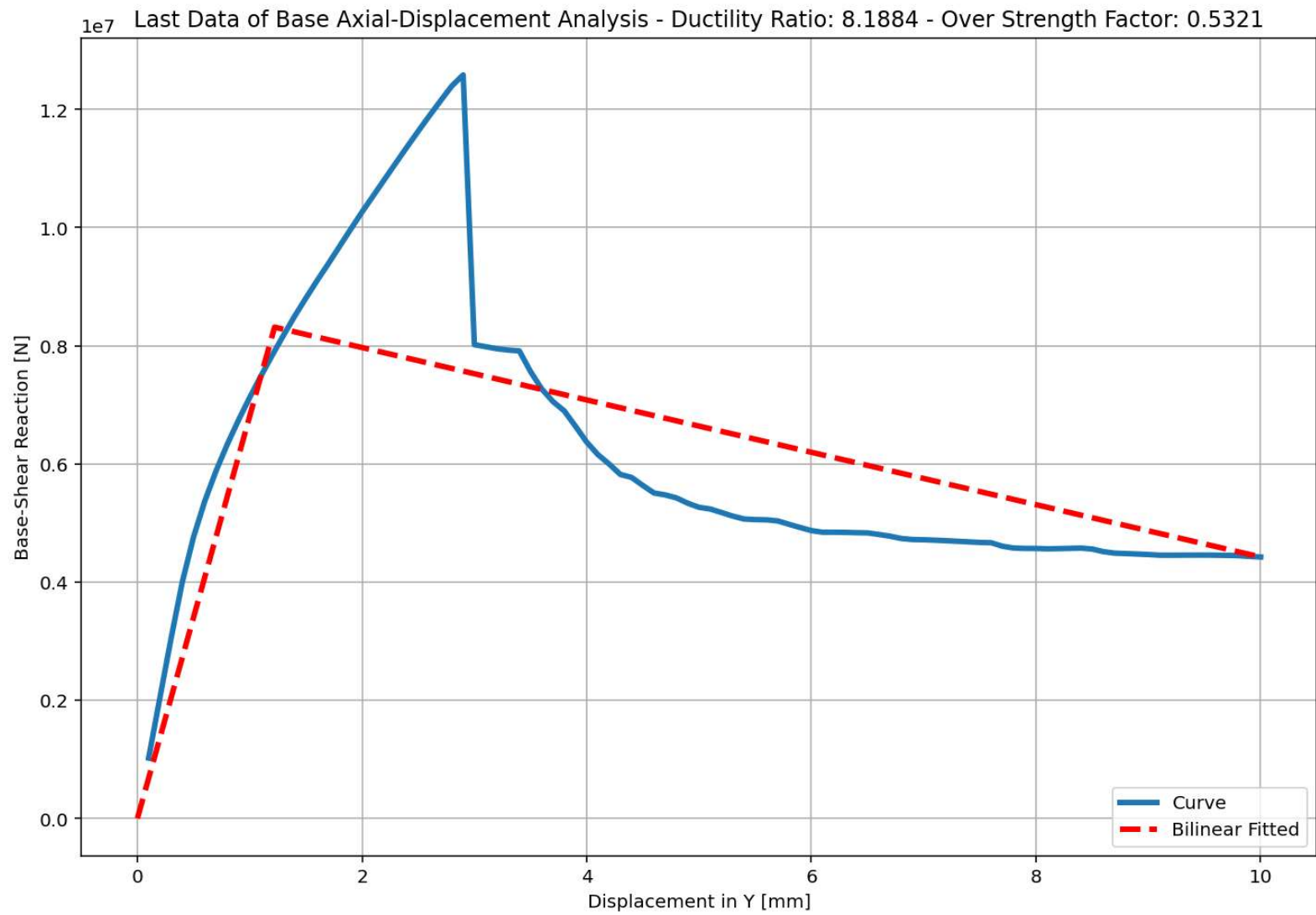
Behavior of beam - Stress-strain of Top and Bottom Fibers Curve











+=====+

= Analysis curve fitted =

Disp Base Shear

-----

[[0.00000000e+00 0.00000000e+00]

[1.06003955e+00 7.29206905e+06]

[1.00000000e+01 4.36432689e+06]]

+=====+

+-----+

Structure Elastic Stiffness : 6879053.73

Structure Plastic Stiffness : 436432.69

Structure Tangent Stiffness : -327489.39

Structure Ductility Ratio : 9.43

Structure Over Strength Factor: 0.60

+-----+

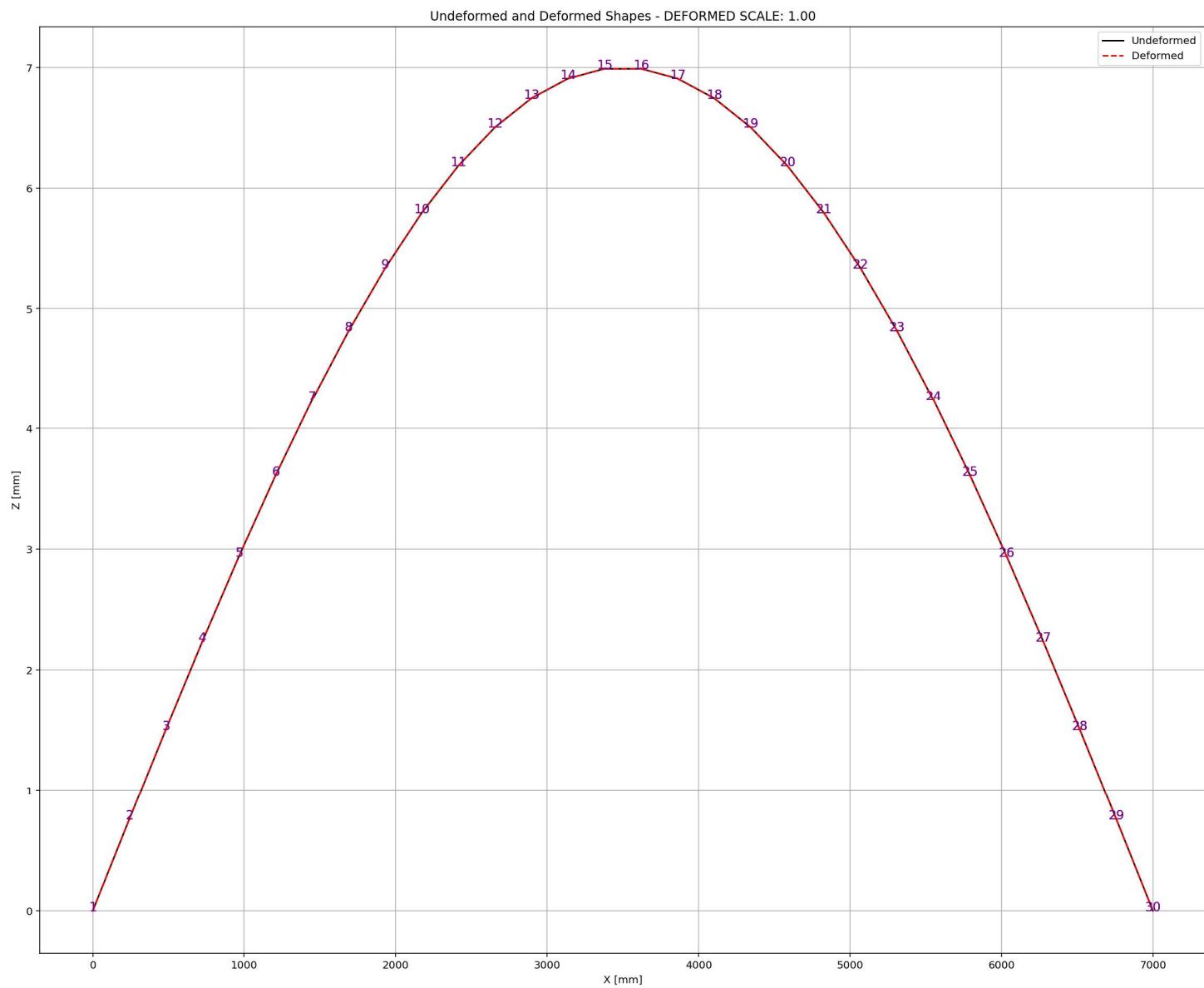
Over Strength Coefficient ( $\Omega$ ): 0.5985

Displacement Ductility Ratio ( $\mu$ ): 9.4336

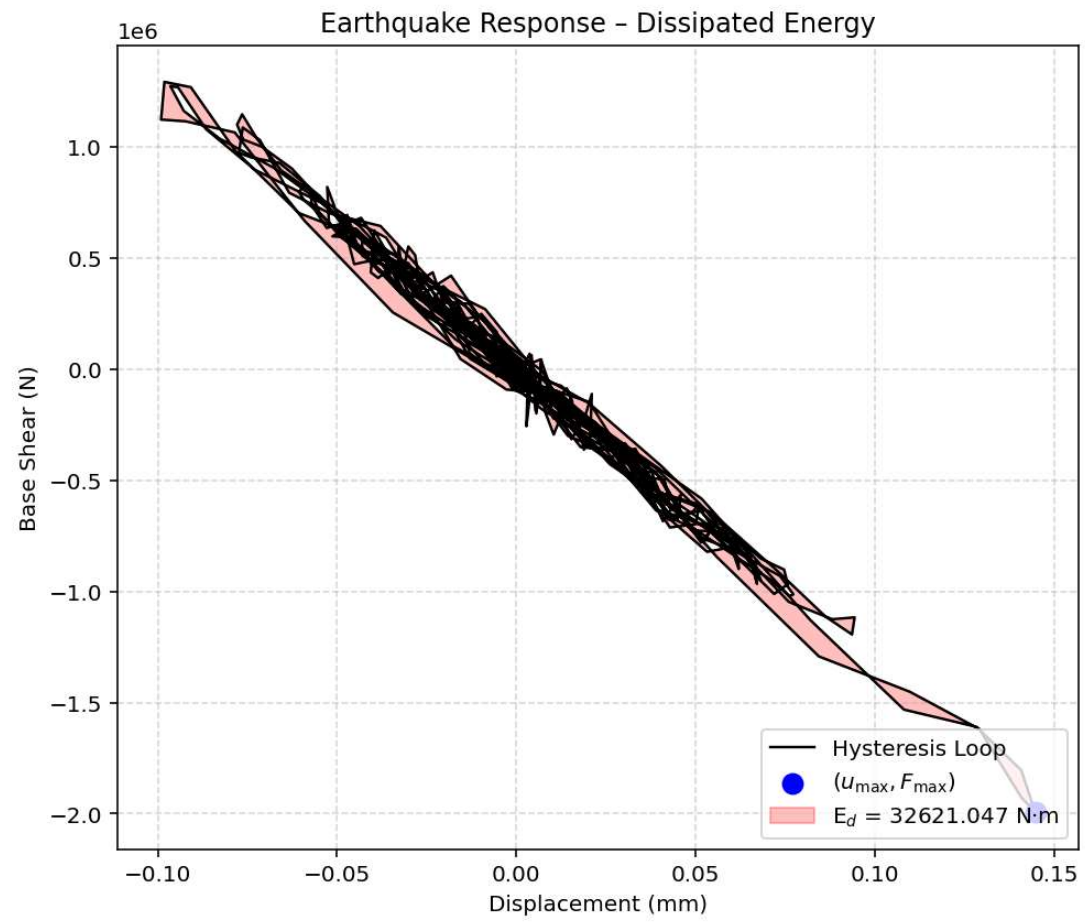
Ductility Coefficient ( $R\mu$ ): 9.4336

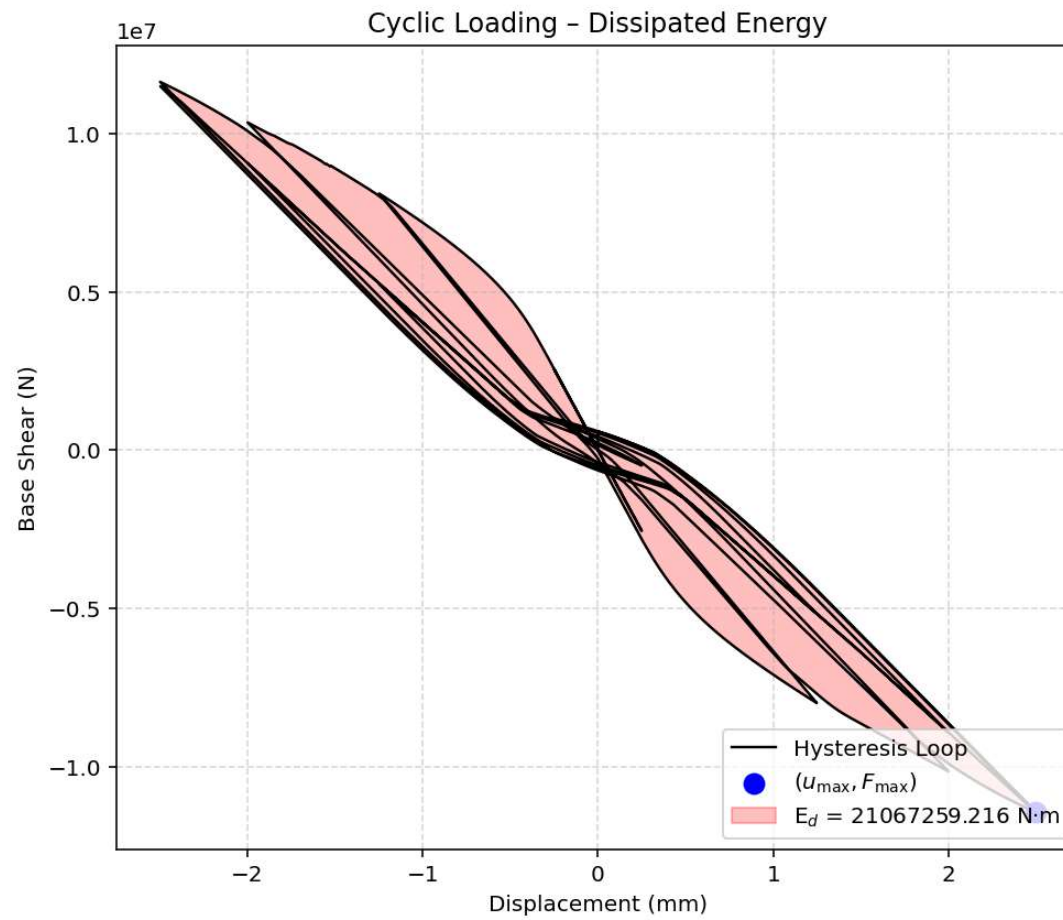
Structural Behavior Coefficient (R): 5.6460

Structural Ductility Damage Index in Y Direction: -10.2375 (%)









**Dissipated Energy from Earthquake= 32621.05 N·mm**

**Dissipated Energy from Cyclic Displacement= 21067259.22 N·mm**

**DISSIPATED ENERGY CAPACITY INDEX: 0.155 [%]**

**ZONE 1: VERY MINOR DAMAGE**

